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Optimal Quantized Compressed Sensing via Projected Gradient Descent

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Abstract

This paper provides a unified treatment to the recovery of structured signals living in a star-shaped set from general quantized measurements $\mathcal{Q}(\mathbf{Ax}-\boldsymbol{\tau})$, where $\mathbf{A}$ is a sensing matrix, $\boldsymbol{\tau}$ is a vector of (possibly random) quantization thresholds, and $\mathcal{Q}$ denotes an L-level quantizer. The ideal estimator of hamming distance minimization (HDM) is optimal but typically infeasible to compute. Under sub-Gaussian design, we study the projected gradient descent (PGD) algorithm with respect to the one-sided ℓ_1-loss and identify the conditions under which PGD achieves the same error rate as HDM, up to logarithmic factors. These conditions include estimates of the separation probability, small-ball probability and some moment bounds that are easy to validate. For multi-bit case, we also develop a complementary approach based on product embedding to show global convergence. When applied to popular models such as 1-bit compressed sensing with Gaussian $\mathbf{A}$ and zero $\boldsymbol{\tau}$ and the dithered 1-bit/multi-bit models with sub-Gaussian $\mathbf{A}$ and uniform dither $\boldsymbol{\tau}$, our unified treatment yields error rates that improve on or match the sharpest results in all instances. Particularly, PGD achieves the information-theoretic optimal rate $\tilde{O}(\frac{k}{mL})$ for recovering k-sparse signals, and the rate $\tilde{O}((\frac{k}{mL})^{1/3})$ for effectively sparse signals. For 1-bit compressed sensing of sparse signals, our result recovers the optimality of normalized binary iterative hard thresholding (NBIHT) that was proved very recently.

Keywords: Quantized compressed sensing, Nonconvex optimization, Nonlinear observations, Gradient descent, Concentration inequality

MSC Numbers: 94A12, 90C26, 49N30

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Contents

1 Introduction

Consider an L-level quantizer $\mathcal{Q}$ which quantizes $q \in \mathbb{R}$ to

$$\mathcal{Q}(q) = \begin{cases} q_1, & \text{if } \ q < b_1 \\ q_2, & \text{if } \ b_1 \le q < b_2 \\ \cdots & \\ q_{L-1}, & \text{if } \ b_{L-2} \le q < b_{L-1} \\ q_L, & \text{if } \ q \ge b_{L-1} \end{cases} \tag{1}$$

for some quantization thresholds $b_1 < b_2 < \cdots < b_{L-1}$ and certain values $q_1 < q_2 < \cdots < q_L$. If $L \ge 3$, we define

$$\Delta := \min_{j=1,\cdots,L-2} |b_{j+1} - b_j| \tag{2}$$

as the resolution of the quantizer $\mathcal{Q}$. For the 1-bit quantizer that outputs either 1 or -1, we let $\Delta = 2$ for technical convenience.[1] Note that $(q_i)_{i=1}^L$ are only used to indicate which quantization cell the unquantized input falls in, hence their exact values are not important; we can thus assume without loss of generality that

$$q_{i+1} = q_i + \Delta, \quad i = 1, 2, \cdots, L-1. \tag{3}$$

See also Figure 1 for an illustration.

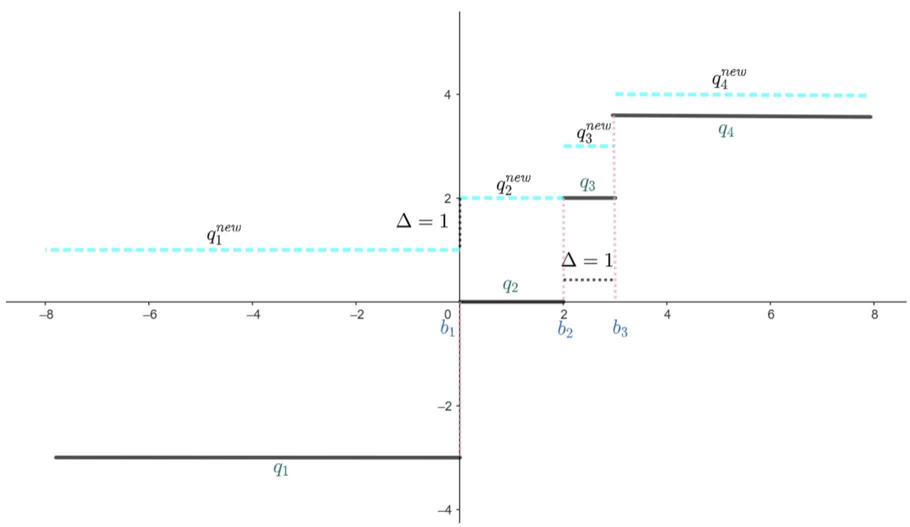

Figure 1: An example of ($L = 4$)-level quantizer. Note that we can assume (3) because, for instance in the right figure, we can always modify $\{q_i\}_{i=1}^4$ to $\{q_i^{new}\}_{i=1}^4$. Similarly, in 1-bit case where $\Delta = 2$, we can always assume $(q_1, q_2) = (-1, 1)$ to fulfill (3).

[1]For (almost) bounded inputs, the 1-bit quantizer can be well approximated by the (shifted) uniform quantizer $\mathcal{Q}_\delta$ in (7) with large enough δ; see, e.g., Figure 2 in [9]. It is thus reasonable to define the resolution of 1-bit quantizer as *some large enough constant*; here, setting $\Delta = 2$ is simply a convenient option for our further development.

Given m sensing vectors $\{\mathbf{a}_i\}_{i=1}^m$ from $\mathbb{R}^n$, a set $\mathcal{K}$ in $\mathbb{R}^n$ that captures signal structure, the goal of quantized compressed sensing is to recover $\mathbf{x}$ from

$$y_i = \mathcal{Q}(\mathbf{a}_i^\top \mathbf{x} - \tau_i), \qquad i = 1, 2, \cdots, m, \tag{4}$$

where τ_i denotes either fixed quantization threshold or randomly drawn *dither*. For the latter case, as with [13, 22, 36, 39], we focus on uniform dithers $\{\tau_i\}_{i=1}^m$ that are i.i.d. uniformly distributed over $[-\Lambda, \Lambda]$ and independent of everything else. We shall refer to Λ as the dithering level and simply set $\Lambda = 0$ to return setting without dithering. In the vector form, we can write

$$\mathbf{y} = \mathcal{Q}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau}),$$

where $\mathbf{A} = [\mathbf{a}_1, \cdots, \mathbf{a}_m]^\top \in \mathbb{R}^{m \times n}$, $\boldsymbol{\tau} = [\tau_1, \cdots, \tau_m]^\top$, and the quantizer is applied in an elementwise fashion.

On the signal structure, we shall assume that $\mathcal{K}$ is star-shaped set satisfying

$$t\mathcal{K} \subset \mathcal{K}, \quad \forall t \in [0, 1],$$

which means that $\mathbf{u} \in \mathcal{K}$ implies $t\mathbf{u} \in \mathcal{K}$ for any $t \in [0, 1]$. This general assumption was adopted in prior works such as [33, 34], covering many interesting examples, including sparse vectors in

$$\mathcal{K} = \Sigma_k^n = \{\mathbf{u} \in \mathbb{R}^n : \mathbf{u} \text{ is } k\text{-sparse}\},$$

low-rank matrices in

$$\mathcal{K} = M_{\bar{r}}^{n_1, n_2} = \{\mathbf{M} \in \mathbb{R}^{n_1 \times n_2} : \mathrm{rank}(\mathbf{M}) \le \bar{r}\},$$

and effectively sparse signals in

$$\mathcal{K} = \sqrt{k}\mathbb{B}_1^n := \{\mathbf{u} \in \mathbb{R}^n : \|\mathbf{u}\|_1 \le \sqrt{k}\},$$

just to name a few. We additionally use

$$\mathbb{A}_\alpha^\beta = \{\mathbf{u} \in \mathbb{R}^n : \alpha \le \|\mathbf{u}\|_2 \le \beta\}$$

with $\beta \ge \alpha \ge 0$ to capture the norm constraint on the signals. Reserving $\mathcal{X}$ for the signal space, we are interested in the recovery of signals $\mathbf{x}$ in

$$\mathcal{X} := \mathcal{K} \cap \mathbb{A}_\alpha^\beta$$

from $\mathbf{y} = \mathcal{Q}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau})$.

Our general framework accommodates some of the most common quantized compressed sensing models as special cases.

1-bit compressed sensing (1bCS). The most common and well-understood model is the so-called 1-bit compressed sensing (e.g., [2, 5, 17, 19, 21, 24, 30–34]) where we seek to recover $\mathbf{x} \in \mathcal{K}$ from

$$\mathbf{y} = \operatorname{sign}(\mathbf{A}\mathbf{x}). \tag{5}$$

In this case, we have $\Delta = 2$ (by our convention), $\Lambda = 0$, and (3) is satisfied. Note that the observations provide no information on the signal norm $\|\mathbf{x}\|_2$, thus we set $\alpha = \beta = 1$ and hence $\mathbb{A}_\alpha^\beta = \mathbb{S}^{n-1} := \{\mathbf{u} \in \mathbb{R}^n : \|\mathbf{u}\|_2 = 1\}$ restricts our attention to the signals in $\mathcal{X} := \mathcal{K} \cap \mathbb{S}^{n-1}$. It is well known that accurate reconstruction can be achieved by efficient algorithms when $\mathbf{A}$ is an i.i.d. Gaussian ensemble (e.g., [30, 31, 33]). These results, however, cannot be extended to sub-gaussian measurements without additional assumptions. See, e.g., [1, 20].

Dithered 1-bit compressed sensing (D1bCS). Suppose that $\boldsymbol{\tau}$ is uniformly distributed over $[-\lambda, \lambda]^m$ for some $\lambda > 0$, we seek to recover $\mathbf{x}$ from measurements

$$\mathbf{y} = \operatorname{sign}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau}). \tag{6}$$

In this case, we have $\Delta = 2$ (by convention), $\Lambda = \lambda$, and (3) holds. It was also commonly assumed that the signals have ℓ_2-norms bounded by R for some $R > 0$. See, e.g., [12–14, 22, 36]. Without loss of generality, we shall set $(\alpha, \beta) = (0, 1)$ so that $\mathbb{A}_\alpha^\beta = \mathbb{B}_2^n := \{\mathbf{u} \in \mathbb{R}^n : \|\mathbf{u}\|_2 \le 1\}$ and consider signals in $\mathcal{X} := \mathcal{K} \cap \mathbb{B}_2^n$. Accurate reconstruction with norm can be achieved under sub-Gaussian $\mathbf{A}$ (as precised by Assumption 1). See, e.g., [13, 22, 36].

Dithered multi-bit compressed sensing (DMbCS). More generally, we can consider a resolution-δ uniform quantizer

$$\mathcal{Q}_\delta(a) = \delta\Big(\Big\lfloor \frac{a}{\delta} \Big\rfloor + \frac{1}{2}\Big). \tag{7}$$

Under the uniform dither $\boldsymbol{\tau} \sim \mathscr{U}([-\frac{\delta}{2}, \frac{\delta}{2}]^m)$ and sub-Gaussian $\mathbf{A}$, we can accurately recover structured signals from

$$\mathbf{y} = \mathcal{Q}_\delta(\mathbf{A}\mathbf{x} - \boldsymbol{\tau}). \tag{8}$$

See, e.g., [19, 36, 39]. While $\mathcal{Q}_\delta(a)$ does not immediately sample *finite bits* from a (note that the range of $\mathcal{Q}_\delta$ is the infinite set $\{(l - \frac{1}{2})\delta : l \in \mathbb{Z}\}$), it is often more practical to use a uniform quantizer with saturation in applications. Specifically, for some even integer $L \ge 4$, we consider the less informative version of $\mathcal{Q}_\delta$ defined as

$$\mathcal{Q}_{\delta,L}(a) = \mathcal{Q}_\delta(a) \cdot \mathbb{1}\Big(|a| < \frac{L\delta}{2}\Big) + \frac{(L-1)\delta}{2}\mathbb{1}\Big(a \ge \frac{L\delta}{2}\Big) + \frac{(1-L)\delta}{2}\mathbb{1}\Big(a \le \frac{-L\delta}{2}\Big). \tag{9}$$

Again we set $\mathbb{A}_\alpha^\beta = \mathbb{B}_2^n$. In this case, we have $\Delta = \delta$, $\Lambda = \frac{\delta}{2}$, and (3) is satisfied. See Figure 2 for an intuitive illustration of $\mathcal{Q}_\delta$ and $\mathcal{Q}_{\delta,L}$. Our goal is to reconstruct $\mathbf{x} \in \mathcal{X} := \mathcal{K} \cap \mathbb{B}_2^n$ from

$$\mathbf{y} = \mathcal{Q}_{\delta,L}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau}). \tag{10}$$

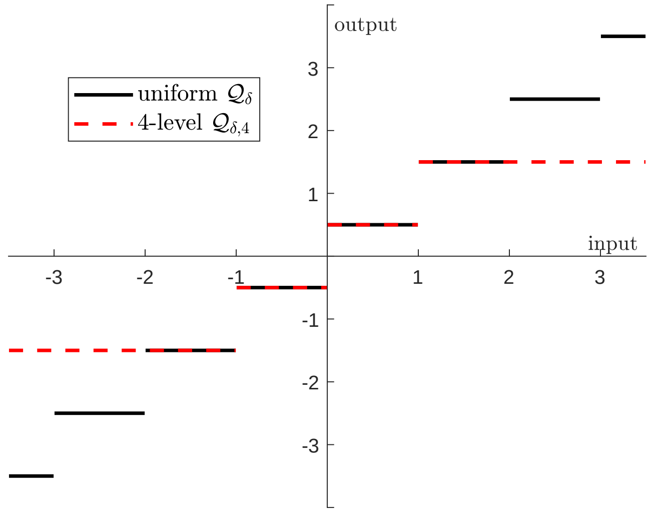

Figure 2: The uniform quantizer $\mathcal{Q}_\delta$ and its saturated version $\mathcal{Q}_{\delta,4}$ under the resolution $\delta = 1$.

For recovering k-sparse signals from $\mathcal{Q}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau})$ with any fixed $(\mathbf{A}, \boldsymbol{\tau}) \in \mathbb{R}^{m\times n} \times \mathbb{R}^{m\times 1}$, the information-theoretic lower bound is known to be $\Omega(\frac{k}{mL})$. On the other hand, an ideal decoding strategy is to return an estimate having quantized measurements coincident with the observations,[2] which can be formulated as constrained hamming distance minimization (HDM). This ideal decoder attains the optimal rate $\tilde{O}(\frac{k}{m})$ for 1bCS of k-sparse signals with Gaussian $\mathbf{A}$ and D1bCS with sub-Gaussian matrix. See [13, 21, 28]. Moreover, the exact sparsity rarely occurs and may not be a realistic signal structure for practitioners, hence effectively sparse signals residing in a scaled ℓ_1-ball are of particular interest. It was also known [13, 28] that HDM achieves uniform error rate $\tilde{O}((\frac{k}{m})^{1/3})$ for recovering signals in $\sqrt{k}\mathbb{B}_1^n$ in 1bCS and D1bCS. As a side contribution, we will unify these information-theoretic bounds on the converse or attainability in Section 2. Particularly, we reproduce the lower bound $\Omega(\frac{k}{mL})$ in Theorem 1, and then in Theorem 3, we show HDM achieves the error rates stated above for any quantized compressed sensing systems with sub-Gaussian sensing vectors and the following two essential components:

- (Small-ball probability) an upper bound on the probability of the unquantized measurement (after dithering) being close to some b_j;
- (Separation probability) a lower bound on the probability of two signals having different quantized measurement.[3]

The main goal of this work is to devise efficient algorithms that can achieve the optimal reconstruction error rate. Let us first review some of the computationally efficient and theoretically guaranteed algorithms in this area.

For noiseless 1bCS, the linear programming approach that achieves $\tilde{O}((\frac{k}{m})^{1/5})$ appeared as the first such an algorithm [31], then the convex relaxation [30] was shown to achieve $\tilde{O}((\frac{k}{m})^{1/4})$ as well as the robustness to a set of noise patterns. The recent work [11] studied a more general binary classification context and showed that Adaboost achieves an error rate $\tilde{O}((\frac{k}{m})^{1/3})$. While these algorithms are

[2] Let us focus on the noiseless case for the ease of presentation; we will demonstrate in Section 6 that the robustness to adversarial corruption can be obtained by slightly more work.

[3] See Appendix C.1 for the reason why this is referred to as separation probability.

capable of recovering effectively sparse signals, when restricted to k-sparse signals, generalized Lasso and projected-back projection (PBP) achieve the faster rate of $\tilde{O}((\frac{k}{m})^{1/2})$ [33, 34].

For D1bCS and DMbCS that are compatible with *sub-Gaussian* sensing matrix, generalized Lasso achieves the non-uniform error rate of $\tilde{O}((\frac{\Delta k}{m})^{1/2})$ for a fixed signal in Σ_k^n [36]. Note that $\Delta = \delta$ in DMbCS, and δ in the results for $\mathcal{Q}_\delta$ could be (roughly) thought of as $1/L$—the reason is that $\mathcal{Q}_{\delta,L}$ can cover inputs bounded by $\frac{L\delta}{2}$ without saturation, and $L\delta = \tilde{O}(1)$ is typically sufficient under sub-Gaussian matrix. Uniformly for all effectively sparse signals living in an ℓ_2-ball, minimizing the one-sided ℓ_1-loss over the signal space achieves $\tilde{O}((\frac{\Delta k}{m})^{1/3})$ [22]. This is sharper than $\tilde{O}((\frac{k}{m})^{1/4})$ achieved by an earlier convex relaxation approach [13] for D1bCS, which is based on a loss function being ignorant of the dithers $(\tau_i)_{i=1}^m$. For DMbCS, PBP surprisingly provides accurate estimate under RIP matrices [39], but note that its sharpest error rate $\tilde{O}((1+\delta)(\frac{k}{m})^{1/2})$ for k-sparse signals exhibits a worse leading factor than generalized Lasso when $\delta \ll 1$. While the results for generalized Lasso in [36] are nonuniform, the recent work [19] established the uniform error rate $\tilde{O}(\sqrt{\Delta}(\frac{k}{m})^{1/4})$ over all effectively sparse signals for 1bCS, D1bCS and the uniform quantized model (8).

In summary, all the results reviewed above are no faster than $\tilde{O}(\Delta(\frac{k}{m})^{1/2})$ in recovering k-sparse signals, thereby essentially inferior to the optimal rate $O(\frac{k}{mL})$; and are no faster than $\tilde{O}((\frac{\Delta k}{m})^{1/3})$ in recovering effectively sparse signals, for which the optimal rate, to our best knowledge, remains unclear (see Remark 2 for further discussion).

Following some prior attempts [17, 23], normalized binary iterative hard thresholding (NBIHT) was recently shown by [24] to be optimal in recovering all $\mathbf{x} \in \Sigma_k^n \cap \mathbb{S}^{n-1}$ from the 1bCS measurements $\mathbf{y} = \text{sign}(\mathbf{A}\mathbf{x})$. That is, NBIHT achieves a uniform error rate of $\tilde{O}(\frac{k}{m})$ in ℓ_2-norm. Similar proof ideas were used to establish a similar efficient and optimal algorithm for one-bit phase retrieval [10]. Nonetheless, similar results are still lacking for other models such as D1bCS and DMbCS, as well as other signal structures like low-rank matrices and effectively sparse signals. Our results fill in this void.

In particular, we propose a *projected gradient descent* (PGD) algorithm for the general quantized compressed sensing problem. The algorithm performs (sub)gradient descent with respect to the one-sided ℓ_1-loss, followed by the sequential projections onto the star-shaped set $\mathcal{K}$ and the norm constraint set $\mathbb{A}_\alpha^\beta$. Note that the computational efficiency of PGD only hinges on the projection onto $\mathcal{K}$, which can be computed efficiently for (effectively) sparse vectors and low-rank matrices. Our major theoretical finding is as follows: under suitable conditions, including those for analyzing HDM and some *additional moment bounds* which convey a type of independence between the marginals of the sensing vector and the separation event (see Assumptions 5–7 and relevant interpretations),

PGD achieves essentially the same error rate as HDM, up to logarithmic factors.

Our proof strategies are inspired by arguments in [10, 24], while as will be clear, we provide significant generalizations (in terms of signal structure captured by a star-shaped set, the model, the algorithm per se) and new technical contributions (especially in the multi-bit case).

The convergence of PGD follows from a type of *restricted approximate invertibility condition* (RAIC) with respect to the interplay among several parties: the quantized sampler $(\mathcal{Q}, \mathbf{A}, \boldsymbol{\tau})$, the star-shaped set $\mathcal{K}$, and the step size η. With true signal $\mathbf{x}$, loss function $\mathcal{L}(\mathbf{u})$ and step size η, RAIC provides a uniform bound on the dual norm of $\mathbf{u} - \mathbf{x} - \eta\partial\mathcal{L}(\mathbf{u})$ with respect to certain low-complexity set. Thus,

Table 1: A Partial Review of Efficient Algorithms for Recovering (Effectively) Sparse Signals

1bCS algorithm	error rate	signal space	uniformity
Linear Program [31]	$\tilde{O}((\frac{k}{m})^{1/5})$	$\sqrt{k}\mathbb{B}_1^n \cap \mathbb{S}^{n-1}$	✓
Convex Relaxation [30]	$\tilde{O}((\frac{k}{m})^{1/4})$	$\sqrt{k}\mathbb{B}_1^n \cap \mathbb{S}^{n-1}$	✗
Generalized Lasso [33]	$\tilde{O}((\frac{k}{m})^{1/2})$	$\Sigma_k^n \cap \mathbb{S}^{n-1}$	✗
PBP [34]	$\tilde{O}((\frac{k}{m})^{1/2})$	$\Sigma_k^n \cap \mathbb{S}^{n-1}$	✗
Adaboost [11]	$\tilde{O}((\frac{k}{m})^{1/3})$	$\sqrt{k}\mathbb{B}_1^n \cap \mathbb{S}^{n-1}$	✓
PGD (a.k.a. NBIHT) [24]	$\tilde{O}(\frac{k}{m})$	$\Sigma_k^n \cap \mathbb{S}^{n-1}$	✓
PGD	$\tilde{O}((\frac{k}{m})^{1/3})$	$\sqrt{k}\mathbb{B}_1^n \cap \mathbb{S}^{n-1}$	✓
D1bCS algorithm	**error rate**	**signal space**	**uniformity**
Convex Relaxation [13]	$\tilde{O}((\frac{k}{m})^{1/4})$	$\sqrt{k}\mathbb{B}_1^n \cap \mathbb{B}_2^n$	✓
Con. Rel. w.r.t. (17) [22]	$\tilde{O}((\frac{k}{m})^{1/3})$	$\sqrt{k}\mathbb{B}_1^n \cap \mathbb{B}_2^n$	✓
Generalized Lasso [36]	$\tilde{O}((\frac{k}{m})^{1/2})$	$\Sigma_k^n \cap \mathbb{B}_2^n$	✗
PGD	$\tilde{O}(\frac{k}{m})$	$\Sigma_k^n \cap \mathbb{B}_2^n$	✓
PGD	$\tilde{O}((\frac{k}{m})^{1/3})$	$\sqrt{k}\mathbb{B}_1^n \cap \mathbb{B}_2^n$	✓
DMbCS algorithm	**error rate**	**signal space**	**uniformity**
Con. Rel. w.r.t. (17) [22]	$\tilde{O}((\frac{k}{mL})^{1/3})$	$\sqrt{k}\mathbb{B}_1^n \cap \mathbb{B}_2^n$	✓, $\mathcal{Q}_{\delta,L}$
Generalized Lasso [8, 36]	$\tilde{O}(\delta(\frac{k}{m})^{1/2})$	$\Sigma_k^n \cap \mathbb{B}_2^n$	[36, ✗, $\mathcal{Q}_\delta$] [8, ✓, $\mathcal{Q}_\delta$]
PBP [39]	$\tilde{O}((1+\delta)(\frac{k}{m})^{1/2})$	$\Sigma_k^n \cap \mathbb{B}_2^n$	✓, $\mathcal{Q}_\delta$
PGD	$\tilde{O}(\frac{k}{mL})$	$\Sigma_k^n \cap \mathbb{B}_2^n$	✓, $\mathcal{Q}_{\delta,L}$
PGD	$\tilde{O}((\frac{k}{mL})^{1/3})$	$\sqrt{k}\mathbb{B}_1^n \cap \mathbb{B}_2^n$	✓, $\mathcal{Q}_{\delta,L}$

* We only review results for 1bCS under standard Gaussian $\mathbf{A}$, D1bCS/DMbCS under sub-Gaussian $\mathbf{A}$; we review the *sharpest* result for a work containing multiple guarantees of different flavours.

† The result is said to be uniform (non-uniform, resp.) if it holds for all signals (a fixed signal, resp.) in $\mathcal{X}$ with a single draw of the sensing ensemble.

‡ For DMbCS, we enclose under "uniformity" whether the result is for $\mathcal{Q}_\delta$ (8) or the less informative $\mathcal{Q}_{\delta,L}$ (10).

RAIC essentially conveys a message that the actual subgradient descent step $\eta\mathcal{L}(\mathbf{u})$ approximates the "ideal" step $\mathbf{u}-\mathbf{x}$, up to some error terms. It is interesting to note that such message departs from the *regularity condition* commonly used in the literature (e.g., [7]), and we expect that RAIC is interesting in its own right and will prove useful in many other nonconvex optimization problems.

To establish RAIC under the proposed conditions, we invoke a covering argument similar to [10, 24] with notable generalizations in that our analysis is built upon generic assumptions, and with a set of refinements such as we no longer need to separately treat the large-distance regime and small-distance regime. New and fundamental challenges arise in the multi-bit setting where the more intricate gradient adds great challenge to the above-described analysis and forbids a unified treatment to the general context of quantized compressed sensing. To overcome these difficulties, we propose to *clip* the gradient to that in the 1-bit case and then control the deviation caused by gradient clipping. This idea proves effective if we restrict our attention to two points having distance much smaller than the quantization resolution. The main intuition is that *the multi-bit quantizer behaves similarly to the 1-bit quantizer in terms of its ability to distinguish such two close enough signals.* Nonetheless, this approach only leads to local convergence for the multi-bit case. We then introduce a complementary approach, which is based on a type of *product embedding* (or *projection distortion*) property seen in the literature for analyzing other estimators, to establish global convergence.

To apply our unified framework to a specific model, it then suffices to validate the conditions, which can be accomplished by elementary calculations and estimations. We will particularly specialize our theory to the popular models 1bCS, D1bCS and DMbCS, showing that PGD achieves error rates which *improve on or match the sharpest existing results in all contexts.* In particular, we summarize in Table 1 both existing results and the ones we establish for recovering (effectively) sparse signals.

While prior works [17, 24] are only for sparse recovery, our works also straightforwardly yield low-rank recovery guarantees: for all three models (1bCS, D1bCS and DMbCS), under appropriate Frobenius norm constraint, our PGD algorithm can recover exactly low-rank matrices in $M_{\bar{r}}^{n_1,n_2}$ with an error rate of $\tilde{O}(\frac{\bar{r}(n_1+n_2)}{mL})$, and the effectively low-rank matrices with nuclear norm bounded by $\sqrt{\bar{r}}$ to an error rate of $\tilde{O}((\frac{\bar{r}(n_1+n_2)}{mL})^{1/3})$.

We introduce some generic notation used throughout the work. Regular letters denote scalars, and boldface letters denote vectors and matrices. We denote by $\mathbb{B}_2^n(\mathbf{u};r)$ the ℓ_2-ball with radius r and center $\mathbf{u}$, shortened to $\mathbb{B}_2^n(r)$ if $\mathbf{u}=0$. For $a,b\in\mathbb{R}$ we write $a\vee b=\max\{a,b\}$ and $a\wedge b=\min\{a,b\}$. For $\mathcal{U},\mathcal{V}\subset\mathbb{R}^n$, $\mathcal{P}_{\mathcal{U}}(\cdot)$ denotes the projection onto $\mathcal{U}$ under ℓ_2-norm, and we let $t\mathcal{U}=\{t\mathbf{u}:\mathbf{u}\in\mathcal{U}\}$ and $\mathcal{U}-\mathcal{V}=\{\mathbf{u}-\mathbf{v}:\mathbf{u}\in\mathcal{U},\mathbf{v}\in\mathcal{V}\}$. The localization of $\mathcal{U}$ to the radius of r is defined as $\mathcal{U}_{(r)}:=(\mathcal{U}-\mathcal{U})\cap\mathbb{B}_2^n(r)$. We work with the dual norm $\|\mathbf{u}\|_{\mathcal{U}^\circ}=\sup_{\mathbf{w}\in\mathcal{U}}\mathbf{w}^\top\mathbf{u}$.

Two natural quantities are used to characterize the complexity of a set $\mathcal{U}$: the Gaussian width $\omega(\mathcal{U})=\mathbb{E}\sup_{\mathbf{u}\in\mathcal{U}}\langle\mathbf{g},\mathbf{u}\rangle$ where $\mathbf{g}\sim\mathcal{N}(0,\mathbf{I}_n)$ and $\langle\mathbf{g},\mathbf{u}\rangle=\mathbf{g}^\top\mathbf{u}$, and the metric entropy $\mathscr{H}(\mathcal{U},r)=\log\mathscr{N}(\mathcal{U},r)$ where $\mathscr{N}(\mathcal{U},r)$ is the minimal number of radius-r ℓ_2-ball needed to cover $\mathcal{U}$, that is, the covering number. We also denote by $|\mathcal{U}|$ the cardinality of a finite set $\mathcal{U}$. The sub-Gaussian norm of a random variable X is defined as $\|X\|_{\psi_2}=\inf\{t>0:\mathbb{E}\exp(\frac{X^2}{t^2})\leq 2\}$, while $\|X\|_{L^p}=(\mathbb{E}|X|^p)^{1/p}$ denotes the L_p-norm. The sub-Gaussian norm of a random vector $\mathbf{X}$ is defined as $\|\mathbf{X}\|_{\psi_2}=\sup_{\mathbf{v}\in\mathbb{S}^{n-1}}\|\mathbf{v}^\top\mathbf{X}\|_{\psi_2}$.

We use $C_0,C_1,C_2,\cdots,c_0,c_1,c_2,\cdots$ to denote constants that may be universal or depend on other quantities that will be specified. Their values may vary from line to line. We write $T_1\lesssim T_2$ or $T_1=O(T_2)$ if $T_1\leq C_1T_2$ for some C_1, and write $T_1\gtrsim T_2$ or $T_1=\Omega(T_2)$ if $T_1\geq c_1T_1$ for some c_1. Note that $\tilde{O}(\cdot)$ and $\tilde{\Omega}(\cdot)$ are the less precise versions of $O(\cdot)$ and $\Omega(\cdot)$ which ignore logarithmic factors. The symbol "$\sim$" will be used to connect identical distributions (that may be represented by certain random variables).

The rest of the paper is organized as follows. We first present the information-theoretic bounds in Section 2. Section 3 formally introduces PGD and its unified analysis. In Section 4 we specialize the theory to 1bCS, D1bCS and DMbCS. We further complement our theoretical results via a number of numerical simulations in Section 5. We close the paper by a set of discussions in Section 6. Most of the proofs are relegated to the appendix.

2 Information-Theoretic Bounds

We focus on non-adaptive memoryless quantization where both $\mathbf{A}$ and $\boldsymbol{\tau}$ are drawn before observing any measurement. Note that the information-theoretic lower bounds are well established in the literature (see, e.g., [6, 10, 21]) and follow from a simple counting argument. We include a proof in Appendix B for the sake of completeness.

Theorem 1 (Information-theoretic lower bound)**.** *Suppose there is a K-dimensional linear subspace V_K contained in $\mathcal{K}$. Given $\mathbf{A}\in\mathbb{R}^{m\times n}$, $\boldsymbol{\tau}\in\mathbb{R}^m$ and $\beta>\alpha\geq 0$, our goal is to recover $\mathbf{x}\in\mathcal{W}:=\mathcal{K}\cap\mathbb{A}_{\alpha,\beta}$ from $\mathbf{y}=\mathcal{Q}(\mathbf{A}\mathbf{x}-\boldsymbol{\tau})$ where $\mathcal{Q}$ is the L-level quantizer as per (1), or recover $\mathbf{x}\in\mathcal{W}:=\mathcal{K}\cap\mathbb{S}^{n-1}$ from $\mathbf{y}=\operatorname{sign}(\mathbf{A}\mathbf{x})$ as per 1bCS (5). If $(L-1)m\geq K$, then for any decoder that takes $\mathbf{y}$ as input and returns $\hat{\mathbf{x}}$ as an estimate of $\mathbf{x}$, there exists some constant $c>0$ that may depend on (α,β) such that*

$$\sup_{\mathbf{x}\in\mathcal{W}}\|\hat{\mathbf{x}}-\mathbf{x}\|_2\geq\frac{cK}{mL}.$$

Remark 1. *Theorem 1 suggests that for recovering k-sparse signals from $\mathcal{Q}(\mathbf{A}\mathbf{x}-\boldsymbol{\tau})$, no algorithm can achieve an error rate faster than $O(\frac{k}{mL})$. Similarly, because there is a $\bar{r}(n_1\vee n_2)$-dimensional subspace contained in $M_{\bar{r}}^{n_1,n_2}$, no algorithm achieves error rate faster than $O(\frac{\bar{r}(n_1+n_2)}{mL})$ for recovering rank-$\bar{r}$ matrices in $\mathbb{R}^{n_1\times n_2}$.*

Remark 2. *Relatively little is known about the information-theoretic lower bound for recovering effectively sparse signals in $\mathcal{K}=\sqrt{k}\mathbb{B}_1^n$. The closest development is probably [15] who showed for D1bCS that $\Omega(\delta^{-3})$ Gaussian measurements are needed to achieve the uniform random hyperplane tessellation over $\sqrt{k}\mathbb{B}_1^n\cap\mathbb{B}_2^n$ with distortion δ, which is nonetheless not necessary for signal reconstruction to δ ℓ_2-error. In fact, the optimality of $\tilde{O}((\frac{k}{m})^{1/3})$ remains an open question (see, e.g., [11]).*

For 1bCS with standard Gaussian $\mathbf{A}$ and D1bCS with sub-Gaussian $\mathbf{A}$ and uniform $\boldsymbol{\tau}$, it was known that the optimal rates in Remark 1 are attainable up to a logarithmic factor by (constrained) hamming distance minimization (HDM):

$$\hat{\mathbf{x}}_{\text{hdm}}=\arg\min_{\mathbf{u}\in\mathcal{X}}\ d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{u}-\boldsymbol{\tau}),\mathbf{y}\big),\tag{11}$$

where

$$d_H(\mathbf{u},\mathbf{v})=\sum_{i=1}^m\mathbb{1}(u_i\neq v_i).$$

See [21, Theorem 2], [28, Theorem 2.5] and [13, Theorem 1.9]. To unify these existing performance bounds, we shall develop a useful notion referred to as *quantized embedding property* to relate the Euclidean distance of two signals and the hamming distance between their quantized measurements. To establish the quantized embedding property, we only need to make three generic assumptions.

As mentioned, we work with sub-Gaussian design throughout this work.

Assumption 1 (Sensing vectors)**.** *The sensing vectors $\mathbf{a}_1,\mathbf{a}_2,\cdots,\mathbf{a}_m$ are i.i.d. isotropic sub-Gaussian, i.e., they satisfy $\mathbb{E}(\mathbf{a}_i\mathbf{a}_i^\top)=\mathbf{I}_n$ and $\|\mathbf{a}_i\|_{\psi_2}\leq C_0$ for some absolute constant C_0.*

Remark 3. *We pause to provide some implications of Assumption 1. First, it leads to the tail bounds and the moment bounds (e.g., [38])*

$$\mathbb{P}(|\mathbf{a}_i^\top \mathbf{u}| \geq t) \leq 2\exp(-2\bar{c}_0 t^2), \quad \forall t \geq 0,\ \forall \mathbf{u} \in \mathbb{S}^{n-1}, \tag{12a}$$

$$(\mathbb{E}|\mathbf{a}_i^\top \mathbf{u}|^p)^{1/p} \leq \bar{c}_1 \sqrt{p}, \quad \forall p \in \mathbb{Z}_+,\ \forall \mathbf{u} \in \mathbb{S}^{n-1} \tag{12b}$$

which hold for some absolute constants $\bar{c}_0$ and $\bar{c}_1$. Moreover, there exists some absolute constant L_0 such that (see Lemma 6 and its proof)

$$\mathbb{E}|\mathbf{a}_i^\top \mathbf{u}| \geq L_0, \quad \forall \mathbf{u} \in \mathbb{S}^{n-1}, \tag{13a}$$

$$\mathbb{P}\Big(|\mathbf{a}_i^\top \mathbf{u}| \geq \frac{1}{2}\Big) \geq 2L_0, \quad \forall \mathbf{u} \in \mathbb{S}^{n-1}. \tag{13b}$$

Next, we characterize the small-ball probability of the unquantized measurement $\mathbf{a}_i^\top \mathbf{u} - \tau_i$ being close to some quantization thresholds $(b_j)_{j=1}^{L-1}$. In all our assumptions, we use $c^{(i)}$'s to denote constants that are typically universal when specialized to a specific model. We will use $\varepsilon^{(i)}$'s instead if these constants are extremely small or 0.

Assumption 2 (Small-ball probability)**.** *For some $c^{(1)} > 0$ and for any $\mathbf{u} \in \mathbb{A}_\alpha^\beta$, it holds that*

$$\mathbb{P}\left(\min_{j \in [L-1]} |\mathbf{a}_i^\top \mathbf{u} - \tau_i - b_j| \leq t\right) \leq \frac{c^{(1)} t}{\Delta \vee \Lambda}, \quad \forall t > 0.$$

The final assumption for quantized embedding property is on the probability of two points being distinguished by the quantized sampler. This quantity, referred to as separation probability, is defined as

$$\mathsf{P}_{\mathbf{u},\mathbf{v}} := \mathbb{P}\Big(\mathcal{Q}(\mathbf{a}_i^\top \mathbf{u} - \tau_i) \neq \mathcal{Q}(\mathbf{a}_i^\top \mathbf{v} - \tau_i)\Big) \tag{14}$$

and will be recurring in subsequent developments.

Assumption 3 (Lower bound on $\mathsf{P}_{\mathbf{u},\mathbf{v}}$)**.** *For any $\mathbf{u}, \mathbf{v} \in \mathbb{A}_\alpha^\beta$,*

$$\mathsf{P}_{\mathbf{u},\mathbf{v}} \geq c^{(2)} \min\left\{\frac{\|\mathbf{u} - \mathbf{v}\|_2}{\Delta \vee \Lambda}, 1\right\}$$

holds for some $c^{(2)} > 0$.

Theorem 2 (Quantized embedding property)**.** *Let $\mathcal{W}$ be a set contained in $\mathbb{A}_\alpha^\beta$, for $\epsilon \in (0, c_0(\Delta \vee \Delta))$, we let $\epsilon' = \frac{c_1 \epsilon}{\log^{1/2}(\frac{\Delta \vee \Lambda}{\epsilon})}$ and suppose*

$$m \geq C_2(\Delta \vee \Lambda)\left(\frac{\omega^2(\mathcal{W}_{(3\epsilon'/2)})}{\epsilon^3} + \frac{\mathscr{H}(\mathcal{W}, \epsilon')}{\epsilon}\right), \tag{15}$$

where c_0 and c_1 are small enough, C_2 are large enough. We have the following two statements:

- *Under Assumptions 1–2, for some constants c_i's and C_i's depending on $c^{(1)}$, with probability at* *least* $1-2\exp(-\frac{c_3 m\epsilon}{\Delta\vee\Lambda})$ *the event* E_s *holds: for any* $\mathbf{u},\mathbf{v}\in\mathcal{W}$ *obeying* $\|\mathbf{u}-\mathbf{v}\|_2\le\frac{\epsilon'}{2}$, *we have*

$$d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{u}-\boldsymbol{\tau}),\mathcal{Q}(\mathbf{A}\mathbf{v}-\boldsymbol{\tau})\big)\le\frac{C_4 m\epsilon}{\Delta\vee\Lambda}.$$

- *Under Assumptions 1–3, for some constants* c_i*'s and* C_i*'s depending on* $(c^{(1)},c^{(2)})$, *with probability at least* $1-\exp(-\frac{c_5 m\epsilon}{\Delta\vee\Lambda})$ *the event* E_l *holds: for any* $\mathbf{u},\mathbf{v}\in\mathcal{W}$ *obeying* $\|\mathbf{u}-\mathbf{v}\|_2\ge 2\epsilon$, *we have*

$$d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{u}-\boldsymbol{\tau}),\mathcal{Q}(\mathbf{A}\mathbf{v}-\boldsymbol{\tau})\big)\ge c_6 m\min\left\{\frac{\|\mathbf{u}-\mathbf{v}\|_2}{\Delta\vee\Lambda},1\right\}. \tag{16}$$

The proof of Theorem 2 is similar to that from [10, Theorem 2.1] (see also earlier works [13, 28, 32]) and is included in Appendix C for completeness. Theorem 2 immediately implies the following performance bound for HDM; this is an information-theoretic upper bound due to the computational intractability of HDM.

Theorem 3 (Information-theoretic upper bound)**.** *Under Assumptions 1–3, for some star-shaped set* $\mathcal{K}$, *we recover* $\mathbf{x}\in\mathcal{X}=\mathcal{K}\cap\mathbb{A}_\alpha^\beta$ *from* $\mathbf{y}=\mathcal{Q}(\mathbf{A}\mathbf{x}-\boldsymbol{\tau})$ *by solving HDM as per (11). There exist constants* c_i*'s and* C_i*'s depending on* $(c^{(1)},c^{(2)})$, *for any* $r\in(0,c_0(\Delta\vee\Lambda))$ *and the corresponding* $r'=\frac{c_1 r}{\log^{1/2}(\frac{\Delta\vee\Lambda}{r})}$, *if*

$$m\ge C_2(\Delta\vee\Lambda)\left(\frac{\omega^2(\mathcal{K}_{(r')})}{r^3}+\frac{\mathscr{H}(\mathcal{X},r')}{r}\right),$$

then with probability at least $1-\exp(-c_3\mathscr{H}(\mathcal{X},r'))$ *we have*

$$\|\hat{\mathbf{x}}_{\rm hdm}-\mathbf{x}\|_2\le 2r,\quad\forall\mathbf{x}\in\mathcal{X}.$$

Proof. Setting $\mathcal{W}=\mathcal{X}=\mathcal{K}\cap\mathbb{A}_\alpha^\beta$ in Theorem 2 and observing $\mathcal{W}_{(3r'/2)}=(\mathcal{W}-\mathcal{W})\cap(\frac{3r'}{2}\mathbb{B}_2^n)\subset(\mathcal{K}-\mathcal{K})\cap(\frac{3r'}{2}\mathbb{B}_2^n)\subset\frac{3}{2}\mathcal{K}_{(r')}$, we find that under the assumed sample size, we have

$$d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{u}-\boldsymbol{\tau}),\mathcal{Q}(\mathbf{A}\mathbf{v}-\boldsymbol{\tau})\big)>0,\quad\forall(\mathbf{u},\mathbf{v})\in\mathcal{X}\ \text{ obeying }\ \|\mathbf{u}-\mathbf{v}\|_2\ge 2r$$

with the promised probability. On the other hand, by the definition of HDM, for any $\mathbf{x}\in\mathcal{X}$, we have $\hat{\mathbf{x}}_{\rm hdm}\in\mathcal{X}$ and $d_H\big(\mathcal{Q}(\mathbf{A}\hat{\mathbf{x}}_{\rm hdm}-\boldsymbol{\tau}),\mathbf{y}\big)\le d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{x}-\boldsymbol{\tau}),\mathbf{y}\big)=0$, giving $d_H\big(\mathcal{Q}(\mathbf{A}\hat{\mathbf{x}}_{\rm hdm}-\boldsymbol{\tau}),\mathcal{Q}(\mathbf{A}\mathbf{x}-\boldsymbol{\tau})\big)=0$. We therefore arrive at $\|\hat{\mathbf{x}}_{\rm hdm}-\mathbf{x}\|_2\le 2r$ ($\forall\mathbf{x}\in\mathcal{X}$), as claimed. $\square$

Remark 4. *Theorem 3 specializes to the following:*

- *(Exactly sparse vectors or low-rank matrices) Setting* $\mathcal{K}=\Sigma_k^n$ *in Theorem 3 finds that HDM achieves error rate* $\tilde{O}(\frac{(\Delta\vee\Lambda)k}{m})$, *and* $\mathcal{K}=M_{\bar r}^{n_1,n_2}$ *shows that HDM achieves* $\tilde{O}(\frac{(\Delta\vee\Lambda)\bar r(n_1+n_2)}{m})$. *These match the optimal rate implied by Theorem 1. (In the sequel, we will validate Assumptions 2–3 under* $L(\Delta\vee\Lambda)=O(1)$ *for 1bCS, D1bCS and DMbCS, thus we can replace* $\Delta\vee\Lambda$ *in the above error rates with* L^{-1}.*)*

- *(Effectively sparse vectors) By* $\omega^2(\mathcal{K}_{(r')})\lesssim\omega^2(\mathcal{K}\cap\mathbb{B}_2^n)$ *and* $\mathscr{H}(\mathcal{X},r')\le\frac{\omega^2(\mathcal{X})}{(r')^2}$, *the error rate* *of HDM is no worst than $\tilde{O}((\frac{(\Delta\vee\Lambda)\omega^2(\mathcal{X})}{m})^{1/3})$. Therefore, HDM achieves $\tilde{O}((\frac{(\Delta\vee\Lambda)k}{m})^{1/3})$ for effectively sparse signals living in $\sqrt{k}\mathbb{B}_1^n$.*

3 Projected Gradient Descent

The main focus of this paper is to develop a computationally efficient procedure to attain the error rates of HDM. We propose to do so via projected gradient descent (PGD). We first derive a slightly different hamming distance loss. Under the L-level quantizer, the measurement $y_i = \mathcal{Q}(\mathbf{a}_i^\top \mathbf{x} - \tau_i)$ can be equivalently viewed as $L-1$ binary measurements:

$$y_{ij} = \mathrm{sign}(\mathbf{a}_i^\top \mathbf{x} - \tau_i - b_j), \quad j = 1, \cdots, L-1.$$

In total, we have $(L-1)m$ binary measurements $\{\mathrm{sign}(\mathbf{a}_i^\top \mathbf{x} - \tau_i - b_j : j \in [L-1],\ i \in [m])\}$, and it is natural to consider the (scaled) hamming distance loss

$$\mathcal{L}(\mathbf{u}) := \frac{\Delta}{m}\sum_{i=1}^m\sum_{j=1}^{L-1}\mathbb{1}\big(\mathrm{sign}(\mathbf{a}_i^\top \mathbf{u} - \tau_i - b_j) \neq y_{i,j}\big) = \frac{\Delta}{m}\sum_{i=1}^m\sum_{j=1}^{L-1}\mathbb{1}\big(-y_{ij}(\mathbf{a}_i^\top \mathbf{u} - \tau_i - b_j) \geq 0\big).$$

While this loss function is discrete and nonconvex, we shall relax $\mathcal{L}(\mathbf{u})$ to the one-sided ℓ_1-loss

$$\begin{aligned}\mathcal{L}_1(\mathbf{u}) &= \frac{\Delta}{m}\sum_{i=1}^m\sum_{j=1}^{L-1}\max\big\{-y_{ij}(\mathbf{a}_i^\top \mathbf{u} - \tau_i - b_j), 0\big\}\\ &= \frac{\Delta}{2m}\sum_{i=1}^m\sum_{j=1}^{L-1}\Big[|\mathbf{a}_i^\top \mathbf{u} - \tau_i - b_j| - y_{ij}(\mathbf{a}_i^\top \mathbf{u} - \tau_i - b_j)\Big],\end{aligned} \tag{17}$$

in light of the idea of replacing $\mathbb{1}(t > 0)$ by $\max\{t, 0\}$. The (sub-)gradient of $\mathcal{L}_1(\mathbf{u})$ at $\mathbf{x}^{(t-1)}$ takes the form

$$\begin{aligned}\partial\mathcal{L}_1(\mathbf{x}^{(t-1)}) &= \frac{\Delta}{2m}\sum_{i=1}^m\sum_{j=1}^{L-1}\big(\mathrm{sign}(\mathbf{a}_i^\top \mathbf{u} - \tau_i - b_j) - y_{ij}\big)\mathbf{a}_i\\ &= \frac{1}{m}\sum_{i=1}^m\big(\mathcal{Q}(\mathbf{a}_i^\top \mathbf{x}^{(t-1)} - \tau_i) - \mathcal{Q}(\mathbf{a}_i^\top \mathbf{x} - \tau_i)\big)\mathbf{a}_i,\end{aligned}$$

where the second equality holds because we have assumed $q_{i+1} = q_i + \Delta$ for $i \in [L-1]$; see (3).

As we are interested in *uniform reconstruction*, it is beneficial to introduce a notation for the gradient which also accounts for the underlying signal. Let us define

$$\mathbf{h}(\mathbf{u}, \mathbf{v}) := \frac{1}{m}\sum_{i=1}^m\big(\mathcal{Q}(\mathbf{a}_i^\top \mathbf{u} - \tau_i) - \mathcal{Q}(\mathbf{a}_i^\top \mathbf{v} - \tau_i)\big)\mathbf{a}_i = \frac{1}{m}\mathbf{A}^\top\big(\mathcal{Q}(\mathbf{A}\mathbf{u} - \boldsymbol{\tau}) - \mathcal{Q}(\mathbf{A}\mathbf{v} - \boldsymbol{\tau})\big) \tag{18}$$

for this purpose. After performing gradient descent to get $\mathbf{x}^{(t-1)} - \eta \cdot \mathbf{h}(\mathbf{x}^{(t-1)}, \mathbf{x})$, the most natural choice is to project onto $\mathcal{X} = \mathcal{K} \cap \mathbb{A}_\alpha^\beta$, but this is not immediately efficient even for convex $\mathcal{K}$ (e.g., $\mathcal{X} = \sqrt{k}\mathbb{B}_1^n \cap \mathbb{S}^{n-1}$). Therefore, we propose to project $\mathbf{x}^{(t-1)} - \eta \cdot \mathbf{h}(\mathbf{x}^{(t-1)}, \mathbf{x})$ onto $\mathcal{K}$ and $\mathbb{A}_\alpha^\beta$ sequentially.

It should be noted that for cone $\mathcal{K}$, the two ways are equivalent in light of $\mathcal{P}_{\mathcal{X}} = \mathcal{P}_{\mathbb{A}_\alpha^\beta} \circ \mathcal{P}_{\mathcal{K}}$.

We formally outline the proposed recovery procedure in Algorithm 1.

Algorithm 1: Projected Gradient Descent for Quantized Compressed Sensing

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1 **Input**: Quantized observations $\mathbf{y} = \mathcal{Q}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau})$, sensing ensemble $(\mathbf{A}, \boldsymbol{\tau})$, a set $\mathcal{K}$ capturing signal structure, a set $\mathbb{A}_\alpha^\beta$ constraining signal norm, initialization $\mathbf{x}^{(0)}$, step size η

2 **For** $t = 1, 2, 3, \cdots$ **do**

$$\tilde{\mathbf{x}}^{(t)} = \mathcal{P}_{\mathcal{K}}\big(\mathbf{x}^{(t-1)} - \eta \cdot \mathbf{h}(\mathbf{x}^{(t-1)}, \mathbf{x})\big), \tag{19}$$

$$\mathbf{x}^{(t)} = \mathcal{P}_{\mathbb{A}_\alpha^\beta}\big(\tilde{\mathbf{x}}^{(t)}\big). \tag{20}$$

Remark 5. *For 1bCS of k-sparse signals, we shall set $\mathcal{K} = \Sigma_k^n$ and $\mathbb{A}_\alpha^\beta = \mathbb{S}^{n-1}$, then Algorithm 1 becomes NBIHT [17, 21, 24].*

3.1 Convergence via RAIC

Our analysis of Algorithm 1 is based on the following deterministic property called *restricted approximate invertibility condition* (RAIC). It generalizes similar notions in [10, 17, 24] from k-sparse signals ($\mathcal{K} = \Sigma_k^n$) to general signal structures captured by star-shaped set.

Definition 1 (RAIC)**.** *Under some quantizer $\mathcal{Q}$, for some given $\mathcal{D} \subset \mathbb{R}^n$ and $\boldsymbol{\mu} = (\mu_1, \mu_2, \mu_3, \mu_4)$ with non-negative scalars $(\mu_i)_{i=1}^4$, we say $(\mathcal{Q}, \mathbf{A}, \boldsymbol{\tau}, \mathcal{K}, \eta)$ respects $(\mathcal{D}, \boldsymbol{\mu})$-RAIC at scale $\phi > 0$ if*

$$\frac{1}{\phi}\|\mathbf{u} - \mathbf{v} - \eta \cdot \mathbf{h}(\mathbf{u}, \mathbf{v})\|_{\mathcal{K}_{(\phi)}^\circ} \le \mu_1\|\mathbf{u} - \mathbf{v}\|_2 + \sqrt{\mu_2 \cdot \|\mathbf{u} - \mathbf{v}\|_2} + \mu_3 \tag{21}$$

holds for any $\mathbf{u}, \mathbf{v} \in \mathcal{D}$ obeying $\|\mathbf{u} - \mathbf{v}\|_2 \le \mu_4$, where $\mathbf{h}(\mathbf{u}, \mathbf{v})$ is defined as per (18).

In light of

$$\frac{1}{\phi}\|\mathbf{u}\|_{\mathcal{K}_{(\phi)}^\circ} = \frac{1}{\phi}\sup_{\mathbf{w} \in \mathcal{K}_{(\phi)}} \langle \mathbf{w}, \mathbf{u} \rangle = \sup_{\mathbf{w} \in (\frac{\mathcal{K}}{\phi})_{(1)}} \langle \mathbf{w}, \mathbf{u} \rangle = \|\mathbf{u}\|_{(\frac{\mathcal{K}}{\phi})_{(1)}^\circ},$$

RAIC delivers a uniform bound on (the dual norm of) $\mathbf{u} - \mathbf{v} - \eta \cdot \mathbf{h}(\mathbf{u}, \mathbf{v})$ with regard to $(\frac{\mathcal{K}}{\phi})_{(1)}$; thus, on the current iterate $\mathbf{u} = \mathbf{x}^{(t)}$ and underlying signal $\mathbf{v} = \mathbf{x}$, RAIC gives the message that the "ideal descent step" $\mathbf{x}^{(t)} - \mathbf{x}$ (in the sense of $\mathbf{x}^{(t)} - (\mathbf{x}^{(t)} - \mathbf{x}) = \mathbf{x}$) is well approximated by the "actual subgradient step" $\eta \cdot \mathbf{h}(\mathbf{x}^{(t)}, \mathbf{x})$, up to a few error terms. See Figure 3 for some intuition. It is interesting to note that such message departs from the one given by *regularity condition* which has been widely adopted in the statistical nonconvex optimization literature (see, e.g., [7, 37]) and indeed stems from convex optimization (see [27, Theorem 2.1.15]). We expect that RAIC will prove useful in many other nonconvex optimization problems; we already witness in the present paper the potentials of RAIC in dealing with projection onto nonconvex sets and obtaining quadratic convergence rate.

Remark 6 (Selection of ϕ)**.** *Because $\mathcal{K}$ is a star-shaped set, we have $\frac{\mathcal{K}}{\phi_1} \subset \frac{\mathcal{K}}{\phi_2}$ if $\phi_1 > \phi_2$, and hence $\frac{1}{\phi}\|\mathbf{u}\|_{\mathcal{K}_{(\phi)}^\circ} = \|\mathbf{u}\|_{(\frac{\mathcal{K}}{\phi})_{(1)}^\circ}$ decreases with ϕ. Therefore, RAIC with smaller ϕ is stronger and accordingly harder to establish. However, in controlling the error after projection, we will rely on Lemma 9 which*

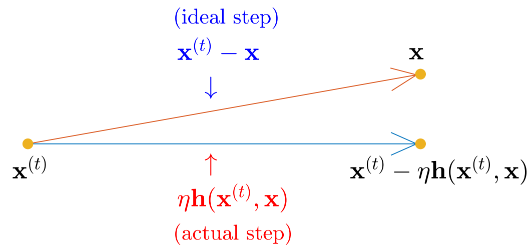

Figure 3: RAIC indicates closeness of the ideal step and the actual subgradient step.

puts ϕ and $\frac{1}{\phi}\|\mathbf{u}-\mathbf{v}-\eta\cdot\mathbf{h}(\mathbf{u},\mathbf{v})\|_{\mathcal{K}^{\circ}_{(\phi)}}$ on an equal footing, making RAIC with overly large ϕ useless. It is easy to note that a reasonable tradeoff is to select ϕ at an order comparable to $\frac{1}{\phi}\|\mathbf{u}-\mathbf{v}-\eta\cdot\mathbf{h}(\mathbf{u},\mathbf{v})\|_{\mathcal{K}^{\circ}_{(\phi)}}$. In the sequel, we will choose $\phi = r$ if RAIC with $\mu_3 = \tilde{O}(r)$ is sought after.

Remark 7 (Conic RAIC)**.** *If $\mathcal{K}$ is a cone, then the choice of $\phi > 0$ becomes inessential because (21) is always equivalent to*

$$\|\mathbf{u}-\mathbf{v}-\eta\cdot\mathbf{h}(\mathbf{u},\mathbf{v})\|_{\mathcal{K}^{\circ}_{(1)}} \le \mu_1\|\mathbf{u}-\mathbf{v}\|_2 + \sqrt{\mu_2\cdot\|\mathbf{u}-\mathbf{v}\|_2} + \mu_3,$$

which is very similar to the definitions of RAIC in [17, 24].

We shall refer to it as local RAIC if the localized parameter μ_4 is essentially smaller than the diameter of $\mathcal{D}$. As we shall see in Theorem 4, under small μ_1, RAIC implies that PGD linearly converges to estimates with error rate $O(\mu_2+\mu_3)$. Under extremely small μ_1, RAIC even implies a quadratic convergence rate to a neighborhood of $\mathbf{x}$ with radius $O(\mu_2+\mu_3)$.

To keep our theorem statement concise, we pause to elaborate on some technical assumptions used in Theorem 4.

Auxiliary assumptions for Theorem 4: First, we need to define

$$\kappa_\alpha := \begin{cases} 1\ , & \text{if } \alpha = 0 \\ 2\ , & \text{if } \alpha > 0 \end{cases}$$

to distinguish the nature of a projection onto $\mathbb{A}^{\beta}_{\alpha}$, which is convex when $\alpha = 0$ but nonconvex when $\alpha > 0$; similar idea appeared in previous analysis of projected gradient descent (e.g., [29, 35]). Recall that we utilize sequential projections as per (19)–(20) to resolve potential computational issue, while this leads to the new issue that $\mathbf{x}^{(t)}$ may not live in $\mathcal{K}$. Our remedy is to define a slightly larger signal set $\overline{\mathcal{X}} := (2\mathcal{K}) \cap \mathbb{A}^{\beta}_{\alpha}$ and show $\mathbf{x}^{(t)} \in 2\mathcal{K}$. Taken collectively, we suppose that $(\mathcal{Q}, \mathbf{A}, \boldsymbol{\tau}, \mathcal{K}, \eta)$ respects $(\overline{\mathcal{X}}, \boldsymbol{\mu})$-RAIC for some $\boldsymbol{\mu} = (\mu_1,\mu_2,\mu_3,\mu_4)$ at some scale ϕ obeying $\phi \le \mu_3$, where the non-negative $(\mu_i)_{i=1}^4$ satisfy $\mu_1 < \frac{1}{2\kappa_\alpha}$ and

$$\underbrace{\left(\frac{\kappa_\alpha\sqrt{\mu_2} + \sqrt{\kappa_\alpha^2\mu_2 + 2\kappa_\alpha(1-2\mu_1\kappa_\alpha)\mu_3}}{1-2\mu_1\kappa_\alpha}\right)^2}_{:=\bar{\kappa}} < \mu_4. \tag{22}$$

If $\mathcal{K}$ is not a cone and $\alpha > 0$, we suppose

$$\mu_1\mu_4 + \sqrt{\mu_2\mu_4} + \mu_3 \le \frac{\alpha}{4}. \tag{23}$$

to ensure $\mathbf{x}^{(t)} \in 2\mathcal{K}$.

Theorem 4. *Under the above assumptions, for any $\mathbf{x} \in \mathcal{X} = \mathcal{K} \cap \mathbb{A}_\alpha^\beta$, we run Algorithm 1 with step size η and some $\mathbf{x}^{(0)} \in \mathbb{B}_2^n(\mathbf{x};\mu_4) \cap \mathcal{X}$ to obtain $\{\mathbf{x}^{(t)}\}_{t=0}^\infty$. Then we have the following statements:*

- *For all $\mathbf{x} \in \mathcal{X}$ and any $t \ge 0$, we have*

$$\|\mathbf{x}^{(t)} - \mathbf{x}\|_2 \le \left(\kappa_\alpha\mu_1 + \frac{1}{2}\right)^t \mu_4 + 4\kappa_\alpha \left[\frac{\kappa_\alpha\mu_2}{(1-2\kappa_\alpha\mu_1)^2} + \frac{\mu_3}{1-2\kappa_\alpha\mu_1}\right]. \tag{24}$$

- *If additionally $\mu_1 \le \sqrt{\frac{\mu_2}{\mu_4}} + \frac{\mu_3}{\mu_4}$, we define $\hat{\kappa} := 4(\kappa_\alpha\sqrt{\mu_2} + \sqrt{\kappa_\alpha^2\mu_2 + \kappa_\alpha\mu_3})^2$ and assume $\mu_4 \ge \hat{\kappa}$. Then for all $\mathbf{x} \in \mathcal{X}$ and any $t \ge 0$, we have*

$$\|\mathbf{x}^{(t)} - \mathbf{x}\|_2 \le \mu_4^{2^{-t}} \hat{\kappa}^{1-2^{-t}}. \tag{25}$$

Proof. The proof can be found in Appendix D. $\square$

3.2 A Sharp Approach to RAIC

In light of Theorem 4, it suffices to focus on controlling $\phi^{-1}\|\mathbf{u}-\mathbf{v}-\eta\cdot\mathbf{h}(\mathbf{u},\mathbf{v})\|_{\mathcal{K}^\circ_{(\phi)}}$ (for some $\phi \in (0,\mu_3]$) for all $\mathbf{u},\mathbf{v} \in 2\mathcal{K} \cap \mathbb{A}_\alpha^\beta$ satisfying $\|\mathbf{u}-\mathbf{v}\|_2 \le \mu_4$. We will use the shorthand $\overline{\mathcal{X}} := 2\mathcal{K} \cap \mathbb{A}_\alpha^\beta$. In this subsection, we present a sharp approach to RAIC through generalizing techniques in [10, 24] with new developments.

The overall framework is a covering argument. For some

$$r \in \left(0, \frac{\Delta \vee \Lambda}{4} \wedge \frac{\mu_4}{2}\right),$$

we let $\mathcal{N}_r$ be a minimal r-net of $\overline{\mathcal{X}}$ such that $\log|\mathcal{N}_r| = \mathscr{H}(\overline{\mathcal{X}}, r)$. Then, for any $\mathbf{u},\mathbf{v} \in \overline{\mathcal{X}}$ obeying $\|\mathbf{u}-\mathbf{v}\|_2 \le \mu_4$, we find their closest points in $\mathcal{N}_r$:

$$\mathbf{u}_1 := \arg\min_{\mathbf{w}\in\mathcal{N}_r} \|\mathbf{w}-\mathbf{u}\|_2 \quad\text{and}\quad \mathbf{v}_1 := \arg\min_{\mathbf{w}\in\mathcal{N}_r} \|\mathbf{w}-\mathbf{v}\|_2. \tag{26}$$

It follows that $\|\mathbf{u}_1-\mathbf{v}_1\|_2 \le \|\mathbf{u}_1-\mathbf{u}\|_2 + \|\mathbf{u}-\mathbf{v}\|_2 + \|\mathbf{v}-\mathbf{v}_1\|_2 \le \mu_4 + 2r < 2\mu_4$. We will ultimately achieve RAIC with $\mu_3 = \tilde{O}(r)$, so we simply choose $\phi = r$ and seek to control $\frac{1}{r}\|\mathbf{u}-\mathbf{v}-\eta\cdot\mathbf{h}(\mathbf{u},\mathbf{v})\|_{\mathcal{K}^\circ_{(r)}}$ (see Remark 6).

We define the constraint sets

$$\mathcal{D}_r^{(2)} := \{(\mathbf{p},\mathbf{q}) \in \overline{\mathcal{X}} : \|\mathbf{p}-\mathbf{q}\|_2 \le r\},$$
$$\mathcal{N}_{r,\mu_4}^{(2)} := \{(\mathbf{p},\mathbf{q}) \in \mathcal{N}_r \times \mathcal{N}_r : 0 < \|\mathbf{p}-\mathbf{q}\|_2 \le 2\mu_4\}.$$

Observing from (18) the following

$$\begin{aligned}
&\mathbf{h}(\mathbf{u}_1,\mathbf{v}_1)-\mathbf{h}(\mathbf{u},\mathbf{v})\\
&=\frac{1}{m}\sum_{i=1}^{m}\big[\mathcal{Q}(\mathbf{a}_i^\top\mathbf{u}-\tau_i)-\mathcal{Q}(\mathbf{a}_i^\top\mathbf{v}-\tau_i)-\mathcal{Q}(\mathbf{a}_i^\top\mathbf{u}_1-\tau_i)+\mathcal{Q}(\mathbf{a}_i^\top\mathbf{v}_1-\tau_i)\big]\mathbf{a}_i\\
&=\mathbf{h}(\mathbf{u}_1,\mathbf{u})+\mathbf{h}(\mathbf{v},\mathbf{v}_1)
\end{aligned}$$

we can proceed as

$$\begin{aligned}
&r^{-1}\big\|\mathbf{u}-\mathbf{v}-\eta\cdot\mathbf{h}(\mathbf{u},\mathbf{v})\big\|_{\mathcal{K}^\circ_{(r)}}\\
&\le r^{-1}\big\|\mathbf{u}_1-\mathbf{v}_1-\eta\cdot\mathbf{h}(\mathbf{u}_1,\mathbf{v}_1)\big\|_{\mathcal{K}^\circ_{(r)}}+r^{-1}\|\mathbf{u}-\mathbf{u}_1\|_{\mathcal{K}^\circ_{(r)}}+r^{-1}\|\mathbf{v}-\mathbf{v}_1\|_{\mathcal{K}^\circ_{(r)}}\\
&\qquad+\frac{\eta}{r}\Big(\|\mathbf{h}(\mathbf{u}_1,\mathbf{u})\|_{\mathcal{K}^\circ_{(r)}}+\|\mathbf{h}(\mathbf{v},\mathbf{v}_1)\|_{\mathcal{K}^\circ_{(r)}}\Big)\\
&\le r^{-1}\big\|\mathbf{u}_1-\mathbf{v}_1-\eta\cdot\mathbf{h}(\mathbf{u}_1,\mathbf{v}_1)\big\|_{\mathcal{K}^\circ_{(r)}}+2r+\frac{2\eta}{r}\cdot\sup_{(\mathbf{p},\mathbf{q})\in\mathcal{D}_r^{(2)}}\big\|\mathbf{h}(\mathbf{p},\mathbf{q})\big\|_{\mathcal{K}^\circ_{(r)}},
\end{aligned}\tag{27}$$

where in the last line we observe $(\mathbf{u}_1,\mathbf{u}),(\mathbf{v},\mathbf{v}_1)\in\mathcal{D}_r^{(2)}$ and take the supremum over them. Note that $\|\mathbf{u}_1-\mathbf{v}_1\|_2\le 2\mu_4$, hence either $\mathbf{u}_1=\mathbf{v}_1$ or $(\mathbf{u}_1,\mathbf{v}_1)\in\mathcal{N}^{(2)}_{r,\mu_4}$ holds. Therefore, bounding $r^{-1}\|\mathbf{u}_1-\mathbf{v}_1-\eta\cdot\mathbf{h}(\mathbf{u}_1,\mathbf{v}_1)\|_{\mathcal{K}^\circ_{(r)}}$ amounts to uniformly controlling $r^{-1}\|\mathbf{p}-\mathbf{q}-\eta\cdot\mathbf{h}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$ over the discrete constraint set $\mathcal{N}^{(2)}_{r,\mu_4}$. In view of the third term in (27), another task is to bound $\|\mathbf{h}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$ over $(\mathbf{p},\mathbf{q})\in\mathcal{D}_r^{(2)}$.

3.2.1 Gradient clipping

Since $\mathcal{Q}$ is non-decreasing, we can write

$$\begin{aligned}
\mathbf{h}(\mathbf{p},\mathbf{q})&=\frac{1}{m}\sum_{i=1}^{m}\operatorname{sign}(\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q}))|\mathcal{Q}(\mathbf{a}_i^\top\mathbf{p}-\tau_i)-\mathcal{Q}(\mathbf{a}_i^\top\mathbf{q}-\tau_i)|\cdot\mathbf{a}_i\\
&=\frac{1}{m}\sum_{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}}\operatorname{sign}(\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q}))|\mathcal{Q}(\mathbf{a}_i^\top\mathbf{p}-\tau_i)-\mathcal{Q}(\mathbf{a}_i^\top\mathbf{q}-\tau_i)|\cdot\mathbf{a}_i,
\end{aligned}\tag{28}$$

where in the second equality we define the index set

$$\mathbf{R}_{\mathbf{p},\mathbf{q}}:=\big\{i\in[m]:\mathcal{Q}(\mathbf{a}_i^\top\mathbf{p}-\tau_i)\neq\mathcal{Q}(\mathbf{a}_i^\top\mathbf{q}-\tau_i)\big\}.$$

For convenience, we further introduce $E^{(i)}_{\mathbf{p},\mathbf{q}}$ to denote the event that $\mathbf{p}$ and $\mathbf{q}$ have different i-th measurements

$$E^{(i)}_{\mathbf{p},\mathbf{q}}:=\{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}\}.$$

In the 1-bit case, we have $|\mathcal{Q}(\mathbf{a}_i^\top\mathbf{p}-\tau_i)-\mathcal{Q}(\mathbf{a}_i^\top\mathbf{q}-\tau_i)|=\Delta$ for $i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}$, hence $\mathbf{h}(\mathbf{p},\mathbf{q})$ reduces to

$$\mathbf{h}(\mathbf{p},\mathbf{q})=\frac{\Delta}{m}\sum_{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}}\operatorname{sign}(\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q}))\mathbf{a}_i.$$

This is not true for the multi-bit setting where $|\mathcal{Q}(\mathbf{a}_i^\top \mathbf{p} - \tau_i) - \mathcal{Q}(\mathbf{a}_i^\top \mathbf{q} - \tau_i)|$, if being non-zero, can take values in $\{\ell\Delta : \ell = 1, 2, \cdots, L-1\}$. The multiple values greatly add to the difficulty in deriving sharp concentration bounds for $r^{-1}\|\mathbf{p} - \mathbf{q} - \eta \cdot \mathbf{h}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$ because the moments of the marginals of $|\mathcal{Q}(\mathbf{a}_i^\top \mathbf{p} - \tau_i) - \mathcal{Q}(\mathbf{a}_i^\top \mathbf{q} - \tau_i)|\mathbf{a}_i$ could have different behaviours.[4] This issue also precludes a unified treatment to the multi-bit case and 1-bit case and increases the difficulty in formulating the forthcoming Assumptions 5–7.

Our remedy is to *clip* the gradient $\mathbf{h}(\mathbf{p}, \mathbf{q})$ to

$$\hat{\mathbf{h}}(\mathbf{p}, \mathbf{q}) := \frac{\Delta}{m} \sum_{i \in \mathbf{R}_{\mathbf{p},\mathbf{q}}} \operatorname{sign}(\mathbf{a}_i^\top (\mathbf{p} - \mathbf{q}))\mathbf{a}_i \tag{29}$$

and then work with $\hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})$. Nonetheless, this induces some deviation (or bias); in particular, triangle inequality gives

$$r^{-1}\|\mathbf{p} - \mathbf{q} - \eta \cdot \mathbf{h}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}} \le r^{-1}\|\mathbf{p} - \mathbf{q} - \eta \cdot \hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}} + \frac{\eta}{r}\|\mathbf{h}(\mathbf{p}, \mathbf{q}) - \hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}, \tag{30a}$$

$$\frac{\eta}{r} \cdot \|\mathbf{h}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}} \le \frac{\eta}{r} \cdot \|\hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}} + \frac{\eta}{r}\|\mathbf{h}(\mathbf{p}, \mathbf{q}) - \hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}, \tag{30b}$$

where $\frac{\eta}{r}\|\mathbf{h}(\mathbf{p}, \mathbf{q}) - \hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$ captures the deviation.

Our hope is that this deviation term is uniformly small over $\mathcal{N}^{(2)}_{r,\mu_4}$ and $\mathcal{D}^{(2)}_r$, which ensures that the gradient clipping has minimal impact on our goals of bounding $r^{-1}\|\mathbf{p} - \mathbf{q} - \eta \cdot \mathbf{h}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$ over $\mathcal{N}^{(2)}_{r,\mu_4}$ and $\frac{\eta}{r}\|\mathbf{h}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$ over $\mathcal{D}^{(2)}_{(r)}$. Intuitively, this is plausible if μ_4 for RAIC is much smaller than Δ. For instance, in DMbCS (10), $|\mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top \mathbf{p} - \tau_i) - \mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top \mathbf{q} - \tau_i)| \ge 2\delta$ implies $|\mathbf{a}_i^\top (\mathbf{p} - \mathbf{q})| \ge \delta$, which is evidently a rare event for $\|\mathbf{p} - \mathbf{q}\|_2 \le \mu_4 \ll \delta$ under sub-Gaussian $\mathbf{a}_i$; in other words, if $i \in \mathbf{R}_{\mathbf{p},\mathbf{q}}$ and $\|\mathbf{p} - \mathbf{q}\|_2 \ll \delta$, then $|\mathcal{Q}(\mathbf{a}_i^\top \mathbf{p} - \tau_i) - \mathcal{Q}(\mathbf{a}_i^\top \mathbf{q} - \tau_i)| = \delta$ holds with very high probability.

More precisely, we need the deviation term to be well controlled as in the following assumption.

Assumption 4 (Deviation of gradient clipping)**.** *For some $c^{(3)}, c^{(4)}$ and some small enough $\varepsilon^{(1)}$,*

$$\frac{\eta}{r}\|\mathbf{h}(\mathbf{p}, \mathbf{q}) - \hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}} \le \frac{\eta\Delta[\varepsilon^{(1)}\|\mathbf{p} - \mathbf{q}\|_2 + c^{(3)}r]}{\Delta \vee \Lambda}, \quad \forall (\mathbf{p}, \mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4}, \tag{31a}$$

$$\frac{\eta}{r}\|\mathbf{h}(\mathbf{p}, \mathbf{q}) - \hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}} \le \frac{\eta\Delta c^{(4)} r}{\Delta \vee \Lambda}, \quad \forall (\mathbf{p}, \mathbf{q}) \in \mathcal{D}^{(2)}_r. \tag{31b}$$

Bound the first term on the right-hand side of (30b): The dual norm of the clipped gradient $r^{-1}\|\hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$ can be uniformly controlled over $(\mathbf{p}, \mathbf{q}) \in \mathcal{D}^{(2)}_r$ by generic arguments: given that $\|\mathbf{p} - \mathbf{q}\|_2 \le r$, we first invoke E_s in Theorem 2 to uniformly control the number of contributors and then apply Lemma 8. This type of arguments will be invoked multiple times in subsequent developments.

Proposition 1 (Bounding $r^{-1}\|\hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$)**.** *Under Assumptions 1–2, for some constants c_i's and*

[4] More specifically, $|\mathcal{Q}(\mathbf{a}_i^\top \mathbf{p} - \tau_i) - \mathcal{Q}(\mathbf{a}_i^\top \mathbf{q} - \tau_i)|$ behaves similarly to $\Delta \cdot \mathbb{1}(i \in \mathbf{R}_{\mathbf{p},\mathbf{q}})$ if $\|\mathbf{p} - \mathbf{q}\|_2 \ll \Delta$, while it is more comparable to $|\mathbf{a}_i^\top (\mathbf{p} - \mathbf{q})|$ if $\|\mathbf{p} - \mathbf{q}\|_2 \gg \Delta$.

C_i's depending on $c^{(1)}$, if $r \le c_0(\Delta \vee \Lambda)$ with sufficiently small c_0, and it holds that

$$m \ge C_1(\Delta \vee \Lambda)\left(\frac{\omega^2(\mathcal{K}_{(r)})}{r^3 \log^{3/2}(\frac{\Delta\vee\Lambda}{r})} + \frac{\mathscr{H}(\overline{\mathcal{X}}, 2r)}{r \log^{1/2}(\frac{\Delta\vee\Lambda}{r})}\right), \tag{32}$$

then with probability at least $1 - \exp(-\frac{c_2 mr}{\Delta\vee\Lambda})$ we have

$$\frac{1}{r}\|\hat{\mathbf{h}}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}} \le \frac{C_3 \Delta r}{\Delta \vee \Lambda}\log^{1/2}\Big(\frac{\Delta\vee\Lambda}{r}\Big), \quad \forall (\mathbf{p},\mathbf{q}) \in \mathcal{D}^{(2)}_r.$$

Proof. See the complete proof in Appendix E.1. □

3.2.2 Orthogonal decomposition

All that remains is to control $r^{-1}\|\mathbf{p} - \mathbf{q} - \eta \cdot \hat{\mathbf{h}}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$ over $(\mathbf{p},\mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4}$. Note that $\mathcal{N}^{(2)}_{r,\mu_4}$ is a finite set, we can accomplish the goal by first bounding it for a fixed pair $(\mathbf{p},\mathbf{q})$ and then taking a union bound over $\mathcal{N}^{(2)}_{r,\mu_4}$.

For fixed $(\mathbf{p},\mathbf{q}) \in \overline{\mathcal{X}} \times \overline{\mathcal{X}}$ obeying $0 < \|\mathbf{p}-\mathbf{q}\|_2 \le 2\mu_4$, we utilize Lemma 10 to establish the following useful parameterization: there exists some orthonormal $(\boldsymbol{\beta}_1 := \frac{\mathbf{p}-\mathbf{q}}{\|\mathbf{p}-\mathbf{q}\|_2}, \boldsymbol{\beta}_2)$ such that[5]

$$\mathbf{p} = u_1\boldsymbol{\beta}_1 + u_2\boldsymbol{\beta}_2 \quad\text{and}\quad \mathbf{q} = v_1\boldsymbol{\beta}_1 + u_2\boldsymbol{\beta}_2 \tag{33}$$

for some coordinates (u_1, u_2, v_1) obeying $u_1 > v_1$ and $u_2 \ge 0$. Now we have the following orthogonal decomposition of the clipped gradient

$$\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}) = \langle\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}),\boldsymbol{\beta}_1\rangle\boldsymbol{\beta}_1 + \langle\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}),\boldsymbol{\beta}_2\rangle\boldsymbol{\beta}_2 + \underbrace{[\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}) - \langle\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}),\boldsymbol{\beta}_1\rangle\boldsymbol{\beta}_1 - \langle\hat{\mathbf{h}},(\mathbf{p},\mathbf{q}),\boldsymbol{\beta}_2\rangle\boldsymbol{\beta}_2]}_{:=\hat{\mathbf{h}}^\perp(\mathbf{p},\mathbf{q})} \tag{34}$$

which allows us to decompose the dual norm as

$$\begin{aligned}
&\frac{1}{r}\big\|\mathbf{p}-\mathbf{q}-\eta\cdot\hat{\mathbf{h}}(\mathbf{p},\mathbf{q})\big\|_{\mathcal{K}^\circ_{(r)}} \\
&= \frac{1}{r}\sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \mathbf{w}^\top\Big[(\mathbf{p}-\mathbf{q}-\eta\cdot\langle\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}),\boldsymbol{\beta}_1\rangle\boldsymbol{\beta}_1) - (\eta\cdot\langle\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}),\boldsymbol{\beta}_2\rangle\boldsymbol{\beta}_2) - (\eta\cdot\hat{\mathbf{h}}^\perp(\mathbf{p},\mathbf{q}))\Big] \\
&\le \big|\|\mathbf{p}-\mathbf{q}\|_2 - \eta\cdot\langle\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}),\boldsymbol{\beta}_1\rangle\big| + \big|\eta\cdot\langle\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}),\boldsymbol{\beta}_2\rangle\big| + r^{-1}\cdot\sup_{\mathbf{w}\in\mathcal{K}_{(r)}}\eta\cdot\mathbf{w}^\top\hat{\mathbf{h}}^\perp(\mathbf{p},\mathbf{q}) \\
:&= \big|\|\mathbf{p}-\mathbf{q}\|_2 - T_1^{\mathbf{p},\mathbf{q}}\big| + |T_2^{\mathbf{p},\mathbf{q}}| + r^{-1}\cdot T_3^{\mathbf{p},\mathbf{q}}.
\end{aligned} \tag{35}$$

We seek to control the three terms in (35).

[5]Note that $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2$ here and some subsequent notation depend on $(\mathbf{p},\mathbf{q})$, while we drop such dependence for succinctness.

3.2.3 Sharp concentration bounds

Substituting $\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}) = \frac{\Delta}{m}\sum_{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}}\operatorname{sign}(\mathbf{a}_i^\top\boldsymbol{\beta}_1)\mathbf{a}_i$, we shall readily find

$$T_1^{\mathbf{p},\mathbf{q}} := \eta\cdot\langle\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}),\boldsymbol{\beta}_1\rangle = \frac{\eta\Delta}{m}\sum_{i=1}^m |\mathbf{a}_i^\top\boldsymbol{\beta}_1|\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)}),$$
$$T_2^{\mathbf{p},\mathbf{q}} := \eta\cdot\langle\hat{\mathbf{h}}(\mathbf{p},\mathbf{q}),\boldsymbol{\beta}_2\rangle = \frac{\eta\Delta}{m}\sum_{i=1}^m \operatorname{sign}(\mathbf{a}_i^\top\boldsymbol{\beta}_1)(\mathbf{a}_i^\top\boldsymbol{\beta}_2)\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)}),$$
$$T_3^{\mathbf{p},\mathbf{q}} := \sup_{\mathbf{w}\in\mathcal{K}_{(r)}}\eta\cdot\mathbf{w}^\top\hat{\mathbf{h}}^\perp(\mathbf{p},\mathbf{q}) = \sup_{\mathbf{w}\in\mathcal{K}_{(r)}}\frac{\eta\Delta}{m}\sum_{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}}\operatorname{sign}(\mathbf{a}_i^\top\boldsymbol{\beta}_1)\big[\mathbf{a}_i-(\mathbf{a}_i^\top\boldsymbol{\beta}_1)\boldsymbol{\beta}_1-(\mathbf{a}_i^\top\boldsymbol{\beta}_2)\boldsymbol{\beta}_2\big]^\top\mathbf{w}.$$

Recall our convention $E_{\mathbf{p},\mathbf{q}}^{(i)} = \{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}\}$.

We seek sharp concentration bounds for $(T_i^{\mathbf{p},\mathbf{q}})_{i=1}^3$. The core ideas are to proceed with arguments aware of the distance between $\mathbf{p}$ and $\mathbf{q}$. For instance, if $\mathbf{p}$ and $\mathbf{q}$ are close enough, the number of effective contributors to $T_i^{\mathbf{p},\mathbf{q}}$ (i.e., $|\mathbf{R}_{\mathbf{p},\mathbf{q}}|$) will be much fewer than m, thus one can hope for tighter concentration bounds. To that end, our particular strategy for bounding $T_3^{\mathbf{p},\mathbf{q}}$ is to first establish the concentration bound by conditioning on $|\mathbf{R}_{\mathbf{p},\mathbf{q}}|$, and then further get rid of the conditioning by analyzing $|\mathbf{R}_{\mathbf{p},\mathbf{q}}|$, a binomial variable, via Chernoff bound. To establish the conditional concentration bound, we shall show the sub-Gaussianity of the conditional distribution $[\mathbf{a}_i-(\mathbf{a}_i^\top\boldsymbol{\beta}_1)\boldsymbol{\beta}_1-(\mathbf{a}_i^\top\boldsymbol{\beta}_2)\boldsymbol{\beta}_2]^\top\mathbf{w}|\{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}\}$ and then use Lemma 11 to obtain a bound relevant to the Gaussian width of $\mathcal{K}_{(r)}$. Similar idea was used in [10, 25]. On the other hand, for the simpler terms $T_1^{\mathbf{p},\mathbf{q}}$ and $T_2^{\mathbf{p},\mathbf{q}}$, we find that bounding the moments as in (36a) and (37a) and then using Bernstein's inequality (see Lemma 7) is sufficient.

We make the following assumptions on these matters.

Assumption 5. *For any* $\mathbf{p},\mathbf{q}\in\overline{\mathcal{X}}$ *obeying* $0<\|\mathbf{p}-\mathbf{q}\|_2\le 2\mu_4$*, we suppose that*

$$\mathbb{E}\Big(|\mathbf{a}_i^\top\boldsymbol{\beta}_1|^p\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big)\le c^{(5)}\mathsf{P}_{\mathbf{p},\mathbf{q}}\cdot\frac{p!}{2},\quad\forall p\ge 2 \tag{36a}$$
$$\left|\mathbb{E}\big(|\mathbf{a}_i^\top\boldsymbol{\beta}_1|\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\big)-\frac{c^{(6)}\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda}\right|\le\frac{\varepsilon^{(2)}\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda}, \tag{36b}$$

hold for some $c^{(5)},c^{(6)}>0$, $\varepsilon^{(2)}\ge 0$.

Assumption 6. *For any* $\mathbf{p},\mathbf{q}\in\overline{\mathcal{X}}$ *obeying* $0<\|\mathbf{p}-\mathbf{q}\|_2\le 2\mu_4$*, we suppose that*

$$\mathbb{E}\Big(|\mathbf{a}_i^\top\boldsymbol{\beta}_2|^p\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big)\le c^{(7)}\mathsf{P}_{\mathbf{p},\mathbf{q}}\cdot\frac{p!}{2},\quad\forall p\ge 2 \tag{37a}$$
$$\left|\mathbb{E}\Big(\operatorname{sign}(\mathbf{a}_i^\top\boldsymbol{\beta}_1)\mathbf{a}_i^\top\boldsymbol{\beta}_2\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big)\right|\le\frac{\varepsilon^{(3)}\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda}, \tag{37b}$$

hold for some $c^{(7)}>0$, $\varepsilon^{(3)}\ge 0$.

For any $\mathbf{w}\in\mathbb{R}^n$, we define

$$J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}} := \operatorname{sign}(\mathbf{a}_i^\top\boldsymbol{\beta}_1)[\mathbf{a}_i-(\mathbf{a}_i^\top\boldsymbol{\beta}_1)\boldsymbol{\beta}_1-(\mathbf{a}_i^\top\boldsymbol{\beta}_2)\boldsymbol{\beta}_2]^\top\mathbf{w}.$$

Assumption 7. *For any $\mathbf{p},\mathbf{q}\in\overline{\mathcal{X}}$ obeying $0<\|\mathbf{p}-\mathbf{q}\|_2\le 2\mu_4$, we suppose that*

$$\mathbb{E}\Big(|J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}}|^p\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big)\le \mathsf{P}_{\mathbf{p},\mathbf{q}}(c^{(8)}\sqrt{p})^p,\quad \forall p\ge 2,\quad \forall \mathbf{w}\in\mathbb{S}^{n-1} \tag{38a}$$

$$\Big|\mathbb{E}\Big(J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}}\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big)\Big|\le\frac{\varepsilon^{(4)}\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda},\quad \forall \mathbf{w}\in\mathbb{S}^{n-1} \tag{38b}$$

hold for some $c^{(8)}>0$, $\varepsilon^{(4)}\ge 0$.

As mentioned, $\varepsilon^{(i)}$'s denote sufficiently small constants, thus (36b), (37b) and (38b) precisely characterize the mean of $T_i^{\mathbf{p},\mathbf{q}}$; for instance, (36b) essentially characterize

$$\mathbb{E}[T_1^{\mathbf{p},\mathbf{q}}]=\eta\Delta\mathbb{E}\big[|\mathbf{a}_i^\top\boldsymbol{\beta}_1|\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\big]\approx\frac{c^{(6)}\eta\Delta\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda}.$$

These are relevant in the selection of the step size η to ensure contraction. While we can validate (37b) and (38b) for 1bCS, D1bCS and DMbCS, which mean that $T_2^{\mathbf{p},\mathbf{q}}$ and $T_3^{\mathbf{p},\mathbf{q}}$ nearly concentrate about 0, in some more intricate problem this may not be true and we will need a more general condition as per (36b). See one-bit phase retrieval [10] for instance.

The moment bounds (36a), (37a) and (38a) are more interesting and, in some sense, convey a type of *independence* between the marginals of $\mathbf{a}_i$ and the separation event $E_{\mathbf{p},\mathbf{q}}^{(i)}$. We shall use (36a) to elaborate on this. In fact, the most natural way to bound $\mathbb{E}(|\mathbf{a}_i^\top\boldsymbol{\beta}_i|^p\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)}))$ is via Cauchy–Schwarz inequality:

$$\mathbb{E}\Big(|\mathbf{a}_i^\top\boldsymbol{\beta}_i|^p\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big)\le\Big(\mathbb{E}\big[|\mathbf{a}_i^\top\boldsymbol{\beta}_1|^{2p}\big]\mathbb{P}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big)^{1/2}\le\sqrt{\mathsf{P}_{\mathbf{p},\mathbf{q}}}(c_1\sqrt{p})^p\le c_2\sqrt{\mathsf{P}_{\mathbf{p},\mathbf{q}}}\cdot\frac{p!}{2} \tag{39}$$

for some absolute constants c_1,c_2. However, this is weaker than what we assumed in (36a) if $\mathsf{P}_{\mathbf{p},\mathbf{q}}=o(1)$. On the other hand, if $\mathbf{a}_i^\top\boldsymbol{\beta}_1$ and $E_{\mathbf{p},\mathbf{q}}^{(i)}$ are independent, then we can easily reach (36a) by

$$\mathbb{E}\Big(|\mathbf{a}_i^\top\boldsymbol{\beta}_i|^p\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big)=\mathsf{P}_{\mathbf{p},\mathbf{q}}\,\mathbb{E}\big[|\mathbf{a}_i^\top\boldsymbol{\beta}_i|^p\big]\le\mathsf{P}_{\mathbf{p},\mathbf{q}}(c_1\sqrt{p})^p\le c_2\,\mathsf{P}_{\mathbf{p},\mathbf{q}}\cdot\frac{p!}{2}. \tag{40}$$

Therefore, roughly put, (36a) indicates a type of independence between $\mathbf{a}_i^\top\boldsymbol{\beta}_i$ and $E_{\mathbf{p},\mathbf{q}}^{(i)}$. As we shall see, such "independence" comes from the rotational invariance of Gaussian sensing vector in 1bCS, while is due to the additional randomness of $\boldsymbol{\tau}$ in D1bCS and DMbCS (it enables us to separately evaluate $\mathbb{E}(\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)}))$ by the randomness of $\boldsymbol{\tau}$).

A more precise description of such independence is that "*the event $E_{\mathbf{p},\mathbf{q}}^{(i)}$ does not significantly affect the sub-Gaussianity of $\mathbf{a}_i^\top\boldsymbol{\beta}_1$, $\mathbf{a}_i^\top\boldsymbol{\beta}_2$ and $J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}}$.*" It is easy to note that (36a), (37a) and (38a) are equivalent to

$$\mathbb{E}\Big(|\mathbf{a}_i^\top\boldsymbol{\beta}_1|^p\big|E_{\mathbf{p},\mathbf{q}}^{(i)}\Big)\le c^{(5)}\cdot\frac{p!}{2}, \tag{41a}$$

$$\mathbb{E}\Big(|\mathbf{a}_i^\top\boldsymbol{\beta}_2|^p\big|E_{\mathbf{p},\mathbf{q}}^{(i)}\Big)\le c^{(7)}\cdot\frac{p!}{2}, \tag{41b}$$

$$\mathbb{E}\Big(|J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}}|^p\big|E_{\mathbf{p},\mathbf{q}}^{(i)}\Big)\le(c^{(8)}\sqrt{p})^p,\quad\forall\mathbf{w}\in\mathbb{S}^{n-1} \tag{41c}$$

for any $p\ge 2$. As $\|X\|_{\psi_2}\asymp\sup_{p\ge1}(\mathbb{E}|X|^p)^{1/p}/\sqrt{p}$ and $\|X\|_{\psi_1}\asymp\sup_{p\ge1}(\mathbb{E}|X|^p)^{1/p}/p$ [38], these conditions essentially characterize the sub-exponential tails of $\mathbf{a}_i^\top \boldsymbol{\beta}_1 | E^{(i)}_{\mathbf{p},\mathbf{q}}$ and $\mathbf{a}_i^\top \boldsymbol{\beta}_2 | E^{(i)}_{\mathbf{p},\mathbf{q}}$

$$\big\|\mathbf{a}_i^\top \boldsymbol{\beta}_1 | E^{(i)}_{\mathbf{p},\mathbf{q}}\big\|_{\psi_1} = O(c^{(5)}) \quad \text{and} \quad \big\|\mathbf{a}_i^\top \boldsymbol{\beta}_2 | E^{(i)}_{\mathbf{p},\mathbf{q}}\big\|_{\psi_1} = O(c^{(7)}), \tag{42}$$

and the sub-Gaussian tail of $J^{\mathbf{p},\mathbf{q}}_{i,\mathbf{w}} | E^{(i)}_{\mathbf{p},\mathbf{q}}$

$$\big\|J^{\mathbf{p},\mathbf{q}}_{i,\mathbf{w}} | E^{(i)}_{\mathbf{p},\mathbf{q}}\big\|_{\psi_2} = O(c^{(8)}), \quad \forall \mathbf{w} \in \mathbb{S}^{n-1}. \tag{43}$$

This perspective was employed in [10, 25] and will be useful in controlling $T_3^{\mathbf{p},\mathbf{q}}$.

With these assumptions in place, we bound $|\|\mathbf{p}-\mathbf{q}\|_2 - T_1^{\mathbf{p},\mathbf{q}}|$, $|T_2^{\mathbf{p},\mathbf{q}}|$ and $T_3^{\mathbf{p},\mathbf{q}}$ in the following propositions.

Proposition 2 (Bounding $|\|\mathbf{p}-\mathbf{q}\|_2 - T_1^{\mathbf{p},\mathbf{q}}|$)**.** *Under Assumption 5, with probability at least* $1 - 2\exp(-2\mathscr{H}(\overline{\mathcal{X}}, r))$*, for all* $(\mathbf{p},\mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4}$ *we have*

$$\big|\|\mathbf{p}-\mathbf{q}\|_2 - T_1^{\mathbf{p},\mathbf{q}}\big| \le 4\eta\Delta \left[\sqrt{\frac{c^{(5)}\mathscr{H}(\overline{\mathcal{X}}, r)\,\mathsf{P}_{\mathbf{p},\mathbf{q}}}{m}} + \frac{\mathscr{H}(\overline{\mathcal{X}}, r)}{m}\right] + \left[\left|1 - \frac{\eta\Delta c^{(6)}}{\Delta \vee \Lambda}\right| + \frac{\eta\Delta\varepsilon^{(2)}}{\Delta \vee \Lambda}\right]\|\mathbf{p}-\mathbf{q}\|_2.$$

Proof. The proof can be found in Appendix E.2. ☐

Proposition 3 (Bounding $|T_2^{\mathbf{p},\mathbf{q}}|$)**.** *Under Assumption 6, with probability at least* $1 - 2\exp(-2\mathscr{H}(\overline{\mathcal{X}}, r))$*, for all* $(\mathbf{p},\mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4}$ *we have*

$$|T_2^{\mathbf{p},\mathbf{q}}| \le 4\eta\Delta \left[\sqrt{\frac{c^{(7)}\mathscr{H}(\overline{\mathcal{X}}, r)\,\mathsf{P}_{\mathbf{p},\mathbf{q}}}{m}} + \frac{\mathscr{H}(\overline{\mathcal{X}}, r)}{m}\right] + \frac{\eta\Delta\varepsilon^{(3)}\|\mathbf{p}-\mathbf{q}\|_2}{\Delta \vee \Lambda}.$$

Proof. The proof can be found in Appendix E.3. ☐

Proposition 4 (Bounding $T_3^{\mathbf{p},\mathbf{q}}$)**.** *Under Assumption 7, with probability at least* $1-4\exp(-2\mathscr{H}(\overline{\mathcal{X}}, r))$*, there exists some absolute constant C such that*

$$T_3^{\mathbf{p},\mathbf{q}} \le Cc^{(8)} r\eta\Delta \sqrt{\Big[\mathsf{P}_{\mathbf{p},\mathbf{q}} + \frac{\mathscr{H}(\overline{\mathcal{X}}, r)}{m}\Big]\Big[\frac{r^{-2}\omega^2(\mathcal{K}_{(r)}) + \mathscr{H}(\overline{\mathcal{X}}, r)}{m}\Big]} + \frac{5r\eta\Delta\varepsilon^{(4)}\|\mathbf{p}-\mathbf{q}\|_2}{\Delta \vee \Lambda}\left(1 + \frac{\mathscr{H}(\overline{\mathcal{X}}, r)}{m\,\mathsf{P}_{\mathbf{p},\mathbf{q}}}\right)$$

holds for all $(\mathbf{p},\mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4}$.

Proof. The proof can be found in Appendix E.4. ☐

3.2.4 RAIC and Main Theorem

Additionally, we assume an upper bound on $\mathsf{P}_{\mathbf{u},\mathbf{v}}$ that complements Assumption 3.

Assumption 8 (Upper bound on $\mathsf{P}_{\mathbf{u},\mathbf{v}}$)**.** *For some* $c^{(9)} > 0$,

$$\mathsf{P}_{\mathbf{u},\mathbf{v}} \le \frac{c^{(9)}\|\mathbf{u}-\mathbf{v}\|_2}{\Delta \vee \Lambda}, \quad \forall \mathbf{u},\mathbf{v} \in \mathbb{A}^\beta_\alpha.$$

Now we can combine all the pieces to obtain our main theorem. We shall pause to present some auxiliary conditions to keep our theorem statement concise.

Auxiliary Assumptions for Theorem 5: We will work with Assumptions 1–8 under $r > 0$ which is a given small enough positive constant obeying $r \lesssim \frac{\mu_4}{\log^{1/2}((\Delta\vee\Lambda)/\mu_4)}$.[6] Recall that $\kappa_\alpha = 1$ if $\alpha = 0$ and $\kappa_\alpha = 2$ if $\alpha > 0$. We choose the step size $\eta = \frac{\Delta\vee\Lambda}{\Delta c^{(6)}}$ inspired by the bound in Proposition 2. Corresponding to $\mu_1 < \frac{1}{2\kappa_\alpha}$, we assume

$$\frac{\sum_{j=1}^3 \varepsilon^{(j)} + 5\varepsilon^{(4)}}{c^{(6)}} \le \frac{0.49}{\kappa_\alpha}. \tag{44}$$

If $\mathcal{K}$ is not a cone and $\alpha > 0$, we also assume

$$\frac{\sum_{j=1}^3 \varepsilon^{(j)} + 5\varepsilon^{(4)}}{c^{(6)}} \le \frac{0.24\alpha}{\mu_4} \tag{45}$$

which stems from (23) in Theorem 4 and ensures $\mathbf{x}^{(t)} \in 2\mathcal{K}$. Furthermore, we assume the mild condition $\mu_4 \le C_0(\Delta \vee \Lambda)$ for some absolute constant C_0.

Theorem 5 (Main Theorem)**.** *Under the above assumptions, we suppose that Assumptions 1–8 hold and $\eta = \frac{\Delta\vee\Lambda}{\Delta c^{(6)}}$. There exist some constants C_i's and c_i's depending on $(c^{(i)})_{i=1}^9$, if*

$$m \ge C_1(\Delta \vee \Lambda)\left(\frac{\omega^2(\mathcal{K}_{(r)})}{r^3} + \frac{\mathscr{H}(\mathcal{X}, r/2)}{r}\right), \tag{46}$$

then with probability at least $1-\exp(-c_2\mathscr{H}(\mathcal{X}, r))$, we have that $(\mathcal{Q}, \mathbf{A}, \boldsymbol{\tau}, \mathcal{K}, \eta = \frac{\Delta\vee\Lambda}{\Delta c^{(6)}})$ respects $(\overline{\mathcal{X}}, \boldsymbol{\mu})$-RAIC at the scale of r with

$$\boldsymbol{\mu} = \left(\frac{(\sum_{j=1}^3 \varepsilon^{(j)} + 5\varepsilon^{(4)})}{c^{(6)}}, C_3 r, C_4 r \log^{1/2}\left(\frac{\Delta\vee\Lambda}{r}\right), \mu_4\right). \tag{47}$$

This RAIC implies the following for any $\mathbf{x} \in \mathcal{X}$ and the corresponding sequence $\{\mathbf{x}^{(t)}\}$ produced by running Algorithm 1 with initialization $\mathbf{x}^{(0)} \in \mathbb{B}_2^n(\mathbf{x}; \mu_4) \cap \mathcal{X}$ and step size $\eta = \frac{\Delta\vee\Lambda}{\Delta c^{(6)}}$:

- *(Linear Convergence) It holds that, for any integer $t \ge 0$,*[7]

$$\|\mathbf{x}^{(t)} - \mathbf{x}\|_2 \le 0.99^t \mu_4 + C_5 r \log^{1/2}\left(\frac{\Delta\vee\Lambda}{r}\right). \tag{48}$$

- *(Quadratic Convergence) If additionally $\mu_1 \le \sqrt{\frac{C_3 r}{\mu_4}}$ holds, then for any $t \ge 0$,*

$$\|\mathbf{x}^{(t)} - \mathbf{x}\|_2 \le \mu_4^{2^{-t}}\left(C_6 r \log^{1/2}\left(\frac{\Delta\vee\Lambda}{r}\right)\right)^{1-2^{-t}}. \tag{49}$$

[6] This is just a technical assumption to ensure $\bar{\kappa} \vee \hat{\kappa} < \mu_4$ in Theorem 4.

[7] We comment that 0.99 is in general improvable (e.g., it can often be a much smaller constant in specific models). This won't be pursued in our paper.

Proof. The constants in this proof, including those hidden behind $\lesssim$, $O(\cdot)$, $\gtrsim$, $\Omega(\cdot)$ and $\asymp$, are allowed to depend on $(c^{(i)})_{i=1}^{9}$ and $(\varepsilon^{(i)})_{i=1}^{4}$. Recall that $\mathcal{N}_r$ is a minimal r-net of $\overline{\mathcal{X}}$, for any $\mathbf{u},\mathbf{v}\in\overline{\mathcal{X}}$ obeying $\|\mathbf{u}-\mathbf{v}\|_2\le\mu_4$, we find $\mathbf{u}_1,\mathbf{v}_1$ as the points in $\mathcal{N}_r$ being closest to $\mathbf{u},\mathbf{v}$ (see (26)) and we have $(\mathbf{u}_1,\mathbf{v}_1)\in\mathcal{N}^{(2)}_{r,\mu_4}$.

Under $\eta=\frac{\Delta\vee\Lambda}{\Delta c^{(6)}}$, we combine (27), (30a), (31a) and (35) to obtain

$$\begin{aligned}
&\|\mathbf{u}-\mathbf{v}-\eta\cdot\mathbf{h}(\mathbf{u},\mathbf{v})\|_{\mathcal{K}^\circ_{(r)}}\\
&\le\|\mathbf{u}_1-\mathbf{v}_1-\eta\cdot\hat{\mathbf{h}}(\mathbf{u}_1,\mathbf{v}_1)\|_{\mathcal{K}^\circ_{(r)}}\\
&\quad+\eta\cdot\|\hat{\mathbf{h}}(\mathbf{u}_1,\mathbf{v}_1)-\mathbf{h}(\mathbf{u}_1,\mathbf{v}_1)\|_{\mathcal{K}^\circ_{(r)}}+2r^2+2\eta\cdot\sup_{(\mathbf{p},\mathbf{q})\in\mathcal{D}_r^{(2)}}\|\mathbf{h}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}\\
&\le r\big|\|\mathbf{u}_1-\mathbf{v}_1\|_2-T_1^{\mathbf{u}_1,\mathbf{v}_1}\big|+r|T_2^{\mathbf{u}_1,\mathbf{v}_1}|+T_3^{\mathbf{u}_1,\mathbf{v}_1}+\frac{r\varepsilon^{(1)}}{c^{(6)}}\|\mathbf{u}_1-\mathbf{v}_1\|_2\\
&\quad+\Big(2+\frac{c^{(3)}}{c^{(6)}}\Big)r^2+2\eta\cdot\sup_{(\mathbf{p},\mathbf{q})\in\mathcal{D}_r^{(2)}}\|\mathbf{h}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}.
\end{aligned}\tag{50}$$

Bound $r\big|\|\mathbf{u}_1-\mathbf{v}_1\|_2-T_1^{\mathbf{u}_1,\mathbf{v}_1}\big|+r|T_2^{\mathbf{u}_1,\mathbf{v}_1}|+T_3^{\mathbf{u}_1,\mathbf{v}_1}$: Note that our assumed sample complexity (46) implies

$$\frac{\omega^2(\mathcal{K}_{(r)})/r^2+\mathscr{H}(\overline{\mathcal{X}},r)}{m}\lesssim\frac{r}{\Delta\vee\Lambda}.$$

Therefore, with promised probability Propositions 2–3 yield

$$\big|\|\mathbf{u}_1-\mathbf{v}_1\|_2-T_1^{\mathbf{u}_1,\mathbf{v}_1}\big|+|T_2^{\mathbf{u}_1,\mathbf{v}_1}|\le\frac{C_1}{c^{(6)}}\sqrt{r(\Delta\vee\Lambda)\,\mathsf{P}_{\mathbf{u}_1,\mathbf{v}_1}}+\frac{(\varepsilon^{(2)}+\varepsilon^{(3)})\|\mathbf{u}_1-\mathbf{v}_1\|_2+C_1r}{c^{(6)}},$$

and Proposition 4 gives

$$T_3^{\mathbf{u}_1,\mathbf{v}_1}\le\frac{C_2r}{c^{(6)}}\sqrt{r\big((\Delta\vee\Lambda)\,\mathsf{P}_{\mathbf{u}_1,\mathbf{v}_1}+r\big)}+\frac{5r\varepsilon^{(4)}\|\mathbf{u}_1-\mathbf{v}_1\|_2}{c^{(6)}}\left(1+O\Big(\frac{r}{(\Delta\vee\Lambda)\,\mathsf{P}_{\mathbf{u}_1,\mathbf{v}_1}}\Big)\right).$$

By $\|\mathbf{u}_1-\mathbf{v}_1\|_2\le C_0(\Delta\vee\Lambda)$ and Assumptions 3, 8 we have $\mathsf{P}_{\mathbf{u}_1,\mathbf{v}_1}\asymp\frac{\|\mathbf{u}_1-\mathbf{v}_1\|_2}{\Delta\vee\Lambda}$. Taken collectively with $\sum_{j=1}^4\varepsilon^{(j)}\lesssim1$ implied by (44), we obtain

$$\begin{aligned}
&r\big|\|\mathbf{u}_1-\mathbf{v}_1\|_2-T_1^{\mathbf{u}_1,\mathbf{v}_1}\big|+r|T_2^{\mathbf{u}_1,\mathbf{v}_1}|+T_3^{\mathbf{u}_1,\mathbf{v}_1}\\
&\le r\left(\frac{(\varepsilon^{(2)}+\varepsilon^{(3)}+5\varepsilon^{(4)})\|\mathbf{u}_1-\mathbf{v}_1\|_2}{c^{(6)}}+C_3\sqrt{r\|\mathbf{u}_1-\mathbf{v}_1\|_2}+C_4r\right)\\
&\le r\left(\frac{(\varepsilon^{(2)}+\varepsilon^{(3)}+5\varepsilon^{(4)})\|\mathbf{u}-\mathbf{v}\|_2}{c^{(6)}}+C_3\sqrt{r\|\mathbf{u}-\mathbf{v}\|_2}+C_5r\right).
\end{aligned}\tag{51}$$

Bound $\sup_{(\mathbf{p},\mathbf{q})\in\mathcal{D}_r^{(2)}}\|\mathbf{h}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$: We combine (30b), (31b) and Proposition 1 to obtain

$$\eta\cdot\|\mathbf{h}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}\le\eta\cdot\|\hat{\mathbf{h}}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}+c_4r^2\le C_4r^2\log^{1/2}\Big(\frac{\Delta\vee\Lambda}{r}\Big)\tag{52}$$

that holds for all $(\mathbf{p},\mathbf{q})\in\mathcal{D}_r^{(2)}$ with the promised probability.

Establish RAIC: Substituting (51) and (52) into (50) and combining with

$$\frac{r\varepsilon^{(1)}\|\mathbf{u}_1-\mathbf{v}_1\|_2}{c^{(6)}} \le \frac{r\varepsilon^{(1)}\|\mathbf{u}-\mathbf{v}\|_2}{c^{(6)}} + \frac{2\varepsilon^{(1)}r^2}{c^{(6)}} = \frac{r\varepsilon^{(1)}\|\mathbf{u}-\mathbf{v}\|_2}{c^{(6)}} + O(r^2),$$

we arrive at the following that holds $\forall \mathbf{u},\mathbf{v}\in\overline{\mathcal{X}}$ obeying $\|\mathbf{u}-\mathbf{v}\|_2\le\mu_4$:

$$\frac{\|\mathbf{u}-\mathbf{v}-\eta\cdot\mathbf{h}(\mathbf{u},\mathbf{v})\|_{\mathcal{K}^\circ_{(r)}}}{r} \le \frac{(\sum_{j=1}^3\varepsilon^{(j)}+5\varepsilon^{(4)})\|\mathbf{u}-\mathbf{v}\|_2}{c^{(6)}} + C_3\sqrt{r\|\mathbf{u}-\mathbf{v}\|_2} + C_6 r\log^{1/2}\Big(\frac{\Delta\vee\Lambda}{r}\Big).$$

This shows that, with the promised probability, $(\mathcal{Q},\mathbf{A},\boldsymbol{\tau},\mathcal{K},\eta)$ respects $(\overline{\mathcal{X}},\boldsymbol{\mu})$-RAIC at the scale of r with $\boldsymbol{\mu}$ given in (47).

Convergence: With RAIC and (44) and (45) that we assume to fulfill $\mu_1<\frac{1}{2\kappa_\alpha}$ and (23), the desired guarantee immediately follows from Theorem 4. $\square$

Our main theorem indicates the following takeaway message of our paper.

Remark 8. *Comparing Theorem 5 and Theorem 3, we find that under the further Assumptions 4–8, the proposed PGD achieves the same error rate as HDM, up to logarithmic factor. While HDM is in general computationally intractable, PGD is efficient so long as the projection onto $\mathcal{K}$ can be executed efficiently, covering some canonical examples such as $\mathcal{K}=\Sigma^n_k$, $M^{n_1,n_2}_{\bar{r}}$, $\sqrt{k}\mathbb{B}^n_1$ and so on.*

3.3 A Complementary Approach

In the multi-bit case, the techniques developed in the last subsection only establish RAIC with $\mu_4\ll\Delta$; in fact, if $\|\mathbf{u}-\mathbf{v}\|_2\ge\Delta$, then $\hat{\mathbf{h}}(\mathbf{u},\mathbf{v})$ deviates too much from the actual gradient $\mathbf{h}(\mathbf{u},\mathbf{v})$. Hence, in general, Assumption 4 does not hold.

In this subsection, we develop a complementary approach for the situation where the sharp approach only yields local convergence. Here, similarly, we introduce r_1 as the accuracy we want to achieve;[8] hence, as per Remark 6, we will let $\phi=r_1$ and seek a different way to uniformly control

$$\frac{1}{r_1}\|\mathbf{u}-\mathbf{v}-\eta\cdot\mathbf{h}(\mathbf{u},\mathbf{v})\|_{\mathcal{K}^\circ_{(r_1)}},\quad \mathbf{u},\mathbf{v}\in\overline{\mathcal{X}}. \tag{53}$$

Let us start with a useful perspective for nonlinear observation (e.g., [18, 33, 34])—to view it as a properly rescaled linear model under a near-centered non-standard noise. In particular, for some properly chosen $\zeta>0$, we decompose the quantized measurement into a re-scaled linear observation $\zeta\cdot\mathbf{a}_i^\top\mathbf{u}$ and an irregular noise $\xi_i(\mathbf{u})$ that hinges on $(\mathbf{a}_i,\tau_i,\mathbf{u})$:

$$\mathcal{Q}(\mathbf{a}_i^\top\mathbf{u}-\tau_i)=\zeta\cdot\mathbf{a}_i^\top\mathbf{u}+\underbrace{\big(\mathcal{Q}(\mathbf{a}_i^\top\mathbf{u}-\tau_i)-\zeta\cdot\mathbf{a}_i^\top\mathbf{u}\big)}_{:=\xi_i(\mathbf{u})}. \tag{54}$$

[8] This is similar to the role of r for the sharp approach in the last subsection.

Utilizing the decomposition, for any $\mathbf{u},\mathbf{v}\in\overline{\mathcal{X}}$ we have

$$\begin{aligned}
&\frac{1}{r_1}\|\mathbf{u}-\mathbf{v}-\eta\cdot\mathbf{h}(\mathbf{u},\mathbf{v})\|_{\mathcal{K}^\circ_{(r_1)}}\\
&=\frac{1}{r_1}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\Big\langle\mathbf{w},\mathbf{u}-\mathbf{v}-\frac{\eta}{m}\sum_{i=1}^m\big[\mathcal{Q}(\mathbf{a}_i^\top\mathbf{u}-\tau_i)-\mathcal{Q}(\mathbf{a}_i^\top\mathbf{v}-\tau_i)\big]\mathbf{a}_i\Big\rangle\\
&=\frac{1}{r_1}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\left[\mathbf{w}^\top\Big(\mathbf{I}_n-\frac{\eta\zeta}{m}\sum_{i=1}^m\mathbf{a}_i\mathbf{a}_i^\top\Big)(\mathbf{u}-\mathbf{v})-\frac{\eta}{m}\sum_{i=1}^m\xi_i(\mathbf{u})\mathbf{a}_i^\top\mathbf{w}+\frac{\eta}{m}\sum_{i=1}^m\xi_i(\mathbf{v})\mathbf{a}_i^\top\mathbf{w}\right]\\
&\le\frac{1}{r_1}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\left|\mathbf{w}^\top\Big(\mathbf{I}_n-\frac{\eta\zeta}{m}\sum_{i=1}^m\mathbf{a}_i\mathbf{a}_i^\top\Big)(\mathbf{u}-\mathbf{v})\right|+2\underbrace{\sup_{\mathbf{u}\in\overline{\mathcal{X}}}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\frac{\eta}{mr_1}\sum_{i=1}^m\xi_i(\mathbf{u})\mathbf{a}_i^\top\mathbf{w}}_{:=\hat{T}},
\end{aligned}\tag{55}$$

The first term can be controlled uniformly for all $\mathbf{u},\mathbf{v}\in\overline{\mathcal{X}}$ by the concentration inequality of product process [26]; see Lemma 12.

Proposition 5. *Under Assumption 1, if $m\ge\frac{\omega^2(\mathcal{K}_{(r_1)})}{r_1^2}$, then there exist some absolute constants C_1,c_2, for any $\mathbf{u},\mathbf{v}\in\overline{\mathcal{X}}$ we have*

$$\begin{aligned}
&\frac{1}{r_1}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\left|\mathbf{w}^\top\Big(\mathbf{I}_n-\frac{\eta\zeta}{m}\sum_{i=1}^m\mathbf{a}_i\mathbf{a}_i^\top\Big)(\mathbf{u}-\mathbf{v})\right|\\
&\le\left(\frac{C_1|\eta\zeta|\cdot\omega(\mathcal{K}_{(r_1)})}{r_1\sqrt{m}}+|\eta\zeta-1|\right)\cdot\|\mathbf{u}-\mathbf{v}\|_2+\frac{C_1|\eta\zeta|\omega(\mathcal{K}_{(r_1)})}{\sqrt{m}}
\end{aligned}\tag{56}$$

with probability at least $1-\exp(-\frac{c_2\omega^2(\mathcal{K}_{(r_1)})}{r_1^2})$.

Proof. See its proof in Appendix E.5. □

In contrast, it takes more technicalities to bound $\hat{T}$ which accounts for the quantization. Indeed, in the literature, the property of $\hat{T}$ being well controlled has proven a major ingredient in the analysis for many other estimators, and is referred to as (global) quantized product embedding (see [8] for $\mathcal{Q}_\delta$), limited projection distortion (see [39] for $\mathcal{Q}_\delta$ or even general non-linearity) or sign product embedding (see [30] for 1bCS).

Auxiliary Assumptions for Theorem 6: As before, we state some secondary assumptions here to ensure cleaner theorem statement. With *arbitrary initialization*, we let $\{\mathbf{x}^{(t)}\}$ be the sequence produced by running Algorithm 1 with initialization $\mathbf{x}^{(0)}\in\mathcal{X}$ and step size η. In a specific nonlinear model, ζ in (54) is often known, and *we simply choose $\eta=\zeta^{-1}$ so that $\eta\zeta=1$ and the bound (56) is "minimized"*. If $\mathcal{K}$ is not a cone and $\alpha>0$, we additionally assume $3r_1\le 0.24\alpha$ to ensure (23). Under these assumptions, we have the following guarantee once $\hat{T}$ defined in (55) is bounded by r_1.

Theorem 6. *Under the above setting and Assumption 1, we assume for some small enough given constant r_1 and $\hat{T}$ defined in (55) is bounded as*

$$\hat{T}\le r_1.$$

For some constants c_i's and C_i's that may depend on (α,β), if

$$m \geq \frac{C_1\omega^2(\mathcal{K}_{(r_1)})}{r_1^2}$$

and $r_1 \leq c_2\beta$, then with probability at least $1-2\exp(-c_3\omega^2(\mathcal{K}))$, $(\mathcal{Q},\mathbf{A},\boldsymbol{\tau},\mathcal{K},\eta)$ respects $(\overline{\mathcal{X}},\boldsymbol{\mu})$-RAIC with $\boldsymbol{\mu}=(r_1,0,2r_1,2\beta)$, and for any $\mathbf{x}\in\mathcal{X}$ and any $t\geq 0$ we have

$$\|\mathbf{x}^{(t)}-\mathbf{x}\|_2 \leq \left(\frac{C_4\omega(\mathcal{K}_{(r_1)})}{r_1\sqrt{m}}\right)^t \beta + 24r_1.$$

Proof. Under $\eta=\zeta^{-1}$ and $m \gtrsim \frac{\omega^2(\mathcal{K}_{(r_1)})}{r_1^2}$, we substitute (56) and $\hat{T}\leq r_1$ into (55) and obtain that $(\mathcal{Q},\mathbf{A},\boldsymbol{\tau},\mathcal{K},\eta)$ respects $(\overline{\mathcal{X}},\boldsymbol{\mu})$-RAIC at the scale r_1 with

$$\boldsymbol{\mu}=(\mu_1,\mu_2,\mu_3,\mu_4)^\top = \left(\frac{C_1\omega(\mathcal{K}_{(r_1)})}{r_1\sqrt{m}},0,3r_1,2\beta\right)^\top,$$

for some absolute constant C_1. Note that we have $\mu_1<\frac{1}{8}$ and $r_1\lesssim\beta$, so by the proof of Theorem 4 we have $\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\leq\varphi_t$, where $\{\varphi_t\}_{t=0}^\infty$ is defined by $\varphi_0=2\beta$ and the recurrence $\varphi_t=4\mu_1\varphi_{t-1}+12r_1$. It is easy to see $\varphi_t\leq(4\mu_1)^t\varphi_0+\frac{12r_1}{1-4\mu_1}$. Substituting $\mu_1=\frac{C_1\omega_{(r_1)}}{r_1\sqrt{m}}$ and $\mu_1<\frac{1}{8}$, the claim follows. ☐

4 Instantiation for Specific Models

In this section, we validate Assumptions 1–8 to establish the optimality of PGD for 1bCS, D1bCS and DMbCS. Our results reproduce the recent result [25] on the optimality of NBIHT for 1bCS of sparse vectors. They also represent the first efficient near-optimal solvers for the uniformly dithered models and low-rank matrix recovery from quantized measurements.

For a specific model, the validation of Assumptions 5–7 consists of elementary probability and moment estimations. It will often be convenient to work with the orthonormal $(\boldsymbol{\beta}_1,\boldsymbol{\beta}_2)$ such that $\mathbf{p}=u_1\boldsymbol{\beta}_1+u_2\boldsymbol{\beta}_2$ and $\mathbf{q}=v_1\boldsymbol{\beta}_1+u_2\boldsymbol{\beta}_2$ for some $u_1>v_1$ and $u_2\geq 0$; also note that $\|\mathbf{p}-\mathbf{q}\|_2=u_1-v_1$; see Section 3.2.2. This enables the expression

$$E_{\mathbf{p},\mathbf{q}}^{(i)}=\{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}\}=\big\{\mathcal{Q}(u_1\mathbf{a}_i^\top\boldsymbol{\beta}_1+u_2\mathbf{a}_i^\top\boldsymbol{\beta}_2-\tau_i)\neq\mathcal{Q}(v_1\mathbf{a}_i^\top\boldsymbol{\beta}_1+u_2\mathbf{a}_i^\top\boldsymbol{\beta}_2-\tau_i)\big\}. \tag{57}$$

Observe that we can simply focus on sufficiently small $\mathsf{P}_{\mathbf{p},\mathbf{q}}$ when we seek to validate (36a), (37a) and (38a). This is because the cases when $\mathsf{P}_{\mathbf{p},\mathbf{q}}\geq c$ (where c is any given small absolute constant) can be addressed by Cauchy–Schwarz inequality; see (39) for instance. Therefore, upon validating Assumption 3 for some absolute constant $c^{(2)}$, we can restrict our attention to $\|\mathbf{p}-\mathbf{q}\|_2\leq c(\Delta\vee\Lambda)$ where c is any given small absolute constant.

4.1 1-Bit Compressed Sensing (1bCS)

We consider the recovery of $\mathbf{x}\in\mathcal{X}:=\mathcal{K}\cap\mathbb{S}^{n-1}$ from $\mathbf{y}=\mathrm{sign}(\mathbf{A}\mathbf{x})$ where $\mathbf{A}$ has i.i.d. $\mathcal{N}(0,1)$ entries. In this case, we have

$$\Delta=2,\quad \Lambda=0,\quad \alpha=1,\quad \beta=1,\quad b_1=0.$$

We seek a global RAIC and set $\mu_4 = 2$; this brings no restriction to $\mathbf{u}, \mathbf{v} \in \mathbb{S}^{n-1}$.

Assumptions 1–4, 8: Note that Assumption 1 is automatic for Gaussian $\mathbf{A}$, and in the 1-bit sensing regime Assumption 4 holds trivially with $\varepsilon^{(1)} = c^{(3)} = c^{(4)} = 0$. Since $\mathbf{a}_i^\top \mathbf{u} \sim \mathcal{N}(0,1)$ for any $\mathbf{u} \in \mathbb{S}^{n-1}$, given $r \in (0, \frac{1}{2})$ we have

$$\mathbb{P}\big(|\mathbf{a}_i^\top \mathbf{u}| \le r\big) \le \sqrt{2/\pi} \cdot r$$

that verifies Assumption 2 with $c^{(1)} = \sqrt{8/\pi}$. Moreover, it is well-documented (e.g., [21,32]) that $\mathsf{P}_{\mathbf{u},\mathbf{v}}$ equals to the Geodesic distance:

$$\mathsf{P}_{\mathbf{u},\mathbf{v}} = \mathbb{P}(\operatorname{sign}(\mathbf{a}_i^\top \mathbf{u}) \ne \operatorname{sign}(\mathbf{a}_i^\top \mathbf{v})) = \frac{\arccos(\mathbf{u}^\top \mathbf{v})}{\pi}, \tag{58}$$

then it follows from some algebra that

$$\frac{1}{\pi}\|\mathbf{u}-\mathbf{v}\|_2 \le \mathsf{P}_{\mathbf{u},\mathbf{v}} \le \frac{1}{2}\|\mathbf{u}-\mathbf{v}\|_2, \quad \forall \mathbf{u}, \mathbf{v} \in \mathbb{S}^{n-1}, \tag{59}$$

which establishes Assumption 3 with $c^{(2)} = \frac{2}{\pi}$ and Assumption 8 with $c^{(9)} = 1$.

Assumptions 5–7: All that remains is to validate Assumptions 5–7. In the parameterization of $(\mathbf{p}, \mathbf{q})$ via $(\boldsymbol{\beta}_1, \boldsymbol{\beta}_2)$, we additionally have $v_1 = -u_1$ due to $\mathbf{p}, \mathbf{q} \in \mathbb{S}^{n-1}$ (see Lemma 10). As discussed, we can focus on sufficiently small $\|\mathbf{p}-\mathbf{q}\|_2 = u_1 - v_1 = 2u_1$ in the validation of (36a), (37a) and (38a). In the following, we validate Assumption 5 as an example.

Fact 1 (Assumption 5 for 1bCS)**.** *For $\mathbf{a}_i \sim \mathcal{N}(0, \mathbf{I}_n)$, Assumption 5 holds for some absolute constant $c^{(5)}$ and*

$$c^{(6)} = \sqrt{2/\pi} \quad \textit{and} \quad \varepsilon^{(2)} = 0.$$

Proof. First we show (36a) holds for some absolute constant $c^{(5)}$. As we observed, we can simply focus on small enough u_1, say, $|u_1| \le \frac{1}{2}$; this implies $|u_2| \ge \frac{1}{2}$ due to $u_1^2 + u_2^2 = \|\mathbf{p}\|_2^2 = 1$. We note that $\mathbf{a}_i^\top \boldsymbol{\beta}_1 := a_1$ and $\mathbf{a}_i^\top \boldsymbol{\beta}_2 := a_2$ are independent $\mathcal{N}(0,1)$ variables. (57) thus specializes to

$$E_{\mathbf{p},\mathbf{q}}^{(i)} = \big\{\operatorname{sign}(u_1 a_1 + u_2 a_2) \ne \operatorname{sign}(-u_1 a_1 + u_2 a_2)\big\} = \{u_1|a_1| > u_2|a_2|\}.$$

Therefore, for any $p \ge 2$ we have

$$\begin{aligned}
&\mathbb{E}\Big(|\mathbf{a}_i^\top \boldsymbol{\beta}_1|^p \mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big) \\
&= \mathbb{E}\Big(|a_1|^p \mathbb{1}\big(|u_1 a_1| > |u_2 a_2|\big)\Big) \\
&\le \mathbb{E}\left(|a_1|^p \sqrt{\frac{2}{\pi}}\Big|\frac{u_1 a_1}{u_2}\Big|\right) \\
&\le |2u_1| \cdot \sqrt{\frac{2}{\pi}} \mathbb{E}\big[|a_1|^{p+1}\big] \le c^{(5)}\,\mathsf{P}_{\mathbf{p},\mathbf{q}} \cdot \frac{p!}{2},
\end{aligned}$$

where the first inequality is due to the randomness of a_2, and the last two inequalities hold because $|u_2| \geq \frac{1}{2}$ and $2u_1 = \|\mathbf{p}-\mathbf{q}\|_2 \asymp \mathsf{P}_{\mathbf{p},\mathbf{q}}$.

Next we establish (36b) with $c^{(6)} = \sqrt{2/\pi}$ and $\varepsilon^{(2)} = 0$. Note that the joint distribution of $(|a_1|,|a_2|)$ is $\frac{2}{\pi}\exp(-\frac{a_1^2+a_2^2}{2})$, we have

$$
\begin{aligned}
&\mathbb{E}\Big(|\mathbf{a}_i^\top \boldsymbol{\beta}_1|\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\Big) \\
&= \mathbb{E}\Big(|a_1|\mathbb{1}\big(u_1|a_1| > u_2|a_2|\big)\Big) \\
&= \frac{2}{\pi}\int_0^\infty \exp\Big(-\frac{w^2}{2}\Big)\left[\int_{u_2w/u_1}^\infty z\exp\Big(-\frac{z^2}{2}\Big)\,\mathrm{d}z\right]\,\mathrm{d}w \\
&= \frac{2}{\pi}\int_0^\infty \exp\Big(-\frac{w^2}{2u_1^2}\Big)\,\mathrm{d}w = \sqrt{\frac{2}{\pi}}u_1.
\end{aligned}
$$

The proof is complete. $\square$

We can similarly validate Assumptions 6–7, see Fact 2 below.

Fact 2 (Assumptions 6, 7 for 1bCS). *For $\mathbf{a}_i \sim \mathcal{N}(0,\mathbf{I}_n)$, Assumption 6 holds for some absolute constant $c^{(7)}$ and $\varepsilon^{(3)} = 0$; Assumption 7 holds for some absolute constant $c^{(8)}$ and $\varepsilon^{(4)} = 0$.*

Proof. The proof can be found in Appendix F.1. $\square$

Now we invoke Theorem 5 and come to the following result.

Theorem 7 (1bCS via PGD). *We recover $\mathbf{x} \in \mathcal{X} = \mathcal{K}\cap\mathbb{S}^{n-1}$ for some star-shaped set $\mathcal{K}$ from $\mathbf{y} = \operatorname{sign}(\mathbf{A}\mathbf{x})$ with standard Gaussian $\mathbf{A}$ by running Algorithm 1 with any $\mathbf{x}^{(0)} \in \mathcal{X}$ and $\eta = \sqrt{\frac{\pi}{2}}$, which produces a sequence $\{\mathbf{x}^{(t)}\}_{t=0}^\infty$. There exist some absolute constants c_i's and C_i's, for any $r \in (0,c_0)$, if*

$$
m \geq C_1\left(\frac{\omega^2(\mathcal{K}_{(r)})}{r^3} + \frac{\mathscr{H}(\mathcal{X},r/2)}{r}\right),
$$

then with probability at least $1-\exp(-c_2\mathscr{H}(\mathcal{X},r/2))$, for any $\mathbf{x}\in\mathcal{X}$ we have

$$
\|\mathbf{x}^{(t)}-\mathbf{x}\|_2 \leq C_3 r\sqrt{\log(r^{-1})}, \quad \forall t \geq C_4\log(\log(r^{-1})).
$$

Proof. We set $\mu_4 = 2$ and consider sufficiently small r. As in Facts 1–2, we have $\eta = \sqrt{\frac{\pi}{2}} = \frac{\Delta\vee\Lambda}{\Delta c^{(6)}}$ and $\varepsilon^{(i)} = 0$ $(i = 1,2,3,4)$, which justify the the conditions in Theorem 5. Therefore, we have $\|\mathbf{x}^{(t)}-\mathbf{x}\|_2 \leq 2^{2^{-t}}(C_1 r\sqrt{\log(r^{-1})})^{1-2^{-t}}$ for any $t \geq 0$ which implies the desired claim. $\square$

Remark 9. *Specializing Theorem 7 to $\mathcal{K} = \Sigma^n_k$ recovers the optimality of NBIHT proved in the recent work [24]. Moreover, letting $\mathcal{K} = M^{n_1,n_2}_{\bar{r}}$ leads to the first efficient and optimal algorithm for 1bCS of low-rank matrices. In addition, letting $\mathcal{K} = \sqrt{k}\mathbb{B}^n_1$ shows PGD achieves $\tilde{O}((\frac{k}{m})^{1/3})$ in 1bCS of effectively sparse signals, which matches with Adaboost [11] (the fastest algorithm in existing works).*

4.2 Dithered 1-Bit Compressed Sensing (D1bCS)

We consider the recovery of $\mathbf{x} \in \mathcal{X} := \mathcal{K} \cap \mathbb{B}_2^n$ from $\mathbf{y} = \operatorname{sign}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau})$ under sub-Gaussian $\mathbf{A}$ with rows $\{\mathbf{a}_i\}_{i=1}^m$ satisfying Assumption 1 and uniform dither $\boldsymbol{\tau} \sim \mathscr{U}([-\lambda, \lambda]^m)$ with some $\lambda \geq 2$; here, $\mathbf{A}$ and $\boldsymbol{\tau}$ are independent. In this case, we have

$$\Delta = 2, \quad \Lambda = \lambda, \quad \alpha = 0, \quad \beta = 1, \quad b_1 = 0.$$

We seek to establish a global RAIC with $\mu_4 = 2$.

Assumptions 2–4, 8: For any $\mathbf{u} \in \mathbb{B}_2^n$ and small enough $r > 0$, we utilize the randomness of $\tau_i \sim \mathscr{U}([-\lambda, \lambda])$ to obtain

$$\mathbb{P}\big(|\mathbf{a}_i^\top \mathbf{u} - \tau_i| \leq r\big) \leq \frac{r}{\lambda},$$

which establishes Assumption 2 with $c^{(1)} = 1$. Also, Assumption 4 holds trivially for 1-bit models with $\varepsilon^{(1)} = c^{(3)} = c^{(4)} = 0$.

Now let us work on Assumptions 3 and 8. Under $\lambda \geq \Delta = 2$, we only need to find absolute constants c_1 and C_2 such that

$$\frac{c_1 \|\mathbf{u} - \mathbf{v}\|_2}{\lambda} \leq \mathsf{P}_{\mathbf{u},\mathbf{v}} \leq \frac{C_2 \|\mathbf{u} - \mathbf{v}\|_2}{\lambda}, \quad \forall \mathbf{u}, \mathbf{v} \in \mathbb{B}_2^n.$$

To this end, in view of

$$\mathsf{P}_{\mathbf{u},\mathbf{v}} = \mathbb{E}\big[\mathbb{1}(\operatorname{sign}(\mathbf{a}_i^\top \mathbf{u} - \tau_i) \neq \operatorname{sign}(\mathbf{a}_i^\top \mathbf{v} - \tau_i))\big],$$

we can first utilize the randomness of τ_i to obtain an upper bound

$$\mathsf{P}_{\mathbf{u},\mathbf{v}} \leq \mathbb{E}\left[\frac{|\mathbf{a}_i^\top (\mathbf{u} - \mathbf{v})|}{2\lambda}\right] \tag{60}$$

and also a lower bound

$$\begin{aligned}
\mathsf{P}_{\mathbf{u},\mathbf{v}} &\geq \mathbb{E}\big[\mathbb{1}(\operatorname{sign}(\mathbf{a}_i^\top \mathbf{u} - \tau_i) \neq \operatorname{sign}(\mathbf{a}_i^\top \mathbf{v} - \tau_i))\mathbb{1}(|\mathbf{a}_i^\top \mathbf{u}| \vee |\mathbf{a}_i^\top \mathbf{v}| \leq \lambda)\big] \\
&= \mathbb{E}\left[\frac{|\mathbf{a}_i^\top (\mathbf{u} - \mathbf{v})|}{2\lambda} \cdot \mathbb{1}\big(|\mathbf{a}_i^\top \mathbf{u}| \vee |\mathbf{a}_i^\top \mathbf{v}| \leq \lambda\big)\right].
\end{aligned} \tag{61}$$

Now we introduce $\mathbf{n}_0 := \frac{\mathbf{u} - \mathbf{v}}{\|\mathbf{u} - \mathbf{v}\|_2} \in \mathbb{S}^{n-1}$ and further deal with the randomness of $\mathbf{a}_i$. By (60) we obtain Assumption 8 with $c^{(9)} = \frac{1}{2}$:

$$\mathsf{P}_{\mathbf{u},\mathbf{v}} \leq \frac{\|\mathbf{u} - \mathbf{v}\|_2 \mathbb{E}[|\mathbf{a}_i^\top \mathbf{n}_0|]}{2\lambda} \leq \frac{\|\mathbf{u} - \mathbf{v}\|_2}{2\lambda}.$$

Conversely, if λ is large enough, then we proceed from (61) as

$$\begin{aligned}
\mathsf{P}_{\mathbf{u},\mathbf{v}} &\geq \frac{\|\mathbf{u} - \mathbf{v}\|_2}{2\lambda} \mathbb{E}\Big[|\mathbf{a}_i^\top \mathbf{n}_0| - |\mathbf{a}_i^\top \mathbf{n}_0| \mathbb{1}(|\mathbf{a}_i^\top \mathbf{u}| > \lambda) - |\mathbf{a}_i^\top \mathbf{n}_0| \mathbb{1}(|\mathbf{a}_i^\top \mathbf{v}| > \lambda)\Big] \\
&\geq \frac{\|\mathbf{u} - \mathbf{v}\|_2}{2\lambda} \Big[L_0 - 4\exp(-\bar{c}_0 \lambda^2)\Big] \geq \frac{L_0 \|\mathbf{u} - \mathbf{v}\|_2}{4\lambda},
\end{aligned}$$

where the second inequality follows from (13a) and (12a) that gives

$$\mathbb{E}\big[|\mathbf{a}_i^\top\mathbf{n}_0|\mathbb{1}(|\mathbf{a}_i^\top\mathbf{w}|>\lambda)\big]\le\sqrt{\mathbb{E}[|\mathbf{a}_i^\top\mathbf{n}_0|^2]\cdot\mathbb{P}(|\mathbf{a}_i^\top\mathbf{w}|>\lambda)}\le 2\exp(-\bar{c}_0\lambda^2),\quad\forall\mathbf{w}\in\mathbb{B}_2^n,$$

and the last inequality holds when λ is large enough. Thus, Assumption 3 holds with absolute constant $c^{(2)}=\frac{L_0}{4}$.

Assumptions 5–7: Because $(\mathbf{a}_i^\top\boldsymbol{\beta}_1,\mathbf{a}_i^\top\boldsymbol{\beta}_2,J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}})$ are independent of τ_i, we can always expect τ_i first and, as with (60) and (61), we have the two-sided bound

$$\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{2\lambda}\cdot\mathbb{1}\big(|\mathbf{a}_i^\top\mathbf{p}|\vee|\mathbf{a}_i^\top\mathbf{q}|\le\lambda\big)\le\mathbb{E}_{\tau_i}\big[\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\big]\le\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{2\lambda}.\tag{62}$$

This enables us to essentially treat $\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})$ as $\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{2\lambda}=\frac{\|\mathbf{p}-\mathbf{q}\|_2}{2\lambda}|\mathbf{a}_i^\top\boldsymbol{\beta}_1|$, as $\{|\mathbf{a}_i^\top\mathbf{p}|\vee|\mathbf{a}_i^\top\mathbf{q}|\le\lambda\}$ holds with overwhelming probability under sufficiently large λ.

Fact 3 (Assumptions 5–7 for D1bCS)**.** *In our D1bCS setting, suppose that $\mathbf{a}_i$ satisfies Assumption 1 which leads to (12) with some absolute constants $\bar{c}_0,\bar{c}_1$. Then we have the following:*

- *Assumption 5 holds for some absolute constant $c^{(5)}$ and*

$$c^{(6)}=\frac{1}{2}\quad\textit{and}\quad\varepsilon^{(2)}=8(\bar{c}_1)^2\exp(-\bar{c}_0\lambda^2);$$

- *Assumption 6 holds for some absolute constant $c^{(7)}$ and $\varepsilon^{(3)}=8(\bar{c}_1)^2\exp(-\bar{c}_0\lambda^2)$;*
- *Assumption 7 holds for some absolute constant $c^{(8)}$ and $\varepsilon^{(4)}=8(\bar{c}_1)^2\exp(-\bar{c}_0\lambda^2)$.*

Proof. The proof can be found in Appendix F.2. □

With all assumptions being validated, Theorem 5 yields the following statement.

Theorem 8 (D1bCS via PGD)**.** *Consider the recovery of $\mathbf{x}\in\mathcal{X}:=\mathcal{K}\cap\mathbb{B}_2^n$ for some star-shaped set $\mathcal{K}$ from $\mathbf{y}=\mathrm{sign}(\mathbf{A}\mathbf{x}-\boldsymbol{\tau})$ where the m rows of $\mathbf{A}$ satisfy Assumption 1 and $\boldsymbol{\tau}\sim\mathscr{U}([-\lambda,\lambda]^m)$. To this end, we obtain $\{\mathbf{x}^{(t)}\}_{t=0}^\infty$ by running Algorithm 1 with $\mathbf{x}^{(0)}\in\mathcal{X}$ and $\eta=\lambda$. There exist some absolute constants c_i's and C_i's, given any $r\in(0,c_1)$, if $\lambda\ge C_2$ and*

$$m\ge C_3\lambda\left(\frac{\omega^2(\mathcal{K}_{(r)})}{r^3}+\frac{\mathscr{H}(\mathcal{X},r/2)}{r}\right),$$

then with probability at least $1-\exp(-c_4\mathscr{H}(\mathcal{X},r))$, for any $\mathbf{x}\in\mathcal{X}$ we have

$$\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\le C_5r\log^{1/2}\left(\frac{\lambda}{r}\right),\quad\forall\ t\ge C_6\log(r^{-1}).$$

Proof. Under $\eta=\lambda=\frac{\Delta\vee\lambda}{\Delta c^{(6)}}$ and $\sum_{j=1}^3\varepsilon^{(j)}+5\varepsilon^{(4)}\le 56(\bar{c}_1)^2\exp(-\bar{c}_0\lambda^2)$, we use $\lambda\ge C_2$ with sufficiently large C_2 to ensure (44). We set $\mu_4=2$ and then Theorem 5 implies $\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\le 2(0.99^t)+C_3r\log^{1/2}(\frac{\lambda}{r})$, which immediately leads to the desired claim. □

Remark 10. *Similarly to Remark 9, Theorem 8 implies that PGD is the first efficient and optimal algorithm for D1bCS of sparse vectors or low-rank matrices. Also, PGD achieves $\tilde{O}((\frac{k}{m})^{1/3})$ for D1bCS of $\mathbf{x} \in \sqrt{k}\mathbb{B}_1^n \cap \mathbb{B}_2^n$, matching with the fastest known algorithm (i.e., the convex program in [22]).*

4.3 Dithered Multi-Bit Compressed Sensing (DMbCS)

We proceed to the more intricate multi-bit model DMbCS that concerns the recovery of $\mathbf{x} \in \mathcal{X} := \mathcal{K} \cap \mathbb{B}_2^n$ from $\mathbf{y} = \mathcal{Q}_{\delta,L}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau})$, where $\mathbf{A}$ has rows $(\mathbf{a}_i)_{i=1}^m$ satisfying Assumption 1, $\boldsymbol{\tau} \sim \mathscr{U}([-\frac{\delta}{2}, \frac{\delta}{2}]^m)$ is independent of $\mathbf{A}$. In this setting, we have

$$\Delta = \delta, \quad \Lambda = \frac{\delta}{2}, \quad \alpha = 0, \quad \beta = 1$$

and the quantization thresholds $\{b_j : j \in [L-1]\} = \{j\delta : j = 1 - \frac{L}{2}, \cdots, -1, 0, 1, \cdots, \frac{L}{2} - 1\}$.

4.3.1 The Sharp Local Approach

We choose $\mu_4 \leq c_1\delta$ for sufficiently small c_1 and aim to show PGD converges to an error rate much sharper than μ_4. We thus also suppose $r \lesssim \delta$.

Assumption 2: For any $\mathbf{u} \in \mathbb{B}_2^n$, we utilize the randomness of τ_i to obtain

$$\mathbb{P}\left(\min_{|j| \leq (L/2)-1} |\mathbf{a}_i^\top \mathbf{u} - \tau_i - j\delta| \leq r\right) = \mathbb{P}\left(-\tau_i \in \bigcup_{|j| \leq (L/2)-1} \left[j\delta - r - \mathbf{a}_i^\top \mathbf{u}, j\delta + r - \mathbf{a}_i^\top \mathbf{u}\right]\right) \leq \frac{2r}{\delta},$$

which establishes Assumption 2 with $c^{(1)} = 2$.

Assumptions 3 and 8: Next, we seek to validate Assumptions 3 and 8 by evaluating

$$\mathsf{P}_{\mathbf{u},\mathbf{v}} = \mathbb{P}\big(\mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top \mathbf{u} - \tau_i) \neq \mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top \mathbf{v} - \tau_i)\big).$$

Compared to D1bCS, this appears more intricate because of the multiple quantization thresholds, while the core idea remains at using the randomness of $\tau_i \sim \mathscr{U}([-\frac{\delta}{2}, \frac{\delta}{2}])$. We have the following.

Fact 4 (Assumptions 3, 8 for DMbCS)**.** *In our DMbCS setting, if $L\delta \geq C_0$ for some absolute constant C_0, then Assumptions 3 and 8 hold with some absolute constant $c^{(2)} > 0$ and $c^{(9)} = 1$.*

Proof. The proof can be found in Appendix F.3. □

Assumption 4: Since we work with the clipped gradient $\hat{\mathbf{h}}(\mathbf{p}, \mathbf{q})$, now we shall validate Assumption 4 to bound the deviation. By (28) and (29) we first formulate

$$\mathbf{h}(\mathbf{p}, \mathbf{q}) - \hat{\mathbf{h}}(\mathbf{p}, \mathbf{q}) = \frac{1}{m} \sum_{i \in \mathbf{R}_{\mathbf{p},\mathbf{q}}} \mathrm{sign}(\mathbf{a}_i^\top(\mathbf{p} - \mathbf{q}))\big(|\mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top \mathbf{p} - \tau_i) - \mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top \mathbf{q} - \tau_i)| - \delta\big)\mathbf{a}_i.$$

Using triangle inequality we can formulate

$$\begin{aligned}
&\eta \cdot \|\mathbf{h}(\mathbf{p},\mathbf{q}) - \hat{\mathbf{h}}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}} \\
&= \eta \cdot \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \left| \frac{1}{m} \sum_{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}} \mathrm{sign}(\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q}))\big(|\mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top\mathbf{p}-\tau_i) - \mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top\mathbf{q}-\tau_i)| - \delta\big)\mathbf{a}_i^\top\mathbf{w} \right| \\
&\le \eta \cdot \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \frac{1}{m} \sum_{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}} \big||\mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top\mathbf{p}-\tau_i) - \mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top\mathbf{q}-\tau_i)| - \delta\big||\mathbf{a}_i^\top\mathbf{w}| \\
&\le \eta \cdot \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \frac{1}{m} \sum_{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}} |\mathbf{a}_i^\top\mathbf{w}||\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|\mathbb{1}\big(|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})| \ge \delta\big), \qquad (63)
\end{aligned}$$

where the last inequality is due to the following Lemma 1.

Lemma 1. *For any $a, b \in \mathbb{R}$ we have*

$$\big||\mathcal{Q}_{\delta,L}(a) - \mathcal{Q}_{\delta,L}(b)| - \delta\big| \cdot \mathbb{1}(\mathcal{Q}_{\delta,L}(a) \neq \mathcal{Q}_{\delta,L}(b)) \le |a-b| \cdot \mathbb{1}(|a-b| \ge \delta).$$

Proof. It will be proved in Appendix F.4. □

The major ideas for showing (31a) is to control the number of nonzero summands in the summation (63), by making use of the factor $\mathbb{1}(|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})| \ge \delta)$. Specifically, by $\|\mathbf{p}-\mathbf{q}\|_2 \le 2\mu_4 \le 2c_1\delta$ with small enough c_1 we manage to show that $|\{i \in [m] : |\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})| \ge \delta\}|$ is uniformly small over $(\mathbf{p},\mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4}$. In contrast, it turns out that a cruder argument which proceeds with all m summands is sufficient for (31b). We have the following statement. It demonstrates that (31a) holds with $\varepsilon^{(1)}$ exponentially decaying with c_1^{-1} if $\mu_4 \le c_1\delta$, thus ensures that the deviation has minimal impact.

Fact 5 (Assumption 4 for DMbCS)**.** *In our DMbCS setting, there exist some absolute constants c_2, C_3, c_4, C_5 such that the following holds: given any sufficiently small $c_1 > 0$, if*

$$\mu_4 \le c_1\delta, \quad m \ge C_3 \exp\Big(\frac{c_2}{c_1^2}\Big)\mathscr{H}(\mathcal{X}, r/2) \quad \textit{and} \quad m \ge \frac{\delta\omega^2(\mathcal{K}_{(r)})}{r^3},$$

then with probability at least $1 - 2\exp(-c_4\mathscr{H}(\mathcal{X}, r))$, Assumption 4 holds with

$$\varepsilon^{(1)} = C_5 \exp\Big(-\frac{c_2}{2c_1^2}\Big)$$

and some absolute constants $c^{(3)}$ and $c^{(4)}$.

Proof. See the complete proof in Appendix F.4. □

Assumptions 5–7: The proof is similar to the proof of Fact 3 but involves more involved calculations. The main intuition is that the randomness of τ_i gives

$$\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{\delta}\mathbb{1}\Big(|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})| < \delta,\ |\mathbf{a}_i^\top\mathbf{p}| \vee |\mathbf{a}_i^\top\mathbf{q}| \le \frac{(L-1)\delta}{2}\Big) \le \mathbb{E}_{\tau_i}\big(\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\big) \le \frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{\delta}, \qquad (64)$$

which we obtain along the proof of Fact 4 in Appendix F.3. This allows us to essentially treat $\mathbb{E}_{\tau_i}(\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}}))$ as $\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{\delta} = \frac{\|\mathbf{p}-\mathbf{q}\|_2|\mathbf{a}_i^\top\boldsymbol{\beta}_1|}{\delta}$ for $(\mathbf{p},\mathbf{q})$ obeying $\|\mathbf{p}-\mathbf{q}\|_2 \le 2\mu_4 \ll \delta$ and $L\delta$ being large enough.

Fact 6 (Assumptions 5–7 for DMbCS)**.** *In our DMbCS setting, assume that $\mu_4 \le c_1\delta$ for some small enough c_1, and that $\mathbf{a}_i$ satisfies (12) for some absolute constants $\bar{c}_0, \bar{c}_1$. We let*

$$\varepsilon' := 8(\bar{c}_1)^2\Big[\exp\Big(-\frac{\bar{c}_0}{4c_1^2}\Big) + \exp\Big(-\frac{\bar{c}_0 L^2\delta^2}{16}\Big)\Big],$$

then the following statements hold:

- *Assumption 5 holds for some absolute constant $c^{(5)}$, $c^{(6)} = 1$ and $\varepsilon^{(2)} = \varepsilon'$.*
- *Assumption 6 holds for some absolute constant $c^{(7)}$, $\varepsilon^{(3)} = \varepsilon'$.*
- *Assumption 7 holds for some absolute constant $c^{(8)}$, $\varepsilon^{(4)} = \varepsilon'$.*

Proof. The proof can be found in Appendix F.5. □

With Assumptions 1–8 being verified, we come to the following statement by invoking Theorem 5.

Corollary 1 (DMbCS via PGD: Phase II)**.** *In our DMbCS setting, we obtain $\{\mathbf{x}^{(t)}\}_{t=0}^\infty$ by running Algorithm 1 with $\mathbf{x}^{(0)} \in \mathcal{X} \cap \mathbb{B}_2^n(\mathbf{x}; c_0\delta)$ and $\eta = 1$, where c_0 is sufficiently small. There exist some absolute constants c_i's and C_i's, if $r \in (0, c_1\delta)$ for small enough c_1, $L\delta \ge C_2$ for large enough C_2, and*

$$m \ge C_3\delta\left(\frac{\omega^2(\mathcal{K}_{(r)})}{r^3} + \frac{\mathscr{H}(\mathcal{X}, r/2)}{r}\right), \tag{65}$$

then with probability at least $1-\exp(-c_4\mathscr{H}(\mathcal{X}, r))$, for any $\mathbf{x} \in \mathcal{X}$ and any $t \ge 0$ it holds that

$$\|\mathbf{x}^{(t)} - \mathbf{x}\|_2 \le c_1 \cdot 0.99^t\delta + C_5 r\log^{1/2}\Big(\frac{\delta}{r}\Big).$$

Proof. We set $\mu_4 = c_0\delta$ with small enough c_0 and only need to check the conditions in Theorem 5. For $r \in (0, c_1\delta)$ with small enough c_1, the sample complexity in Fact 5 (i.e., $m \ge \frac{\delta\omega^2(\mathcal{K}_{(r)})}{r^3}$) can be ensured by (65) in our statement. By Facts 5–6 and $\eta = 1$, we have

$$\frac{\eta\Delta(\sum_{j=1}^4 \varepsilon^{(j)})}{\Delta \vee \Lambda} \le C_5\exp\Big(-\frac{c_3}{2c_0^2}\Big) + 24(\bar{c}_1)^2\left[\exp\Big(-\frac{\bar{c}_0}{4c_0^2}\Big) + \exp\Big(-\frac{\bar{c}_0 L^2\delta^2}{16}\Big)\right]$$

for some absolute constants c_3 and C_5. Therefore, with small enough c_0 and large enough $L\delta$, we can ensure (44). Then we invoke Theorem 5 to establish the result. □

4.3.2 The Complementary Global Approach

The incomplete part of Corollary 1 lies in the condition $\|\mathbf{x}^{(0)} - \mathbf{x}\|_2 \le c_0\delta$ with small enough c_0, which remains non-trivial under fine quantization corresponding to small δ. In this subsection, we invoke the complementary approach to show that *Algorithm 1 with $\mathbf{x}^{(0)} = 0$ enters a region surrounding the underlying signal with radius $c_0\delta$ after a few iterations.*

In the generic development in Section 3.3, r_1 is the (order of) reconstruction accuracy we want to attain, so we should let

$$r_1 = c_1\delta$$

for small constant c_1 to be chosen, in order to fulfill the initialization requirement in Corollary 1. Moreover, we should choose $\zeta = 1$ and use $\mathbf{a}_i^\top \mathbf{u}$ to approximate $\mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top \mathbf{u} - \tau_i)$ [8, 22, 36, 39], which leads to the decomposition $\mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top \mathbf{u} - \tau_i) = \mathbf{a}_i^\top \mathbf{u} + \xi_i(\mathbf{u})$. Accordingly, we set the step size $\eta = 1$.

Therefore, all that remains is to control $\hat{T}$ in (55), which can be written as

$$\hat{T} := \sup_{\mathbf{u}\in\overline{\mathcal{X}}} \sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}} \left| \frac{1}{mr_1} \big\langle \mathcal{Q}_{\delta,L}(\mathbf{Au} - \boldsymbol{\tau}) - \mathbf{Au}, \mathbf{Aw} \big\rangle \right|.$$

We shall first decompose $\hat{T}$ into

$$\begin{aligned} \hat{T} \le& \underbrace{\sup_{\mathbf{u}\in\overline{\mathcal{X}}} \sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}} \left| \frac{1}{mr_1} \big\langle \mathcal{Q}_{\delta}(\mathbf{Au} - \boldsymbol{\tau}) - \mathbf{Au}, \mathbf{Aw} \big\rangle \right|}_{:=\hat{T}^{(1)}} \\ &+ \underbrace{\sup_{\mathbf{u}\in\overline{\mathcal{X}}} \sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}} \left| \frac{1}{mr_1} \big\langle \mathcal{Q}_{\delta}(\mathbf{Au} - \boldsymbol{\tau}) - \mathcal{Q}_{\delta,L}(\mathbf{Au} - \boldsymbol{\tau}), \mathbf{Aw} \big\rangle \right|}_{:=\hat{T}^{(2)}} \end{aligned} \tag{66}$$

where $\hat{T}^{(1)}$ and $\hat{T}^{(2)}$ capture the distortion arising from the dithered uniform quantization and the impact of saturation, respectively.

To bound $\hat{T}^{(1)}$, we use the global quantized product embedding result for $\mathcal{Q}_\delta$ from [8], which we rephrase in Lemma 4 in Appendix F.6. Then, $\hat{T}^{(2)}$ is bounded in Lemma 5. Our idea to control $\hat{T}^{(2)}$ is similar to that for bounding the deviation from gradient clipping (Fact 5)—first we establish uniform bound on the number of measurements affected by saturation, then we utilize Lemma 8. Taken collectively, we come to the following statement.

Proposition 6 (Bounding $\hat{T}$)**.** *In our DMbCS setting we let $r_1 = c_1\delta$ for some small enough c_1. There exist some constants c_i's and C_i's that may depend on c_1, if $C_1 \log^{1/2}(\frac{m}{\omega^2(\mathcal{X})}) \le L\delta \le C_2\omega(\mathcal{X})$ and*

$$m \ge C_3 \left[\mathscr{H}(\mathcal{X}, c_4\delta) + \frac{\omega^2(\mathcal{K}_{(\delta)})}{\delta^2} + \frac{\omega(\mathcal{X})\omega(\mathcal{K}_{(\delta)})}{\delta^2} \right], \tag{67}$$

then with probability at least $1 - \exp(-\frac{c_5\omega^2(\mathcal{X})}{(L\delta)^2}) - \exp(-c_6\mathscr{H}(\mathcal{X}, 2c_7\delta))$, we have

$$\hat{T} \le r_1.$$

Proof. The proof can be found in Appendix F.6. $\square$

Then we invoke Theorem 6 to obtain the following statement.

Corollary 2 (DMbCS via PGD: Phase I)**.** *In our DMbCS setting, suppose we obtain $\{\mathbf{x}^{(t)}\}_{t=0}^{\infty}$ by runing Algorithm 1 with $\eta = 1$ and $\mathbf{x}^{(0)} = 0$. In the same setting as Proposition 6 (i.e., constraint on*

$L\delta$, sample size and probability), for all $\mathbf{x}\in\mathcal{X}$ we have

$$\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\le\left(\frac{C_8\omega(\mathcal{K}_{(\delta)})}{\delta\sqrt{m}}\right)^t+24c_1\delta.$$

Proof. Under our assumptions, Proposition 6 implies $\hat{T}\le r_1$, and we further note that other conditions needed in Theorem 6 are satisfied. Hence, our statement follows from Theorem 6. $\square$

4.3.3 Final Result

By running Algorithm 1 with $\mathbf{x}^{(0)}=0$, Corollary 2 establishes the first stage that the sequence enters $\mathbb{B}_2^n(\mathbf{x};c_0\delta)$, and then the subsequent refinement to a sharp error rate is characterized by Corollary 1. We thus obtain the following final theorem for DMbCS.

Theorem 9 (DMbCS via PGD)**.** *Consider the recovery of $\mathbf{x}\in\mathcal{X}:=\mathcal{K}\cap\mathbb{B}_2^n$ for some star-shaped set $\mathcal{K}$ from $\mathbf{y}=\mathcal{Q}_{\delta,L}(\mathbf{A}\mathbf{x}-\boldsymbol{\tau})$ where the m rows of $\mathbf{A}$ satisfy Assumption 5 and $\boldsymbol{\tau}\sim\mathscr{U}([-\frac{\delta}{2},\frac{\delta}{2}]^m)$. To this end, we obtain $\{\mathbf{x}^{(t)}\}_{t=0}^{\infty}$ by running Algorithm 1 with $\mathbf{x}^{(0)}=0$ and $\eta=1$. There exist some absolute constants c_i's and C_i's, given any $r\in(0,c_1\delta)$, if $C_2\log^{1/2}(\frac{m}{\omega^2(\mathcal{X})})\le L\delta\le C_3\omega(\mathcal{X})$ and*

$$m\ge C_4\left(\frac{\delta\cdot\omega^2(\mathcal{K}_{(r)})}{r^3}+\frac{\delta\cdot\mathscr{H}(\mathcal{X},r/2)}{r}+\frac{\omega(\mathcal{X})\cdot\omega(\mathcal{K}_{(\delta)})}{\delta^2}\right),\tag{68}$$

then with probability at least $1-\exp(-c_6\mathscr{H}(\mathcal{X},c_7\delta))-\exp(-\frac{c_8\omega^2(\mathcal{X})}{(L\delta)^2})$, for any $\mathbf{x}\in\mathcal{X}$ we have

$$\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\le C_8r\log^{1/2}\left(\frac{\delta}{r}\right),\qquad\forall t\ge C_9\log(r^{-1}).$$

Proof. Because $\mathscr{H}(\mathcal{X},c\delta)\lesssim\frac{\delta}{r}\mathscr{H}(\mathcal{X},\frac{r}{2})$ and $\frac{\omega^2(\mathcal{K}_{(\delta)})}{\delta^2}\lesssim\frac{\delta\omega^2(\mathcal{K}_{(r)})}{r^3}$, (68) can fulfill the sample complexity requirements in Corollaries 1–2. Suppose the small enough c_0 in Corollary 1 has been fixed, and regarding this c_0 we invoke Corollary 2 with small enough c_1 to ensure $\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\le 0.99^t+\frac{c_0\delta}{2}$. Note that this implies $\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\le c_0\delta$ for any $t\ge t_0:=\lceil C_2\log(\delta^{-1})\rceil$ for some absolute constant C_2. Then, we utilize $\|\mathbf{x}^{(t_0+t)}-\mathbf{x}\|_2\le 0.99^t(c_0\delta)+C_3r\log^{1/2}(\frac{\delta}{r})$ for any $t\ge 0$, which implies $\|\mathbf{x}^{(t_0+t)}-\mathbf{x}\|_2\le C_4r\log^{1/2}(\frac{\delta}{r})$ for any $t\ge t_1:=\lceil C_5\log(\frac{\delta}{r})\rceil$. The result follows due to $t_0+t_1\le C_6\log(r^{-1})$. $\square$

Remark 11. *Corollary 1, when specialized to $\mathcal{K}=\Sigma_k^n$ and $M_{\bar{r}}^{n_1,n_2}$, yields the error rates $\tilde{O}(\frac{\delta k}{m})$ and $\tilde{O}(\frac{\delta\bar{r}(n_1+n_2)}{m})$. Under L-level quantizer, since $\delta=\tilde{O}(\frac{1}{L})$ suffices to fulfill $L\delta\gtrsim\log^{1/2}(\frac{m}{\omega^2(\mathcal{K}\cap\mathbb{B}_2^n)})$, PGD achieves error rates $\tilde{O}(\frac{k}{mL})$ and $\tilde{O}(\frac{\bar{r}(n_1+n_2)}{mL})$ that nearly match the lower bounds in Theorem 1. For the recovery of sparse vectors, to our best knowledge, this is the first result for a (memoryless) multi-bit quantized compressed sensing system to be decoded nearly optimally by an efficient algorithm.*

We believe the condition $m\gtrsim\delta^{-2}\omega(\mathcal{X})\omega(\mathcal{K}_{(c_5\delta)})$ is a proof artifact and can be removed. While we leave further investigation to future work, we mention two regimes where we will be able to get rid of this constraint.

Remark 12. *The constraint $m\gtrsim\delta^{-2}\omega(\mathcal{X})\omega(\mathcal{K}_{(c_5\delta)})$, which arises in Lemma 5 (bounding the effect of saturation), is benign for the most interesting cases but precludes fine quantization with very small δ.*

We can get rid of the constraint in either of the following two regimes (where there is no saturation and hence Lemma 5 is no longer needed): (i) We consider the uniform quantizer $\mathcal{Q}_\delta$ without the saturation issue; (ii) We only pursue the non-uniform recovery of a fixed signal $\mathbf{x} \in \mathcal{X}$ as with results in [36]; the reason is that for fixed $\mathbf{x}$ we have $\|\mathbf{A}\mathbf{x} - \boldsymbol{\tau}\|_\infty \lesssim \sqrt{\log m}$ w.h.p.; thus, if $L\delta \gtrsim \sqrt{\log m}$, then w.h.p. we have $\mathbf{y} = \mathcal{Q}_{\delta,L}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau}) = \mathcal{Q}_\delta(\mathbf{A}\mathbf{x} - \boldsymbol{\tau})$.

5 Numerical Simulations

We demonstrate the error rates of Algorithm 1 implied by Theorems 7–9 by simulating the following:

- **1bCS** where we recover $\mathbf{x} \in \mathcal{K} = \Sigma^n_k,\ M^{n_1,n_2}_{\bar{r}},\ \sqrt{k}\mathbb{B}^n_1$ with unit ℓ_2-norm from $\mathbf{y} = \mathrm{sign}(\mathbf{A}\mathbf{x})$ under $\mathbf{A} \sim \mathcal{N}^{m\times n}(0,1)$, by running Algorithm 1 with $\mathbf{x}^{(0)} \in \mathcal{K} \cap \mathbb{S}^{n-1}$ and $\eta = \sqrt{\pi/2}$ for 100 iterations;

- **D1bCS** where we recover $\mathbf{x} \in \mathcal{K} = \Sigma^n_k,\ M^{n_1,n_2}_{\bar{r}},\ \sqrt{k}\mathbb{B}^n_1$ with ℓ_2-norm not greater than 1 from $\mathbf{y} = \mathrm{sign}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau})$, under $\mathbf{A}$ having i.i.d. $\{-1,1\}$-valued Bernoulli entries (Bernoulli design) and $\boldsymbol{\tau} \sim \mathscr{U}([-\Lambda,\Lambda]^m)$, by running Algorithm 1 with $\mathbf{x}^{(0)} = 0$ and $\eta = \Lambda$ for 100 iterations;

- **DMbCS** where we recover $\mathbf{x} \in \mathcal{K} = \Sigma^n_k,\ M^{n_1,n_2}_{\bar{r}},\ \sqrt{k}\mathbb{B}^n_1$ with ℓ_2-norm not greater than 1 from $\mathbf{y} = \mathcal{Q}_{\delta,L}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau})$, under Bernoulli design $\mathbf{A}$ and $\boldsymbol{\tau} \sim \mathscr{U}([-\frac{\delta}{2},\frac{\delta}{2}]^m)$, by running Algorithm 1 with $\mathbf{x}^{(0)} = 0$ and $\eta = 1$ for 100 iterations;

We draw a k-sparse signal from a uniform distribution over $\Sigma^n_k \cap \mathbb{S}^{n-1}$, and generate a rank-$\bar{r}$ matrix in $\mathbb{R}^{n_1\times n_2}$ by drawing $\mathbf{X}_0 \sim \mathcal{N}^{n_1\times n_2}(0,1)$ and then retaining only the top $\bar{r}$ SVD components of $\mathbf{X}_0$. Then, the signal is normalized to unit ℓ_2-norm for 1bCS, rescaled to $\|\mathbf{x}\|_2 \sim \mathscr{U}([0,1])$ for D1bCS and DMbCS. For effectively sparse signals living in $\mathbb{B}^n_1(\sqrt{k})$, we generate it as

$$\mathbf{x} = \big(\underbrace{\pm a, \pm a, \cdots, \pm a}_{\text{first } c \text{ entries}}, \underbrace{\pm b, \pm b, \cdots, \pm b}_{\text{last } n-c \text{ entries}}\big)^\top$$

where $c \sim \mathscr{U}(\{1,\cdots,\lceil 0.6k\rceil\})$, $a = \frac{\sqrt{k}+\sqrt{k+n(n-k-c)/c}}{n}$, $b = \frac{\sqrt{k}-ca}{n-c}$, and the sign of each entry is 1 or -1 with equal probability. It is easy to verify that signals generated in this way have unit ℓ_2-norm and ℓ_1-norm equaling $\sqrt{k}$, respectively. Each data point is averaged over 50 independent trials.

Our experimental results are reported in Figures 4–6. More details are provided in the caption. For the recovery of sparse signals and low-rank matrices (the first two sub-figures), we clearly find that the data points roughly shape a straight line with slope -1, confirming that the errors decay with m in the optimal rate $O(m^{-1})$. For the recovery of effectively sparse signals (the rightmost sub-figures), we observe decay rates that are noticeably slower than $O(m^{-1})$ but faster than $O(m^{-1/3})$. In fact, some curves clearly follow the $m^{-1/3}$ decay rate and suggest that this is tight in recovering effectively sparse signals via PGD. Moreover, simultaneously doubling k (or $\bar{r}$) and the measurement number m (e.g., $(k,m) = (3,400)$ v.s. $(k,m) = (6,800)$) almost maintains the same estimation error, which corroborates the scaling laws $\frac{k}{m}$ and $\frac{\bar{r}}{m}$. It is worth mentioning that Λ for D1bCS should be chosen to fit the signal norm, and not doing so leads to performance degradation, as seen by the curves of $\Lambda = 0.8,\ 3.2$ in Figure 5(Left). This is also true for $L\delta$ in DMbCS, though we only report the results under the appropriate choice $L\delta = 5$.

In the multi-bit model DMbCS we additionally track the role of L to corroborate the scaling law $\frac{1}{mL}$. To this end, we simultaneously double L and halve m to maintain the same mL (e.g., $(L,m)=(4,200)$ v.s. $(L,m)=(8,100)$); we have observed that the estimation errors in these cases almost coincide, which is consistent with our theoretical error rate. This is however not true for overly large L: the curves of $L=32,\ 64$ in Figure 6(Left) and $L=32$ in Figure 6(Middle) fail to maintain the same estimation errors as others, especially under small measurement number. This phenomenon can also be nicely interpreted by our Theorem 9: consider the recovery of k-sparse signals, we will need $m\gtrsim k\log(\frac{en}{k})$ to fulfill (68), namely, the measurement number should exceed a threshold that is independent of L and δ; when $\frac{mL}{k}$ is small, the measurement numbers for the cases with large L may not be numerous enough to fulfill (68).

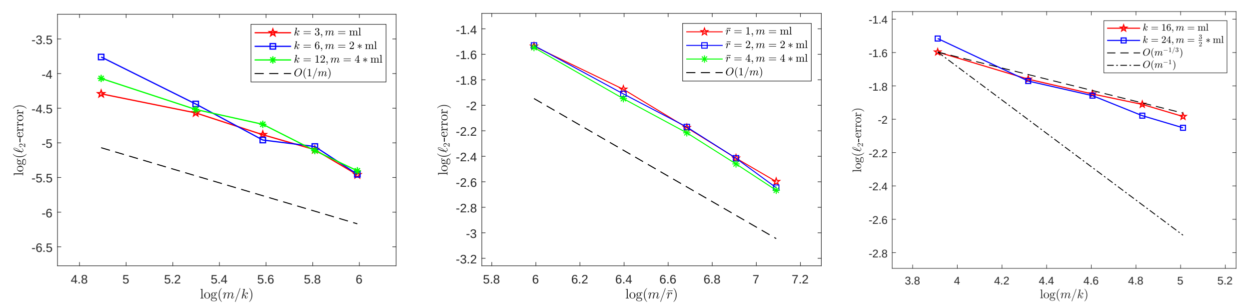

Figure 4: In 1bCS, PGD achieves error rates $\tilde{O}(\frac{k}{m})$ for $\mathbf{x}\in\Sigma_k^n$ (left), $\tilde{O}(\frac{\bar{r}(n_1+n_2)}{m})$ for $\mathbf{x}\in M_{\bar{r}}^{n_1,n_2}$ (middle), and $\tilde{O}((\frac{k}{m})^{1/3})$ for $\mathbf{x}\in\sqrt{k}\mathbb{B}_1^n\cap\mathbb{S}^{n-1}$ (right). In sparse recovery, we recover k-sparse 500-dimensional signals under $m=c\cdot\mathrm{ml}$ with $\mathrm{ml}=400:200:1200$. In low-rank recovery, we recover rank-$\bar{r}$ 25×25 matrices under $m=c\cdot\mathrm{ml}$ with $\mathrm{ml}=400:200:1200$. In recovering effectively sparse signals, we test 300-dimensional signals under $m=c\cdot\mathrm{ml}$ with $\mathrm{ml}=800:400:2400$.

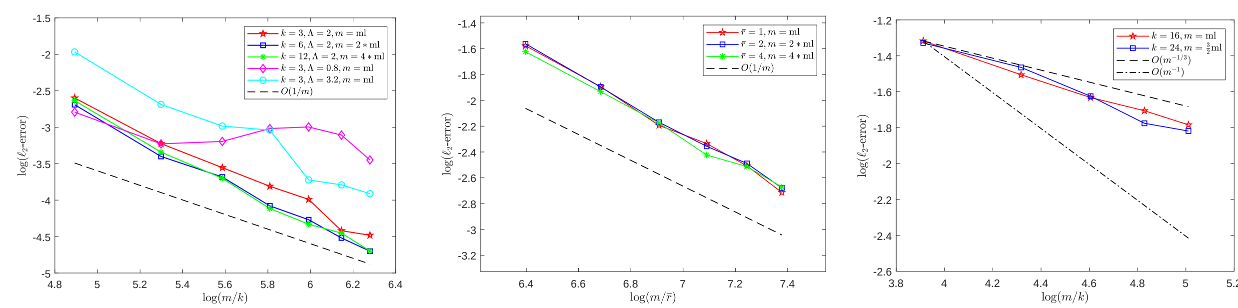

Figure 5: In D1bCS, PGD achieves error rates $\tilde{O}(\frac{k}{m})$ for $\mathbf{x}\in\Sigma_k^n$ (left), $\tilde{O}(\frac{\bar{r}(n_1+n_2)}{m})$ for $\mathbf{x}\in M_{\bar{r}}^{n_1,n_2}$ (middle), and $\tilde{O}((\frac{k}{m})^{1/3})$ for $\mathbf{x}\in\sqrt{k}\mathbb{B}_1^n\cap\mathbb{S}^{n-1}$ (right). In sparse recovery, we recover k-sparse 500-dimensional signals under $m=c\cdot\mathrm{ml}$ with $\mathrm{ml}=400:200:1600$. In low-rank recovery, we recover rank-$\bar{r}$ 25×25 matrices under measurement number $m=c\cdot\mathrm{ml}$ with $\mathrm{ml}=600:200:1600$ with $\Lambda=1.5$. In recovering effectively sparse signals, we test 300-dimensional signals under $m=c\cdot\mathrm{ml}$ with $\mathrm{ml}=800:400:2400$.

6 Discussions

The goal of quantized compressed sensing is to recover structured signals $\mathbf{x}$ from only a few quantized compressive measurements. The information-theoretic lower bounds for recovering k-sparse signal are known as $\tilde{O}(\frac{k}{mL})$ in ℓ_2-norm. However, most efficient algorithms in the literature are sub-optimal

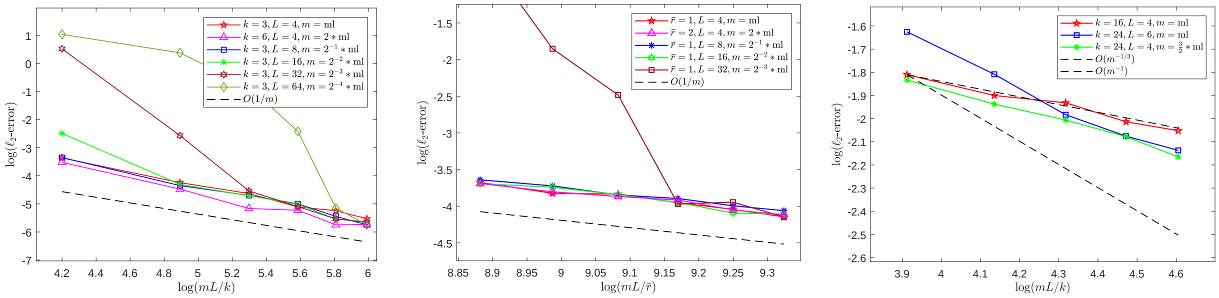

Figure 6: In DMbCS with $\delta=\frac{5}{L}$, PGD achieves error rates $\tilde{O}(\frac{k}{mL})$ for $\mathbf{x}\in\Sigma_k^n$ (left), $\tilde{O}(\frac{\bar{r}(n_1+n_2)}{mL})$ for $\mathbf{x}\in M_{\bar{r}}^{n_1,n_2}$ (middle), and $\tilde{O}((\frac{k}{mL})^{1/3})$ for $\mathbf{x}\in\sqrt{k}\mathbb{B}_1^n\cap\mathbb{S}^{n-1}$ (right). In sparse recovery, we recover k-sparse 500-dimensional signals under measurement number $m=c\cdot\mathrm{ml}$ with $\mathrm{ml}=200:200:1200$. In low-rank recovery, we recover rank-$\bar{r}$ 25×25 matrices under measurement number $m=c\cdot\mathrm{ml}$ with $\mathrm{ml}=1800:200:2800$. In recovering effectively sparse signals, we test 400-dimensional signals under $m=c\cdot\mathrm{ml}$ with $\mathrm{ml}=800:200:1600$.

and only achieve error rates inferior to $\tilde{O}(\frac{1}{L}(\frac{k}{m})^{1/2})$. The only exception is NBIHT that was recently shown [24] to be near-optimal for the very specific problem of recovering k-sparse $\mathbf{x}$ from $\mathbf{y}=\mathrm{sign}(\mathbf{A}\mathbf{x})$. Moreover, for recovering the effectively sparse signals in $\sqrt{k}\mathbb{B}_1^n$, the best known error rate (including that for the intractable HDM) is at the order $\tilde{O}((\frac{k}{mL})^{1/3})$; see Table 1.

In this work, we provide a unified treatment to the general quantized compressed sensing problem by analyzing a PGD algorithm. An intriguing finding is that, under sub-Gaussian design, PGD is a procedure that achieves the same error rate as HDM under the following conditions: *separation probability*, *small-ball probability* and *some moment bounds*. By validating these conditions for the popular models 1bCS, D1bCS and DMbCS, we establish error rates for PGD that improve on or match the fastest error rates in all instances.

It is worth discussing the robustness of PGD to noise or corruption. In a follow-up work of [24], the same authors showed [25] that NBIHT achieves error rate $\tilde{O}(\frac{k}{m}+\bar{\zeta})$ under a $\bar{\zeta}$-fraction of adversarial bit flips, that is, a setting where one observes $\mathbf{y}_{\mathrm{cor}}$ obeying

$$d_H(\mathbf{y}_{\mathrm{cor}},\mathrm{sign}(\mathbf{A}\mathbf{x}))\le\bar{\zeta}m.$$

Under the same corruption pattern, Adaboost achieves $\tilde{O}((\frac{k}{m}+\bar{\zeta})^{1/3})$ for signals in $\sqrt{k}\mathbb{B}_1^n\cap\mathbb{S}^{n-1}$, as shown in [11, Corollary 2.3].

Regarding our PGD algorithm, we shall note that the robustness to adversarial bit flips is in fact a straightforward extension of the current corruption-free results. Suppose that we observe $\mathbf{y}_{\mathrm{cor}}=\mathbf{y}+\mathbf{e}$ under the corruption $\mathbf{e}=(e_1,\cdots,e_m)^\top$, then the update rule of PGD reads

$$\mathbf{x}^{(t)}=\mathcal{P}_{\mathbb{A}_\alpha^\beta}\left(\mathcal{P}_{\mathcal{K}}\Big(\mathbf{x}^{(t-1)}-\eta\cdot\mathbf{h}(\mathbf{x}^{(t-1)},\mathbf{x})-\frac{\eta\cdot\mathbf{A}^\top\mathbf{e}}{m}\Big)\right). \tag{69}$$

Because using an adapted version of RAIC to imply convergence, the only difference on an additional term $O\big(\frac{\eta}{mr}\|\mathbf{A}^\top\mathbf{e}\|_{\mathcal{K}_{(r)}^\circ}\big)$ contributing to the estimation error. Let us characterize the corruption jointly by its sparsity and ℓ_2-norm:

$$\|\mathbf{e}\|_0\le\bar{\zeta}m\quad\text{and}\quad\|\mathbf{e}\|_2\le\bar{\gamma}\sqrt{m}.$$

Then, Lemma 8, together with the sample complexity in Theorem 5, establishes the high-probability event

$$
\begin{aligned}
&\frac{\eta}{mr}\|\mathbf{A}^\top\mathbf{e}\|_{\mathcal{K}^\circ_{(r)}} = \frac{\eta}{mr}\sup_{\mathbf{w}\in\mathcal{K}_{(r)}}\sum_{i=1}^m e_i\mathbf{a}_i^\top\mathbf{w}\\
&\le \frac{\eta}{m}\sup_{\mathbf{w}\in r^{-1}\mathcal{K}_{(r)}}\max_{\substack{I\subset[m]\\|I|\le\bar{\zeta}m}}\sum_{i\in I}e_i|\mathbf{a}_i^\top\mathbf{w}|\\
&\le \frac{\eta\|\mathbf{e}\|_2}{m}\sup_{\mathbf{w}\in r^{-1}\mathcal{K}_{(r)}}\max_{\substack{I\subset[m]\\|I|\le\bar{\zeta}m}}\Big(\sum_{i\in I}|\mathbf{a}_i^\top\mathbf{w}|^2\Big)^{1/2}\\
&\lesssim \frac{\eta\bar{\gamma}\cdot\omega(\mathcal{K}_{(r)})}{\sqrt{m}r} + \eta\bar{\gamma}\sqrt{\bar{\zeta}\log(\bar{\zeta}^{-1})}\\
&\lesssim \eta\bar{\gamma}\sqrt{\frac{r}{\Delta\vee\Lambda}} + \eta\bar{\gamma}\sqrt{\bar{\zeta}\log(\bar{\zeta}^{-1})}.
\end{aligned}
$$

For 1bCS, we can always set $\bar{\gamma} = (2\bar{\zeta})^{1/2}$; combined with $(r\bar{\zeta})^{1/2} \le 2r + 2\bar{\gamma}$, we find that the corruption only increments the estimation error by $O(\bar{\zeta}\log^{1/2}(\bar{\zeta}^{-1}))$. This recovers the major result in [25] regarding the robustness of NBIHT. The robustness to pre-quantization noise which corrupts $\mathbf{y}$ to $\mathbf{y}_{\text{cor}} := \mathcal{Q}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau} - \mathbf{e})$ is perhaps more intricate; we leave this for future work and refer interested readers to the relevant treatments in [13, 22].

As most prior works, we focused on recovering signals that are exactly k-sparse or live in $\sqrt{k}\mathbb{B}_1^n$. To bridge these two cases together, we can characterize the approximate sparsity via ℓ_q-norm ($q \in (0,1]$) and assume

$$
\mathbf{x} \in \mathcal{K} := k^{\frac{1}{q}-\frac{1}{2}}\mathbb{B}_q^n = \left\{\mathbf{u}\in\mathbb{R}^n : \sum_{i\in[n]}|u_i|^q \le k^{1-\frac{q}{2}}\right\}.
$$

By known estimates on the metric entropy of ℓ_q-ball and Dudley's inequality, our Theorem 5 implies that PGD achieves error rate $\tilde{O}((\frac{k}{m})^{\frac{2-q}{2+q}})$ over approximately sparse signals living in the scaled ℓ_q-ball. This establishes an interesting phenomenon: by increasing q from 0 to 1, the PGD error rate continuously decreases while the signal set $k^{\frac{1}{q}-\frac{1}{2}}\mathbb{B}_q^n\cap\mathbb{S}^{n-1}$ continuously "increases" in terms of inclusion. The caveat here is that the projection onto ℓ_q-ball with $q\in(0,1)$ is difficult to compute, but we believe this finding is of interest in light of some existing heuristic for approximating such projection (e.g., [40]). We also note in passing [3] who analyzed the performance of ℓ_q-PGD for linear compressed sensing.

We mention a few questions that need further investigation. An important direction is to develop information-theoretic lower bound for the approximately sparse cases. For example, the rate $\tilde{O}((\frac{k}{m})^{1/3})$ is the best known rate for recovering signals in $\sqrt{k}\mathbb{B}_1^n$, but it remains unclear whether this is tight in any sense, as we discussed in Remark 2. Moreover, the current analysis is built upon sub-Gaussian sensing matrix. It is of great interest to relax this assumption to fit better with the sensing matrix in actual signal processing applications or the features in machine learning problems.

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Appendix

A Roadmap and Notation

Below is the roadmap for the Appendix:

- The main aim of this appendix is to provide a table of notation for the convenience of the readers;
- Appendix B provides the proof of Theorem 1;
- In Appendix C, we first connect quantized compressed sensing with the hyperplane tessellation problem and then prove Theorem 2;
- Appendix D proves Theorem 4 regarding the convergence of PGD implied by RAIC;
- Appendix E collects the missing proofs in the unified analysis for the sharp approach to RAIC in Section 3.2 and the complementary approach in Section 3.3;
- In Appendix F, we show the optimality of PGD for the specific models of 1bCS, D1bCS and DMbCS by validating all the assumptions made in our unified framework;
- The auxiliary lemmas that support our technical proofs are collected in Appendix G.

The recurring non-standard notation can be found in Table 2.

Table 2: List of recurring non-standard notation

$\mathbf{A}, \mathbf{x}, \mathbf{y}$	sub-Gaussian matrix, true signal, vector of quantized measurements
$\mathcal{Q}$	a generic L-level quantizer
Δ	resolution of the quantizer (set as 2 for 1-bit quantizer)
$\boldsymbol{\tau}, \Lambda$	uniform dither $\boldsymbol{\tau} \sim \mathscr{U}[-\Lambda, \Lambda]^m$, dithering scale
$\mathcal{K}$	the star-shaped set that captures signal structure
$\mathbb{A}_\alpha^\beta$	the annulus that captures signal norm
$\mathcal{X}, \overline{\mathcal{X}}$	signal set $\mathcal{X} = \mathcal{K} \cap \mathbb{A}_\alpha^\beta$, enlarged signal set $\overline{\mathcal{X}} = (2\mathcal{K}) \cap \mathbb{A}_\alpha^\beta$
$\mathcal{Q}_\delta, \mathcal{Q}_{\delta,L}$	uniform quantizer (7), L-level uniform quantizer (9)
$\mathsf{P}_{\mathbf{u},\mathbf{v}}$	the probability of $\mathbf{u}, \mathbf{v}$ being distinguished (or separated)
$\mathcal{L}(\mathbf{u}), \mathcal{L}_1(\mathbf{u})$	hamming distance loss, one-sided ℓ_1 loss
$\mathbf{h}(\mathbf{u}, \mathbf{v})$	subgradient of $\mathcal{L}_1(\mathbf{u})$ under true signal $\mathbf{v}$ (18)
$\hat{\mathbf{h}}(\mathbf{u}, \mathbf{v})$	the clipped subgradient (29)
r	desired accuracy for the sharp approach
r_1	desired accuracy for the complementary approach
$\mathcal{D}_r^{(2)}$	$\{(\mathbf{p}, \mathbf{q}) \in \overline{\mathcal{X}} : \\|\mathbf{p} - \mathbf{q}\\|_2 \le r\}$
$\mathcal{N}_{r,\mu_4}^{(2)}$	$\{(\mathbf{p}, \mathbf{q}) \in \mathcal{N}_r \times \mathcal{N}_r : 0 < \\|\mathbf{p} - \mathbf{q}\\|_2 \le 2\mu_4\}$
$\mathbf{R}_{\mathbf{p},\mathbf{q}}$	the index set $\{i \in [m] : \mathcal{Q}(\mathbf{a}_i^\top \mathbf{p} - \tau_i) \neq \mathcal{Q}(\mathbf{a}_i^\top \mathbf{q} - \tau_i)\}$
$E_{\mathbf{p},\mathbf{q}}^{(i)}$	the event $\{i \in \mathbf{R}_{\mathbf{p},\mathbf{q}}\}$
$c^{(i)},\ i \in [9]$	constants in Assumptions 2–8
$\varepsilon^{(i)},\ i \in [4]$	sufficiently small constants in Assumptions 4–7
κ_α	constant that equals 1 when $\alpha = 0$ and equals 2 when $\alpha > 0$
$\zeta, \xi_i(\mathbf{u})$	viewing nonlinear observations as noisy linear ones (54)

B The Proof of Theorem 1 (Lower Bounds)

Proof. We allow the constants in this proof to depend on the given α and β. Because $V_K \subset \mathcal{K}$, we only need to establish the lower bounds for the easier problems of recovering $\mathbf{x} \in V_K \cap \mathbb{A}_\alpha^\beta$ with $\beta > \alpha$ from $\mathbf{y} = \mathcal{Q}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau})$, and recovering $\mathbf{x} \in V_K \cap \mathbb{S}^{n-1}$ from $\mathbf{y} = \mathrm{sign}(\mathbf{A}\mathbf{x})$; these problems are *easier* because of the smaller signal space. For a small $\epsilon > 0$, by standard packing results (e.g., [38]), there exist $P_1 \subset V_K \cap \mathbb{A}_\alpha^\beta$ with $|P_1| \geq (\frac{c_1}{\epsilon})^K$ and $P_2 \subset V_K \cap \mathbb{S}^{n-1}$ with $|P_2| \geq (\frac{c_2}{\epsilon})^{K-1}$, such that the ℓ_2-distance of any two different points in P_i $(i = 1, 2)$ is greater than ϵ. On the other hand, by standard counting results, we have

$$|\{\mathcal{Q}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau}) : \mathbf{x} \in V_K\}| \leq \left|\left\{(\mathrm{sign}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau} - b_1)^\top, \cdots, \mathrm{sign}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau} - b_{L-1})^\top)^\top : \mathbf{x} \in V_K\right\}\right|,$$

which can be bounded by *the number of orthants in* $\mathbb{R}^{(L-1)m}$ *intersected by an affine linear subspace of dimension no greater than* K. (This affine linear subspace can be formally written as $\left\{\left((\mathbf{A}\mathbf{x} - \boldsymbol{\tau} - b_1)^\top, \cdots, (\mathbf{A}\mathbf{x} - \boldsymbol{\tau} - b_{L-1})^\top\right)^\top : \mathbf{x} \in V_K\right\}$.) Therefore, we have (see, e.g., [10, Lemma 3.2])

$$|\{\mathcal{Q}(\mathbf{A}\mathbf{x} - \boldsymbol{\tau}) : \mathbf{x} \in V_K\}| \leq \Big(\frac{C_3(L-1)m}{K}\Big)^K;$$

by similar reasoning, we also have (e.g., [21, Eq. (16)])

$$|\{\mathrm{sign}(\mathbf{A}\mathbf{x}) : \mathbf{x} \in V_K\}| \leq \Big(\frac{C_4 m}{K}\Big)^{K-1}.$$

Notice that both $(\frac{C_3(L-1)m}{K})^K \geq (\frac{c_1}{\epsilon})^K$ and $(\frac{C_4 m}{K})^{K-1} \geq (\frac{c_2}{\epsilon})^{K-1}$ lead to $\epsilon \geq \frac{c_5 K}{mL}$. $\square$

C Quantized Embedding Property

The general L-level quantizer (1) associated with some $(\mathbf{a}, \tau)$ forms the quantized sampler $\mathcal{Q}(\langle \mathbf{a}, \cdot \rangle - \tau)$.

C.1 Preparations

We will need the notions of (well) separation. A hyperplane $\mathcal{H}_{\mathbf{a},\tau} := \{\mathbf{p} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{p} - \tau = 0\}$ separates the whole space $\mathbb{R}^n$ into two sides:

$$\mathcal{H}_{\mathbf{a},\tau}^+ = \{\mathbf{p} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{p} \geq \tau\} \ \ \text{and} \ \ \mathcal{H}_{\mathbf{a},\tau}^- = \{\mathbf{p} \in \mathbb{R}^n : \mathbf{a}^\top \mathbf{p} < \tau\}.$$

We say two points $\mathbf{u}, \mathbf{v}$ are separated by $\mathcal{H}_{\mathbf{a},\tau}$ if they live in different sides of the hyperplane, which is just a geometric statement equivalent to $\mathrm{sign}(\mathbf{a}^\top \mathbf{u} - \tau) \neq \mathrm{sign}(\mathbf{a}^\top \mathbf{v} - \tau)$.

To build this viewpoint for the L-level quantizer (1), we note that $\mathcal{Q}(\mathbf{a}^\top \mathbf{u} - \tau) \neq \mathcal{Q}(\mathbf{a}^\top \mathbf{v} - \tau)$ happens if and only if $\mathrm{sign}(\mathbf{a}^\top \mathbf{u} - \tau - b_j) \neq \mathrm{sign}(\mathbf{a}^\top \mathbf{v} - \tau - b_j)$ holds for some $j \in [L-1]$. Hence, the L-level quantizer corresponds to $L-1$ hyperplanes $\{\mathcal{H}_{\mathbf{a},\tau+b_j} : j \in [L-1]\}$; geometrically, $\mathcal{Q}(\mathbf{a}^\top \mathbf{u} - \tau) \neq \mathcal{Q}(\mathbf{a}^\top \mathbf{v} - \tau)$ holds if and only if $\mathbf{u}, \mathbf{v}$ are separated by $\mathcal{H}_{\mathbf{a},\tau+b_j}$ for some $j \in [L-1]$.

Such hyperplane separation, however, can be unstable: suppose that $\mathbf{u} \in \mathcal{H}_{\mathbf{a},\tau}^+$ and $\mathbf{v} \in \mathcal{H}_{\mathbf{a},\tau}^-$ live very close to the hyperplane itself $\mathcal{H}_{\mathbf{a},\tau}$, then $\mathcal{H}_{\mathbf{a},\tau}$ may no longer separate $\mathbf{u}'$ and $\mathbf{v}$ even though $\mathbf{u}'$ and $\mathbf{u}$ are very close. As with [10, 13, 22, 32], we shall work with θ-well-separation to resolve this issue.

Definition 2. *In the context of quantized compressed sensing under the quantizer (1) with resolution Δ, we say $\mathcal{H}_{\mathbf{a},\tau}$ θ-well-separates $\mathbf{u}$ and $\mathbf{v}$ if $\mathcal{H}_{\mathbf{a},\tau}$ separates $\mathbf{u}$ and $\mathbf{v}$, and it holds additionally that*

$$\min\big\{|\mathbf{a}^\top\mathbf{u}-\tau|,|\mathbf{a}^\top\mathbf{v}-\tau|\big\}\geq\theta\min\{\|\mathbf{u}-\mathbf{v}\|_2,\Delta\vee\Lambda\}.$$

Next, we extend the notion of well (separation) to an L-level quantizer.

Definition 3. *For $\mathcal{Q}$ in (1) and some $(\mathbf{a},\tau)$, we say the quantized sampler $\mathcal{Q}(\langle\mathbf{a},\cdot\rangle-\tau)$ distinguishes $\mathbf{u}$ and $\mathbf{v}$, if there exists some $j\in[L-1]$ such that $\mathcal{H}_{\mathbf{a},\tau+b_j}$ separates $\mathbf{u}$ and $\mathbf{v}$ (this is just equivalent to $\mathcal{Q}(\mathbf{a}^\top\mathbf{u}-\tau)\neq\mathcal{Q}(\mathbf{a}^\top\mathbf{v}-\tau)$). We say the quantized sampler $\mathcal{Q}(\langle\mathbf{a},\cdot\rangle-\tau)$ θ-well-distinguishes $\mathbf{u}$ and $\mathbf{v}$, if there exists some $j\in[L-1]$ such that $\mathcal{H}_{\mathbf{a},\tau+b_j}$ θ-well-separates $\mathbf{u}$ and $\mathbf{v}$.*

In the following, we present two useful facts.

Lemma 2. *Given $\mathbf{u}$ and $\mathbf{v}$, if*

$$\min_{j\in[L-1]}|\mathbf{a}^\top\mathbf{u}-\tau-b_j|>2|\mathbf{a}^\top(\mathbf{u}-\mathbf{v})|,\tag{70}$$

then $\mathbf{u}$ and $\mathbf{v}$ are not distinguished by $\mathcal{Q}(\langle\mathbf{a},\cdot\rangle-\tau)$, i.e., $\mathcal{Q}(\mathbf{a}^\top\mathbf{u}-\tau)=\mathcal{Q}(\mathbf{a}^\top\mathbf{v}-\tau)$.

Proof. We shall assume $\mathbf{u},\mathbf{v}$ are distinguished by $\mathcal{Q}(\langle\mathbf{a},\cdot\rangle-\tau)$ and seek a contradiction. Particularly, suppose $\mathcal{Q}(\mathbf{a}^\top\mathbf{u}-\tau)\neq\mathcal{Q}(\mathbf{a}^\top\mathbf{v}-\tau)$, then there exists some $j\in[L-1]$ such that $\mathcal{H}_{\mathbf{a},\tau+b_j}$ separates $\mathbf{u}$ and $\mathbf{v}$, namely, $\mathrm{sign}(\mathbf{a}^\top\mathbf{u}-\tau-b_j)\neq\mathrm{sign}(\mathbf{a}^\top\mathbf{v}-\tau-b_j)$. This leads to

$$|\mathbf{a}^\top(\mathbf{u}-\mathbf{v})|=\big|(\mathbf{a}^\top\mathbf{u}-\tau-b_j)-(\mathbf{a}^\top\mathbf{v}-\tau-b_j)\big|\geq|\mathbf{a}^\top\mathbf{u}-\tau-b_j|,$$

which contradicts (70). $\square$

Lemma 3. *Given $\mathbf{u},\mathbf{v},\mathbf{p},\mathbf{q}$ and some $\theta>0$, if $\mathcal{Q}(\langle\mathbf{a},\cdot\rangle-\tau)$ θ-well-distinguishes $\mathbf{u}$ and $\mathbf{v}$, and additionally*

$$\max\big\{|\mathbf{a}^\top(\mathbf{p}-\mathbf{u})|,|\mathbf{a}^\top(\mathbf{q}-\mathbf{v})|\big\}<\frac{\theta}{2}\min\big\{\|\mathbf{u}-\mathbf{v}\|_2,\Delta\vee\Lambda\big\},\tag{71}$$

then $\mathbf{p}$ and $\mathbf{q}$ are distinguished by $\mathcal{Q}(\langle\mathbf{a},\cdot\rangle-\tau)$, i.e., $\mathcal{Q}(\mathbf{a}^\top\mathbf{p}-\tau)\neq\mathcal{Q}(\mathbf{a}^\top\mathbf{q}-\tau)$.

Proof. By Definition 3, for some $j\in[L-1]$ we have $\mathrm{sign}(\mathbf{a}^\top\mathbf{u}-\tau-b_j)\neq\mathrm{sign}(\mathbf{a}^\top\mathbf{v}-\tau-b_j)$ and

$$\min\big\{|\mathbf{a}^\top\mathbf{u}-\tau-b_j|,|\mathbf{a}^\top\mathbf{v}-\tau-b_j|\big\}\geq\theta\min\{\|\mathbf{u}-\mathbf{v}\|_2,\Delta\vee\Lambda\}.$$

Combining with (71) we obtain

$$\begin{aligned}2|\mathbf{a}^\top(\mathbf{p}-\mathbf{u})|&<|\mathbf{a}^\top\mathbf{u}-\tau-b_j|,\\2|\mathbf{a}^\top(\mathbf{q}-\mathbf{v})|&<|\mathbf{a}^\top\mathbf{v}-\tau-b_j|.\end{aligned}$$

Now we invoke Lemma 2 with respect to the 1-bit quantized sampler $\mathrm{sign}(\langle\mathbf{a},\cdot\rangle-\tau-b_j)$, yielding that it does not distinguish $\mathbf{u}$ and $\mathbf{p}$, nor $\mathbf{v}$ and $\mathbf{q}$. We hence arrive at $\mathrm{sign}(\mathbf{a}^\top\mathbf{p}-\tau-b_j)=\mathrm{sign}(\mathbf{a}^\top\mathbf{u}-\tau-b_j)$ and $\mathrm{sign}(\mathbf{a}^\top\mathbf{q}-\tau-b_j)=\mathrm{sign}(\mathbf{a}^\top\mathbf{v}-\tau-b_j)$. Taken collectively with $\mathrm{sign}(\mathbf{a}^\top\mathbf{u}-\tau-b_j)\neq\mathrm{sign}(\mathbf{a}^\top\mathbf{v}-\tau-b_j)$, we obtain $\mathrm{sign}(\mathbf{a}^\top\mathbf{p}-\tau-b_j)\neq\mathrm{sign}(\mathbf{a}^\top\mathbf{q}-\tau-b_j)$ that implies the desired $\mathcal{Q}(\mathbf{a}^\top\mathbf{p}-\tau)\neq\mathcal{Q}(\mathbf{a}^\top\mathbf{q}-\tau)$. $\square$

C.2 The Proof of Theorem 2 (Quantized Embedding Property)

Proof. We now delve into the arguments for proving Theorem 2. For some $\epsilon \in (0, c_0(\Delta \vee \Lambda))$ with small enough c_0, we let $\epsilon' = \frac{c_1 \epsilon}{\log^{1/2}(\frac{\Delta \vee \Lambda}{\epsilon})}$ for some small enough constant c_1. We construct $\mathcal{N}_{\epsilon'}$ as a minimal ϵ'-net of $\mathcal{W}$ that satisfies $\log|\mathcal{N}_{\epsilon'}| = \mathscr{H}(\mathcal{W}, \epsilon')$. For any $\mathbf{u}, \mathbf{v} \in \mathcal{W}$, we find their closest points in $\mathcal{N}_{\epsilon'}$ as follows:

$$\mathbf{p} = \arg\min_{\mathbf{w} \in \mathcal{N}_{\epsilon'}} \|\mathbf{w} - \mathbf{u}\|_2 \quad \text{and} \quad \mathbf{q} = \arg\min_{\mathbf{w} \in \mathcal{N}_{\epsilon'}} \|\mathbf{w} - \mathbf{v}\|_2. \tag{72}$$

(i) Establish the event E_s

Given any $\mathbf{u}, \mathbf{v} \in \mathcal{W}$ obeying $\|\mathbf{u}-\mathbf{v}\|_2 \leq \epsilon'/2$, we have $\|\mathbf{u}-\mathbf{p}\|_2 \leq \epsilon'$ and $\|\mathbf{v}-\mathbf{p}\|_2 \leq \|\mathbf{v}-\mathbf{u}\|_2 + \|\mathbf{u}-\mathbf{p}\|_2 \leq 3\epsilon'/2$. By Lemma 2 and a union bound, we can proceed with the following deterministic arguments:

$$\begin{aligned}
& m - d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{u} - \boldsymbol{\tau}), \mathcal{Q}(\mathbf{A}\mathbf{p} - \boldsymbol{\tau})\big) \\
& = \sum_{i=1}^m \mathbb{1}\big(\mathcal{Q}(\mathbf{a}_i^\top \mathbf{u} - \tau_i) = \mathcal{Q}(\mathbf{a}_i^\top \mathbf{p} - \tau_i)\big) \\
& \geq \sum_{i=1}^m \mathbb{1}\Big(\min_{j \in [L-1]} |\mathbf{a}_i^\top \mathbf{p} - \tau_i - b_j| > \epsilon,\ |\mathbf{a}_i^\top (\mathbf{u} - \mathbf{p})| \leq \frac{\epsilon}{2}\Big) \\
& \geq \sum_{i=1}^m \mathbb{1}\Big(\min_{j \in [L-1]} |\mathbf{a}_i^\top \mathbf{p} - \tau_i - b_j| > \epsilon\Big) - \sum_{i=1}^m \mathbb{1}\Big(|\mathbf{a}_i^\top (\mathbf{u} - \mathbf{p})| > \frac{\epsilon}{2}\Big),
\end{aligned}$$

which implies

$$d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{u} - \boldsymbol{\tau}), \mathcal{Q}(\mathbf{A}\mathbf{p} - \boldsymbol{\tau})\big) \leq \sum_{i=1}^m \mathbb{1}\Big(\min_{j \in [L-1]} |\mathbf{a}_i^\top \mathbf{p} - \tau_i - b_j| \leq \epsilon\Big) + \sum_{i=1}^m \mathbb{1}\Big(|\mathbf{a}_i^\top (\mathbf{u} - \mathbf{p})| > \frac{\epsilon}{2}\Big).$$

This remains valid when replacing $\mathbf{u}$ with $\mathbf{v}$:

$$d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{v} - \boldsymbol{\tau}), \mathcal{Q}(\mathbf{A}\mathbf{p} - \boldsymbol{\tau})\big) \leq \sum_{i=1}^m \mathbb{1}\Big(\min_{j \in [L-1]} |\mathbf{a}_i^\top \mathbf{p} - \tau_i - b_j| \leq \epsilon\Big) + \sum_{i=1}^m \mathbb{1}\Big(|\mathbf{a}_i^\top (\mathbf{v} - \mathbf{p})| > \frac{\epsilon}{2}\Big).$$

Noticing $\mathbf{p} \in \mathcal{N}_{\epsilon'}$ and $\mathbf{u} - \mathbf{p}, \mathbf{v} - \mathbf{p} \in \mathcal{W}_{(3\epsilon'/2)}$, by triangle inequality and taking supremums, we arrive at

$$\begin{aligned}
& d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{u} - \boldsymbol{\tau}), \mathcal{Q}(\mathbf{A}\mathbf{v} - \boldsymbol{\tau})\big) \\
& \leq d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{u} - \boldsymbol{\tau}), \mathcal{Q}(\mathbf{A}\mathbf{p} - \boldsymbol{\tau})\big) + d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{v} - \boldsymbol{\tau}), \mathcal{Q}(\mathbf{A}\mathbf{p} - \boldsymbol{\tau})\big) \\
& \leq 2\sum_{i=1}^m \mathbb{1}\Big(\min_{j \in [L-1]} |\mathbf{a}_i^\top \mathbf{p} - \tau_i - b_j| \leq \epsilon\Big) + \sum_{i=1}^m \mathbb{1}\Big(|\mathbf{a}_i^\top (\mathbf{u} - \mathbf{p})| > \frac{\epsilon}{2}\Big) + \sum_{i=1}^m \mathbb{1}\Big(|\mathbf{a}_i^\top (\mathbf{v} - \mathbf{p})| > \frac{\epsilon}{2}\Big) \\
& \leq 2\underbrace{\sup_{\mathbf{p} \in \mathcal{N}_{\epsilon'}} \sum_{i=1}^m \mathbb{1}\Big(\min_{j \in [L-1]} |\mathbf{a}_i^\top \mathbf{p} - \tau_i - b_j| \leq \epsilon\Big)}_{:=\check{T}_1} + 2\underbrace{\sup_{\mathbf{w} \in \mathcal{W}_{(3\epsilon'/2)}} \sum_{i=1}^m \mathbb{1}\Big(|\mathbf{a}_i^\top \mathbf{w}| > \frac{\epsilon}{2}\Big)}_{:=\check{T}_2}.
\end{aligned} \tag{73}$$

Bound $\breve{T}_1$: To bound $\breve{T}_1$, we shall invoke Chernoff bound along with a union bound. By Assumption 2, for any $\mathbf{p}\in\mathcal{N}_{\epsilon'}$, $\min_{j\in[L-1]}|\mathbf{a}_i^\top\mathbf{p}-\tau_i-b_j|\le\epsilon$ holds with probability smaller than $\frac{c^{(1)}\epsilon}{\Delta\vee\Lambda}$, hence we have

$$\begin{aligned}
&\mathbb{P}\left(\sum_{i=1}^m \mathbb{1}\Big(\min_{j\in[L-1]}|\mathbf{a}_i^\top\mathbf{p}-\tau_i-b_j|\le\epsilon\Big)\ge\frac{3mc^{(1)}\epsilon}{2(\Delta\vee\Lambda)}\right)\\
&\le\mathbb{P}\left(\mathrm{Bin}\Big(m,\frac{c^{(1)}\epsilon}{\Delta\vee\Lambda}\Big)\ge\frac{2mc^{(1)}\epsilon}{\Delta\vee\Lambda}\right)\\
&\le\exp\Big(-\frac{c^{(1)}}{12}\frac{m\epsilon}{\Delta\vee\Lambda}\Big),
\end{aligned}$$

where the last inequality is due to Chernoff bound. Hence, a union bound shows that $\breve{T}_1\le\frac{3c^{(1)}m\epsilon}{2(\Delta\vee\Lambda)}$ holds with probability at least

$$1-\exp\Big(\mathscr{H}(\mathcal{W},\epsilon')-\frac{c^{(1)}}{12}\frac{m\epsilon}{\Delta\vee\Lambda}\Big)\ge1-\exp\Big(-\frac{c^{(1)}}{24}\frac{m\epsilon}{\Delta\vee\Lambda}\Big),$$

with the proviso that $m\ge\frac{24(\Delta\vee\Lambda)}{c^{(1)}}\frac{\mathscr{H}(\mathcal{W},\epsilon')}{\epsilon}$, which has been assumed in (15) in our statement.

Bound $\breve{T}_2$: We seek to control $\breve{T}_2$ to a scaling similar to the bound for $\breve{T}_1$. In particular, we seek to show $\breve{T}_2\le\frac{c'm\epsilon}{\Delta\vee\Lambda}$ for some constant c' that is independent of other constants and can be set small enough if needed. To this end, we let $\ell_0:=\lceil\frac{c'm\epsilon}{\Delta\vee\Lambda}\rceil$ and observe that it is sufficient to ensure

$$\sup_{\mathbf{w}\in\mathcal{W}_{(3\epsilon'/2)}}\max_{\substack{I\subset[m]\\|I|\le\ell_0}}\left(\frac{1}{\ell_0}\sum_{i\in I}|\mathbf{a}_i^\top\mathbf{w}|^2\right)^{1/2}\le\frac{\epsilon}{2}.$$

By Lemma 8, with probability at least $1-2\exp(-\frac{c'm\epsilon}{\Delta\vee\Lambda})$, we only need to ensure

$$\frac{\omega(\mathcal{W}_{(3\epsilon'/2)})}{\sqrt{\ell_0}}+\frac{3\epsilon'}{2}\log^{1/2}\Big(\frac{em}{\ell_0}\Big)\le c_3\epsilon$$

for some small enough absolute constant c_3; note that this follows from (15) and our choice $\epsilon'=\frac{c\epsilon}{\log^{1/2}(\frac{\Delta\vee\Lambda}{\epsilon})}$ with small enough c. Substituting the bounds on $\breve{T}_1$ and $\breve{T}_2$ into (73), we have shown that the event E_s holds with the promised probability.

(ii) Establish the event E_l

With Assumptions 2–3, we let $c^*:=\min\{\frac{c^{(2)}}{4c^{(1)}},\frac14\}$ and first establish a lower bound on the probability of $\mathbf{u}$ and $\mathbf{v}$ being c^*-well-distinguished by $\mathcal{Q}(\langle\mathbf{a}_i,\cdot\rangle-\tau_i)$:

$$\begin{aligned}
&\mathbb{P}\Big(\mathcal{Q}(\langle\mathbf{a}_i,\cdot\rangle-\tau_i)\ c^*\text{-well-distinguishes }\mathbf{u}\text{ and }\mathbf{v}\Big)\\
&\ge\mathbb{P}\big(\mathcal{Q}(\mathbf{a}_i^\top\mathbf{u}-\tau_i)\ne\mathcal{Q}(\mathbf{a}_i^\top\mathbf{v}-\tau_i)\big)\\
&\quad-\mathbb{P}\left(\min_{j\in[L-1]}\min\{|\mathbf{a}_i^\top\mathbf{u}-\tau_i-b_j|,|\mathbf{a}_i^\top\mathbf{v}-\tau_i-b_j|\}<c^*\min\{\|\mathbf{u}-\mathbf{v}\|_2,\Delta\vee\Lambda\}\right)
\end{aligned}$$

$$
\begin{aligned}
&\geq \mathbb{P}\big(\mathcal{Q}(\mathbf{a}_i^\top \mathbf{u}-\tau_i)\neq \mathcal{Q}(\mathbf{a}_i^\top \mathbf{v}-\tau_i)\big)\\
&\quad -\mathbb{P}\left(\min_{j\in[L-1]}|\mathbf{a}_i^\top\mathbf{u}-\tau_i-b_j|<c^*\min\{\|\mathbf{u}-\mathbf{v}\|_2,\Delta\vee\Lambda\}\right)\\
&\quad -\mathbb{P}\left(\min_{j\in[L-1]}|\mathbf{a}_i^\top\mathbf{v}-\tau_i-b_j|<c^*\min\{\|\mathbf{u}-\mathbf{v}\|_2,\Delta\vee\Lambda\}\right)\\
&\geq c^{(2)}\min\left\{\frac{\|\mathbf{u}-\mathbf{v}\|_2}{\Delta\vee\Lambda},1\right\}-\frac{2c^{(1)}c^*\min\{\|\mathbf{u}-\mathbf{v}\|_2,\Delta\vee\Lambda\}}{\Delta\vee\Lambda}\\
&\geq \frac{c^{(2)}}{2}\min\left\{\frac{\|\mathbf{u}-\mathbf{v}\|_2}{\Delta\vee\Lambda},1\right\}. \qquad (74)
\end{aligned}
$$

For any $(\mathbf{u},\mathbf{v})\in\mathcal{W}\times\mathcal{W}$ obeying $\|\mathbf{u}-\mathbf{v}\|_2\geq 2\epsilon$ and the corresponding $\mathbf{p},\mathbf{q}$ found by (72), by $\epsilon'<\frac{\epsilon}{2}$ we have

$$
\|\mathbf{p}-\mathbf{q}\|_2\geq\|\mathbf{u}-\mathbf{v}\|_2-\|\mathbf{u}-\mathbf{p}\|_2-\|\mathbf{v}-\mathbf{q}\|_2\geq\frac{\|\mathbf{u}-\mathbf{v}\|_2}{2}\geq\epsilon, \qquad (75)
$$

where the second inequality holds because $\|\mathbf{u}-\mathbf{p}\|_2+\|\mathbf{v}-\mathbf{q}\|_2\leq 2\epsilon'<\epsilon\leq\frac{1}{2}\|\mathbf{u}-\mathbf{v}\|_2$. Then by Lemma 3, along with $\min\{\|\mathbf{p}-\mathbf{q}\|_2,\Delta\vee\Lambda\}\geq\epsilon$, we proceed with the following deterministic arguments:

$$
\begin{aligned}
&d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{u}-\boldsymbol{\tau}),\mathcal{Q}(\mathbf{A}\mathbf{v}-\boldsymbol{\tau})\big)\\
&\geq\sum_{i=1}^m\mathbb{1}\Big(\mathcal{Q}(\langle\mathbf{a}_i,\cdot\rangle-\tau_i)\ c^*\text{-well-distinguishes }\mathbf{p}\text{ and }\mathbf{q},\ \ \max\{|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{p})|,|\mathbf{a}_i^\top(\mathbf{v}-\mathbf{q})|\}\leq\frac{c^*\epsilon}{2}\Big)\\
&\geq\sum_{i=1}^m\mathbb{1}\big(\mathcal{Q}(\langle\mathbf{a}_i,\cdot\rangle-\tau_i)\ c^*\text{-well-distinguishes }\mathbf{p}\text{ and }\mathbf{q}\big) \qquad (76a)\\
&\qquad\qquad -\sum_{i=1}^m\mathbb{1}\Big(|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{p})|\geq\frac{c^*\epsilon}{2}\Big)-\sum_{i=1}^m\mathbb{1}\Big(|\mathbf{a}_i^\top(\mathbf{v}-\mathbf{q})|\geq\frac{c^*\epsilon}{2}\Big). \qquad (76b)
\end{aligned}
$$

Bound (76a): Let us denote the term in (76a) by $\breve{T}_3^{\mathbf{p},\mathbf{q}}$, and now we seek to lower bound $\breve{T}_3^{\mathbf{p},\mathbf{q}}$ uniformly for all $(\mathbf{p},\mathbf{q})\in\mathcal{U}_{\epsilon,\epsilon'}:=\{(\mathbf{p},\mathbf{q})\in\mathcal{N}_{\epsilon'}\times\mathcal{N}_{\epsilon'}:\|\mathbf{p}-\mathbf{q}\|_2\geq\epsilon\}$. We first achieve this for a fixed pair $(\mathbf{p},\mathbf{q})\in\mathcal{U}_{\epsilon,\epsilon'}$. By (74) and Chernoff bound we obtain that, for fixed $(\mathbf{p},\mathbf{q})\in\mathcal{U}_{\epsilon,\epsilon'}$,

$$
\begin{aligned}
&\mathbb{P}\Big(\breve{T}_3^{\mathbf{p},\mathbf{q}}\leq\frac{c^{(2)}m}{4}\min\Big\{\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda},1\Big\}\Big)\\
&\leq\mathbb{P}\left(\mathrm{Bin}\Big(m,\frac{c^{(2)}}{2}\min\Big\{\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda},1\Big\}\Big)\leq\frac{c^{(2)}m}{4}\min\Big\{\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda},1\Big\}\right)\\
&\leq\exp\left(-\frac{c^{(2)}m}{24}\min\Big\{\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda},1\Big\}\right)\\
&\leq\exp\Big(-\frac{c^{(2)}m\epsilon}{24(\Delta\vee\Lambda)}\Big).
\end{aligned}
$$

Taking a union bound, $\breve{T}_3^{\mathbf{p},\mathbf{q}} > \frac{c^{(2)}m}{8}\min\big\{\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda},1\big\}$ holds for any $(\mathbf{p},\mathbf{q})\in\mathcal{U}_{\epsilon,\epsilon'}$ with probability at least

$$1-\exp\Big(2\mathscr{H}(\mathcal{W},\epsilon')-\frac{c^{(2)}m\epsilon}{24(\Delta\vee\Lambda)}\Big)\geq 1-\exp\Big(-\frac{c^{(2)}m\epsilon}{48(\Delta\vee\Lambda)}\Big),$$

with the proviso that $m\geq\frac{96(\Delta\vee\Lambda)}{c^{(2)}\epsilon}\mathscr{H}(\mathcal{W},\epsilon')$; note that this is assumed in (15).

Bound (76b): Next, we bound the terms in (76b). By $\mathbf{u}-\mathbf{p},\mathbf{v}-\mathbf{q}\in\mathcal{W}_{(3\epsilon'/2)}$, we have

$$\sum_{i=1}^m\mathbb{1}\Big(|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{p})|\geq\frac{c^*\epsilon}{2}\Big)+\sum_{i=1}^m\mathbb{1}\Big(|\mathbf{a}_i^\top(\mathbf{v}-\mathbf{q})|\geq\frac{c^*\epsilon}{2}\Big)\leq 2\sup_{\mathbf{w}\in\mathcal{W}_{(3\epsilon'/2)}}\sum_{i=1}^m\Big(|\mathbf{a}_i^\top\mathbf{w}|\geq\frac{c^*\epsilon}{2}\Big):=2\breve{T}_4.$$

By arguments parallel to those for bounding $\breve{T}_2$, under (15) and $\epsilon'=\frac{c_1\epsilon}{\log^{1/2}(\frac{\Delta\vee\Lambda}{\epsilon})}$ with some c_1,C_2 depending on $c^{(1)}$ and $c^{(2)}$, we can show $2\breve{T}_4\leq\frac{c^{(2)}m\epsilon}{16(\Delta\vee\Lambda)}$ with probability at least $1-\exp(-\frac{c''m\epsilon}{\Delta\vee\Lambda})$ for some c'' depending on $c^{(1)}$ and $c^{(2)}$.

Hence, we can substitute the bounds on $\breve{T}_3^{\mathbf{p},\mathbf{q}}$ and $\breve{T}_4$ into (76), yielding the following for any $(\mathbf{u},\mathbf{v})\in\mathcal{W}\times\mathcal{W}$ obeying $\|\mathbf{u}-\mathbf{v}\|_2\geq 2\epsilon$:

$$\begin{aligned}&d_H\big(\mathcal{Q}(\mathbf{A}\mathbf{u}-\boldsymbol{\tau}),\mathcal{Q}(\mathbf{A}\mathbf{v}-\boldsymbol{\tau})\big)\\&\geq\frac{c^{(2)}m}{4}\min\Big\{\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda},1\Big\}-\frac{c^{(2)}m\epsilon}{16(\Delta\vee\Lambda)}\\&\geq\frac{c^{(2)}m}{8}\min\Big\{\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda},1\Big\}\\&\geq\frac{c^{(2)}m}{16}\min\Big\{\frac{\|\mathbf{u}-\mathbf{v}\|_2}{\Delta\vee\Lambda},1\Big\},\end{aligned}$$

where the last inequality follows from $\|\mathbf{p}-\mathbf{q}\|_2\geq\frac{1}{2}\|\mathbf{u}-\mathbf{v}\|_2$, see (75). We have shown that the desired E_l holds with the promised probability. □

D The Proof of Theorem 4 (RAIC Implies Convergence)

Proof. We define a sequence $\{\varphi_t\}_{t=0}^\infty$ by the initial value $\varphi_0=\mu_4$ and the recurrence

$$\varphi_t=2\mu_1\kappa_\alpha\varphi_{t-1}+2\kappa_\alpha\sqrt{\mu_2\varphi_{t-1}}+2\kappa_\alpha\mu_3,\quad t=1,2,\cdots. \tag{77}$$

Notice that $\bar{\kappa}$ defined in (22) is the unique fixed point of the above recurrence. Together with $\bar{\kappa}<\mu_4$, we know that $\{\varphi_t\}_{t=0}^\infty$ is monotonically decreasing and converges to $\bar{\kappa}$ when $t\to\infty$. We omit the proof of this simple fact.

Now we prove that $\mathbf{x}^{(t)}\in 2\mathcal{K}$ and $\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\leq\varphi_t$ for any integer $t\geq 0$. We notice that this is trivially true for $t=0$ and use induction: suppose that $\mathbf{x}^{(t-1)}\in 2\mathcal{K}$ and $\|\mathbf{x}^{(t-1)}-\mathbf{x}\|_2\leq\varphi_{t-1}$ for some $t\geq 1$, we seek to show $\mathbf{x}^{(t)}\in 2\mathcal{K}$ and $\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\leq\varphi_t$. Recall that $\tilde{\mathbf{x}}^{(t)}=\mathcal{P}_{\mathcal{K}}(\mathbf{x}^{(t-1)}-\eta\cdot\mathbf{h}(\mathbf{x}^{(t-1)},\mathbf{x}))$ and $\mathbf{x}^{(t)}=\mathcal{P}_{\mathbb{A}_\alpha^\beta}(\tilde{\mathbf{x}}^{(t)})$. Notice that $\mathbf{x}^{(t-1)}\in\mathbb{A}_\alpha^\beta$ and hence $\mathbf{x}^{(t-1)}\in\overline{\mathcal{X}}=(2\mathcal{K})\cap\mathbb{A}_\alpha^\beta$, and that

$\|\mathbf{x}^{(t-1)}-\mathbf{x}\|_2 \le \varphi_{t-1}\le \mu_4$. Then by Lemma 9 and the RAIC (set $\phi=\mu_3$), we have

$$\begin{aligned}\|\tilde{\mathbf{x}}^{(t)}-\mathbf{x}\|_2 &= \big\|\mathcal{P}_{\mathcal{K}}\big(\mathbf{x}^{(t-1)}-\eta\cdot\mathbf{h}(\mathbf{x}^{(t-1)},\mathbf{x})\big)-\mathbf{x}\big\|_2 &(78)\\ &\le \max\left\{\mu_3,\frac{2}{\mu_3}\big\|\mathbf{x}^{(t-1)}-\mathbf{x}-\eta\cdot\mathbf{h}(\mathbf{x}^{(t-1)},\mathbf{x})\big\|_{\mathcal{K}^{\circ}_{(\mu_3)}}\right\} &(79)\\ &\le 2\mu_1\|\mathbf{x}^{(t-1)}-\mathbf{x}\|_2+2\sqrt{\mu_2\|\mathbf{x}^{(t-1)}-\mathbf{x}\|_2}+2\mu_3 &(80)\\ &\le 2\mu_1\varphi_{t-1}+2\sqrt{\mu_2\varphi_{t-1}}+2\mu_3. &(81)\end{aligned}$$

Let us first show $\mathbf{x}^{(t)}\in 2\mathcal{K}$. Note that $\tilde{\mathbf{x}}^{(t)}\in\mathcal{K}$ and $\mathbf{x}^{(t)}=\mathcal{P}_{\mathbb{A}_\alpha^\beta}(\tilde{\mathbf{x}}^{(t)})=\gamma\tilde{\mathbf{x}}^{(t)}$ for $\gamma=\frac{\|\mathbf{x}^{(t)}\|_2}{\|\tilde{\mathbf{x}}^{(t)}\|_2}$. Thus, $\mathbf{x}^{(t)}\in 2\mathcal{K}$ is evident when $\mathcal{K}$ is a cone. If $\alpha=0$, then the projection onto $\mathbb{A}_\alpha^\beta$ cannot increase norm, hence $\gamma\le 1$ and we obtain $\mathbf{x}^{(t)}\in\mathcal{K}$. If $\mathcal{K}$ is not a cone and $\alpha>0$, we will need (23) to ensure $\mathbf{x}^{(t)}\in 2\mathcal{K}$. Note that we can consider only the case when the projection onto $\mathbb{A}_\alpha^\beta$ increases norm (i.e., $\|\mathbf{x}^{(t)}\|_2\ge\|\tilde{\mathbf{x}}^{(t)}\|_2$), and in this case we necessarily have $\|\mathbf{x}^{(t)}\|_2=\alpha$. By (81) and $\varphi_{t-1}\le\mu_4$ we have $\|\tilde{\mathbf{x}}^{(t)}-\mathbf{x}\|_2\le 2\mu_1\mu_4+2\sqrt{\mu_2\mu_4}+2\mu_3\le\frac{\alpha}{2}$. Therefore, we have $\|\tilde{\mathbf{x}}^{(t)}\|_2\ge\frac{\alpha}{2}$ and $\gamma=\frac{\|\mathbf{x}^{(t)}\|_2}{\|\tilde{\mathbf{x}}^{(t)}\|_2}\le 2$, which gives $\mathbf{x}^{(t)}\in 2\mathcal{K}$. Moreover, we shall show $\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\le\varphi_t$. Note that $\kappa_\alpha=1$ when $\mathbb{A}_\alpha^\beta$ is convex and $\kappa_\alpha=2$ when it is not, so by a well-known bound for projection (e.g., [35, Lemma 15]), we have

$$\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\le\kappa_\alpha\|\tilde{\mathbf{x}}^{(t)}-\mathbf{x}\|_2\le 2\kappa_\alpha\mu_1\varphi_{t-1}+2\kappa_\alpha\sqrt{\mu_2\varphi_{t-1}}+2\kappa_\alpha\mu_3=\varphi_t, \tag{82}$$

as desired. The induction is complete. Due to $\|\mathbf{x}^{(t)}-\mathbf{x}\|_2\le\varphi_t$, all that remains is to control $\{\varphi_t\}_{t=0}^\infty$.

Linear Convergence

By using

$$2\kappa_\alpha\sqrt{\mu_2\varphi_{t-1}}\le\Big(\frac{1}{2}-\kappa_\alpha\mu_1\Big)\varphi_{t-1}+\frac{\kappa_\alpha^2\mu_2}{\frac{1}{2}-\kappa_\alpha\mu_1},$$

we obtain

$$\varphi_t\le\Big(\kappa_\alpha\mu_1+\frac{1}{2}\Big)\varphi_{t-1}+2\kappa_\alpha\Big(\frac{\kappa_\alpha\mu_2}{1-2\kappa_\alpha\mu_1}+\mu_3\Big).$$

Due to $\kappa_\alpha\mu_1<\frac{1}{2}$, by initializing with $\varphi_0=\mu_4$ and iterating the above inequality, we obtain

$$\varphi_t\le\Big(\kappa_\alpha\mu_1+\frac{1}{2}\Big)^t\mu_4+4\kappa_\alpha\left[\frac{\kappa_\alpha\mu_2}{(1-2\kappa_\alpha\mu_1)^2}+\frac{\mu_3}{1-2\kappa_\alpha\mu_1}\right]. \tag{83}$$

The claim immediately follows.

Quadratic Convergence

Combining $\mu_1\le\sqrt{\frac{\mu_2}{\mu_4}}+\frac{\mu_3}{\mu_4}$ and $\varphi_t\le\mu_4$, we have

$$\varphi_t\le 2\Big(\sqrt{\frac{\mu_2}{\mu_4}}+\frac{\mu_3}{\mu_4}\Big)\kappa_\alpha\varphi_{t-1}+2\kappa_\alpha\sqrt{\mu_2\varphi_{t-1}}+2\kappa_\alpha\mu_3\le 4\kappa_\alpha\big(\sqrt{\mu_2\varphi_{t-1}}+\mu_3\big).$$

We observe that $\hat{\kappa}$ is the unique solution to the equation $x = 4\kappa_\alpha(\sqrt{\mu_2 x} + \mu_3)$. We also construct

$$\omega_t := \mu_4^{2^{-t}} \hat{\kappa}^{1-2^{-t}}, \quad \forall t \ge 0$$

and notice that $\{\omega_t\}_{t=0}^{\infty}$ obeys $\omega_0 = \mu_4$, $\omega_t = \sqrt{\hat{\kappa}\omega_{t-1}}$, $\omega_t \ge \hat{\kappa}$. Now let us show $\varphi_t \le \omega_t$ for any integer $t \ge 0$ by induction. This evidently holds when $t = 0$. Suppose $\varphi_{t-1} \le \omega_{t-1}$ for some $t \ge 1$, then we proceed as

$$\varphi_t \le 4\kappa_\alpha(\sqrt{\mu_2\varphi_{t-1}} + \mu_3) \le 4\kappa_\alpha(\sqrt{\mu_2\omega_{t-1}} + \mu_3) \le \sqrt{\hat{\kappa}\omega_{t-1}} = \omega_t,$$

where the second inequality holds because $\omega_{t-1} \ge \hat{\kappa}$ and hence $(\sqrt{\hat{\kappa}} - 4\kappa_\alpha\sqrt{\mu_2})\sqrt{\omega_{t-1}} \ge \hat{\kappa} - 4\kappa_\alpha\sqrt{\mu_2\hat{\kappa}} = 4\kappa_\alpha\mu_3$. The induction is complete and we obtain $\|\mathbf{x}^{(t)} - \mathbf{x}\|_2 \le \varphi_t \le \omega_t$, as claimed. □

E The Proofs in Sections 3.2–3.3 (Unified Analysis)

We collect the missing proofs for our unified analysis for Algorithm 1.

E.1 The Proof of Proposition 1 (Bounding $\|\hat{\mathbf{h}}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}}$)

Proof. The constants in this proof, including those hidden behind $\gtrsim$ and $\lesssim$, are allowed to depend on $c^{(1)}$ in Assumption 2. We invoke the first statement in Theorem 2 to uniformly bound $|\mathbf{R}_{\mathbf{p},\mathbf{q}}|$ over $(\mathbf{p},\mathbf{q}) \in \mathcal{D}_r^{(2)}$. Specializing Theorem 2 to $(\epsilon', \epsilon) = (2r, r'')$ with $r'' := C_1 r \log^{1/2}(\frac{\Delta\vee\Lambda}{r})$ for some sufficiently large C_1 and $\mathcal{W} = \overline{\mathcal{X}} = 2\mathcal{K} \cap \mathbb{A}_\alpha^\beta$, we obtain that with probability at least $1 - 2\exp(-\frac{c_2 mr}{\Delta\vee\Lambda})$, we have

$$|\mathbf{R}_{\mathbf{p},\mathbf{q}}| = d_H(\mathcal{Q}(\mathbf{A}\mathbf{p} - \boldsymbol{\tau}), \mathcal{Q}(\mathbf{A}\mathbf{q} - \boldsymbol{\tau})) \le \frac{c_3 mr''}{\Delta \vee \Lambda}, \quad \forall (\mathbf{p},\mathbf{q}) \in \mathcal{D}_r^{(2)}, \tag{84}$$

as long as

$$m \ge C_4(\Delta \vee \Lambda)\left(\frac{\omega^2(\overline{\mathcal{X}}_{(3r)})}{(r'')^3} + \frac{\mathscr{H}(\overline{\mathcal{X}}, 2r)}{r''}\right). \tag{85}$$

Due to $\overline{\mathcal{X}} \subset 2\mathcal{K}$, we have $\overline{\mathcal{X}}_{(3r)} = (\overline{\mathcal{X}} - \overline{\mathcal{X}}) \cap \mathbb{B}_2^n(3r) \subset (2\mathcal{K} - 2\mathcal{K}) \cap \mathbb{B}_2^n(3r) \subset 3\mathcal{K}_{(r)}$, together with $r'' = C_1 r \log^{1/2}(\frac{\Delta\vee\Lambda}{r})$, (85) can be ensured by (32) we assumed.

On the event (84), by (29), Cauchy-Schwarz inequality and Lemma 8 we could proceed uniformly for all $(\mathbf{p},\mathbf{q}) \in \mathcal{D}_r^{(2)}$ as follows:

$$\begin{aligned}
&\frac{1}{r}\big\|\hat{\mathbf{h}}(\mathbf{p},\mathbf{q})\big\|_{\mathcal{K}^\circ_{(r)}} = \frac{1}{r}\sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \mathbf{w}^\top \hat{\mathbf{h}}(\mathbf{p},\mathbf{q}) \\
&= \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \frac{\Delta}{mr} \sum_{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}} \mathrm{sign}(\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q}))\mathbf{a}_i^\top \mathbf{w} \\
&\le \frac{c_3 r''\Delta}{r(\Delta\vee\Lambda)} \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \max_{\substack{I\subset[m] \\ |I|\le c_3 mr''/(\Delta\vee\Lambda)}} \left(\frac{c_3 mr''}{\Delta\vee\Lambda}\right)^{-1} \sum_{i\in I} |\mathbf{a}_i^\top \mathbf{w}|
\end{aligned}$$

$$
\begin{aligned}
&\leq \frac{c_3 r'' \Delta}{r(\Delta\vee\Lambda)} \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \max_{\substack{I\subset[m] \\ |I|\leq c_3 m r''/(\Delta\vee\Lambda)}} \left[\Big(\frac{c_3 m r''}{\Delta\vee\Lambda}\Big)^{-1} \sum_{i\in I} |\mathbf{a}_i^\top \mathbf{w}|^2\right]^{1/2} \\
&\leq \frac{C_5 r''\Delta}{r(\Delta\vee\Lambda)} \left[\Big(\frac{(\Delta\vee\Lambda)\cdot\omega^2(\mathcal{K}_{(r)})}{c_3 m r''}\Big)^{1/2} + r\cdot \log^{1/2}\Big(\frac{\Delta\vee\Lambda}{r''}\Big)\right] \\
&\leq \frac{C_6 r''\Delta}{\Delta\vee\Lambda} \log^{1/2}\Big(\frac{\Delta\vee\Lambda}{r''}\Big)
\end{aligned}
$$

where the first inequality in the last line holds with probability at least $1-2\exp(-\frac{c_7 m r''}{\Delta\vee\Lambda})$ due to Lemma 8, and the second inequality holds so long as $m \gtrsim \frac{(\Delta\vee\Lambda)\cdot\omega^2(\mathcal{K}_{(r)})}{r'' r^2 \log(\frac{\Delta\vee\Lambda}{r''})}$, and this can be implied by (32) in our statement due to $r'' = C_1 r\log^{1/2}(\frac{\Delta\vee\Lambda}{r})$. Combining everything and substituting r'' conclude the proof. $\square$

E.2 The Proof of Proposition 2 (Bounding $|\|\mathbf{p}-\mathbf{q}\|_2 - T_1^{\mathbf{p},\mathbf{q}}|$)

Proof. We shall first bound $|\|\mathbf{p}-\mathbf{q}\|_2 - T_1^{\mathbf{p},\mathbf{q}}|$ for fixed $(\mathbf{p},\mathbf{q})\in\mathcal{N}^{(2)}_{r,\mu_4}$ and then take a union bound. Recall that $T_1^{\mathbf{p},\mathbf{q}} = \frac{\eta\Delta}{m}\sum_{i=1}^m |\mathbf{a}_i^\top\boldsymbol{\beta}_1|\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})$, and so we have $\mathbb{E}[T_1^{\mathbf{p},\mathbf{q}}] = \eta\Delta\cdot\mathbb{E}\big[|\mathbf{a}_i^\top\boldsymbol{\beta}_1|\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\big]$. Therefore, triangle inequality and (36b) yield

$$
\big|\|\mathbf{p}-\mathbf{q}\|_2 - T_1^{\mathbf{p},\mathbf{q}}\big| \leq \big|T_1^{\mathbf{p},\mathbf{q}} - \mathbb{E}[T_1^{\mathbf{p},\mathbf{q}}]\big| + \left[\Big|1-\frac{\eta\Delta c^{(6)}}{\Delta\vee\Lambda}\Big| + \frac{\eta\Delta\varepsilon^{(2)}}{\Delta\vee\Lambda}\right]\cdot\|\mathbf{p}-\mathbf{q}\|_2. \tag{86}
$$

Next, we use Bernstein's inequality (Lemma 7) to bound

$$
\big|T_1^{\mathbf{p},\mathbf{q}} - \mathbb{E}[T_1^{\mathbf{p},\mathbf{q}}]\big| = \frac{\eta\Delta}{m}\left|\sum_{i=1}^m \Big(|\mathbf{a}_i^\top\boldsymbol{\beta}_1|\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}}) - \mathbb{E}\big[|\mathbf{a}_i^\top\boldsymbol{\beta}_1|\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\big]\Big)\right|.
$$

For any integer $p\geq 2$, (36a) gives

$$
\sum_{i=1}^m \mathbb{E}\Big||\mathbf{a}_i^\top\boldsymbol{\beta}_1|\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\Big|^p \leq \frac{p!}{2}c^{(5)} m\,\mathsf{P}_{\mathbf{p},\mathbf{q}}\,.
$$

Hence, for any $t>0$, Lemma 7 yields

$$
\mathbb{P}\left(\big|T_1^{\mathbf{p},\mathbf{q}} - \mathbb{E}[T_1^{\mathbf{p},\mathbf{q}}]\big| \leq \eta\Delta\Big[\Big(\frac{2c^{(5)}\,\mathsf{P}_{\mathbf{p},\mathbf{q}}\,t}{m}\Big)^{1/2} + \frac{t}{m}\Big]\right) \geq 1-2\exp(-t). \tag{87}
$$

We let $t = 4\mathscr{H}(\overline{\mathcal{X}},r)$ and take a union bound over no more than $|\mathscr{N}(\overline{\mathcal{X}},r)|^2$ pairs of $(\mathbf{p},\mathbf{q})\in\mathcal{N}^{(2)}_{r,\mu_4}$, yielding that the uniform bound

$$
\big|T_1^{\mathbf{p},\mathbf{q}} - \mathbb{E}[T_1^{\mathbf{p},\mathbf{q}}]\big| \leq 4\eta\Delta\left[\Big(\frac{c^{(5)}\,\mathsf{P}_{\mathbf{p},\mathbf{q}}\,\mathscr{H}(\overline{\mathcal{X}},r)}{m}\Big)^{1/2} + \frac{\mathscr{H}(\overline{\mathcal{X}},r)}{m}\right], \quad \forall(\mathbf{p},\mathbf{q})\in\mathcal{N}^{(2)}_{r,\mu_4} \tag{88}
$$

holds with probability at least $1-2\exp(-2\mathscr{H}(\overline{\mathcal{X}},r))$. Substituting this into (86) yields the claim. $\square$

E.3 The Proof of Proposition 3 (Bounding $|T_2^{\mathbf{p},\mathbf{q}}|$)

Proof. Recall that $T_2^{\mathbf{p},\mathbf{q}} = \frac{\eta\Delta}{m}\sum_{i=1}^m \mathrm{sign}(\mathbf{a}_i^\top \boldsymbol{\beta}_1)(\mathbf{a}_i^\top\boldsymbol{\beta}_2)\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})$. Combining with (37b) we have

$$|\mathbb{E}[T_2^{\mathbf{p},\mathbf{q}}]| = \eta\Delta\big|\mathbb{E}[\mathrm{sign}(\mathbf{a}_i^\top\boldsymbol{\beta}_1)(\mathbf{a}_i^\top\boldsymbol{\beta}_2)\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})]\big| \le \frac{\eta\Delta\varepsilon^{(3)}\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda}.$$

Therefore, by triangle inequality we reach

$$|T_2^{\mathbf{p},\mathbf{q}}| \le |T_2^{\mathbf{p},\mathbf{q}} - \mathbb{E}[T_2^{\mathbf{p},\mathbf{q}}]| + \frac{\eta\Delta\varepsilon^{(3)}\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda}.$$

All that remains is to bound $|T_2^{\mathbf{p},\mathbf{q}} - \mathbb{E}[T_2^{\mathbf{p},\mathbf{q}}]|$, which is parallel to bounding $|T_1^{\mathbf{p},\mathbf{q}} - \mathbb{E}[T_1^{\mathbf{p},\mathbf{q}}]|$ in the proof of Proposition 2, following from the moment bound (37a) and Lemma 7. We omit the details. □

E.4 The Proof of Proposition 4 (Bounding $T_3^{\mathbf{p},\mathbf{q}}$)

Proof. The idea is to first condition on $\mathbf{R}_{\mathbf{p},\mathbf{q}}$ to bound the random process over $\mathbf{w}\in\mathcal{K}_{(r)}$ and then get rid of the conditioning by analyzing $|\mathbf{R}_{\mathbf{p},\mathbf{q}}|$. Recall that we defined $J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}} = \mathrm{sign}(\mathbf{a}_i^\top\boldsymbol{\beta}_1)[\mathbf{a}_i - (\mathbf{a}_i^\top\boldsymbol{\beta}_1)\boldsymbol{\beta}_1 - (\mathbf{a}_i^\top\boldsymbol{\beta}_2)\boldsymbol{\beta}_2]^\top\mathbf{w}$ and this is linear on $\mathbf{w}\in\mathbb{R}^n$, and we can write

$$T_3^{\mathbf{p},\mathbf{q}} = \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \frac{\eta\Delta}{m}\sum_{i\in\mathbf{R}_{\mathbf{p},\mathbf{q}}} J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}}.$$

We already have noted that for $\mathbf{w}\in\mathbb{S}^{n-1}$, the conditional distribution $J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}}|E_{\mathbf{p},\mathbf{q}}^{(i)}$ has $O(c^{(8)})$ sub-Gaussian norm; see (43).

Conditioning on $|\mathbf{R}_{\mathbf{p},\mathbf{q}}|$

For any integer $0\le r_{\mathbf{p},\mathbf{q}}\le m$, $T_3^{\mathbf{p},\mathbf{q}}|\{|\mathbf{R}_{\mathbf{p},\mathbf{q}}| = r_{\mathbf{p},\mathbf{q}}\}$ is identically distributed as

$$\sup_{\mathbf{w}\in\mathcal{K}_{(r)}}\frac{\eta\Delta}{m}\sum_{i=1}^{r_{\mathbf{p},\mathbf{q}}} Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}),$$

where $Z_1^{\mathbf{p},\mathbf{q}}(\mathbf{w}), Z_2^{\mathbf{p},\mathbf{q}}(\mathbf{w}),\cdots,Z_{r_{\mathbf{p},\mathbf{q}}}^{\mathbf{p},\mathbf{q}}(\mathbf{w})$ are i.i.d. random variables following the conditional distribution $J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}}|E_{\mathbf{p},\mathbf{q}}^{(i)}$. Therefore, for any $\mathbf{w}\in\mathbb{S}^{n-1}$, $\|Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w})\|_{\psi_2} = O(c^{(8)})$ and hence $\|Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}) - \mathbb{E}[Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w})]\|_{\psi_2} = O(c^{(8)})$ [38, Lemma 2.6.8]. Now we center the random process to come to

$$\sup_{\mathbf{w}\in\mathcal{K}_{(r)}}\frac{\eta\Delta}{m}\sum_{i=1}^{r_{\mathbf{p},\mathbf{q}}} Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}) \le \sup_{\mathbf{w}\in\mathcal{K}_{(r)}}\frac{\eta\Delta}{m}\sum_{i=1}^{r_{\mathbf{p},\mathbf{q}}}\big[Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}) - \mathbb{E}\big(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w})\big)\big] + \sup_{\mathbf{w}\in\mathcal{K}_{(r)}}\frac{\eta\Delta r_{\mathbf{p},\mathbf{q}}\mathbb{E}(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}))}{m}. \tag{89}$$

We seek to control the first term in the right-hand side of (89). For any $\mathbf{w}_1,\mathbf{w}_2\in\mathcal{K}_{(r)}$ we let $\mathbf{w}_3 := \frac{\mathbf{w}_1-\mathbf{w}_2}{\|\mathbf{w}_1-\mathbf{w}_2\|_2}$, then by the linearity of $Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w})$ on $\mathbf{w}$ we have

$$\bigg\|\frac{\eta\Delta}{m}\sum_{i=1}^{r_{\mathbf{p},\mathbf{q}}}\big[Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}_1) - \mathbb{E}\big(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}_1)\big)\big] - \frac{\eta\Delta}{m}\sum_{i=1}^{r_{\mathbf{p},\mathbf{q}}}\big[Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}_2) - \mathbb{E}\big(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}_2)\big)\big]\bigg\|_{\psi_2}$$

$$
\begin{aligned}
&= \left\| \frac{\eta\Delta}{m} \sum_{i=1}^{r_{\mathbf{p},\mathbf{q}}} \left[Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}_3) - \mathbb{E}\big(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}_3)\big) \right] \right\|_{\psi_2} \|\mathbf{w}_1 - \mathbf{w}_2\|_2 \\
&\le \frac{C_1 c^{(8)} \eta\Delta \sqrt{r_{\mathbf{p},\mathbf{q}}} \|\mathbf{w}_1 - \mathbf{w}_2\|_2}{m},
\end{aligned}
$$

where the last inequality holds for some absolute constant C_1, due to [38, Proposition 2.6.1] and $\|Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}) - \mathbb{E}(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}))\|_{\psi_2} = O(c^{(8)})$ for $\mathbf{w} \in \mathbb{S}^{n-1}$.

Therefore, by invoking Lemma 11 with $K = \frac{C_1 c^{(8)} \eta\Delta\sqrt{r_{\mathbf{p},\mathbf{q}}}}{m}$ and $t = 2[\mathscr{H}(\overline{\mathcal{X}}, r)]^{1/2}$, we obtain that

$$
\sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \left| \frac{\eta\Delta}{m} \sum_{i=1}^{r_{\mathbf{p},\mathbf{q}}} \left[Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}) - \mathbb{E}\big(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w})\big) \right] \right| \le \frac{C_2 c^{(8)} \eta\Delta\sqrt{r_{\mathbf{p},\mathbf{q}}}}{m} \left[\omega(\mathcal{K}_{(r)}) + 2r\sqrt{\mathscr{H}(\overline{\mathcal{X}}, r)} \right]
$$

with probability at least $1 - 2\exp(-4\mathscr{H}(\overline{\mathcal{X}}, r))$. Due to

$$
T_3^{\mathbf{p},\mathbf{q}} | \{|\mathbf{R}_{\mathbf{p},\mathbf{q}}| = r_{\mathbf{p},\mathbf{q}}\} \sim \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \frac{\eta\Delta}{m} \sum_{i=1}^{r_{\mathbf{p},\mathbf{q}}} Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w})
$$

and (89), we reach the conditional concentration bound

$$
\left[T_3^{\mathbf{p},\mathbf{q}} \big| \{|\mathbf{R}_{\mathbf{p},\mathbf{q}}| = r_{\mathbf{p},\mathbf{q}}\} \right] \le \frac{C_2 c^{(8)} \eta\Delta\sqrt{r_{\mathbf{p},\mathbf{q}}}}{m} \left[\omega(\mathcal{K}_{(r)}) + 2r\sqrt{\mathscr{H}(\overline{\mathcal{X}}, r)} \right] + \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \frac{\eta\Delta r_{\mathbf{p},\mathbf{q}} \mathbb{E}(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}))}{m} \tag{90}
$$

that holds with probability at least $1 - 2\exp(-4\mathscr{H}(\overline{\mathcal{X}}, r))$.

Unconditional Bound

We further analyze the behaviour of $|\mathbf{R}_{\mathbf{p},\mathbf{q}}| \sim \mathrm{Bin}(m, \mathsf{P}_{\mathbf{p},\mathbf{q}})$ to derive the final claim. Using Lemma 7 (or Chernoff bound), for any $t > 0$ we obtain

$$
\mathbb{P}\Big(\big| |\mathbf{R}_{\mathbf{p},\mathbf{q}}| - m\,\mathsf{P}_{\mathbf{p},\mathbf{q}} \big| \le \sqrt{2m\,\mathsf{P}_{\mathbf{p},\mathbf{q}}\, t} + t \Big) \ge 1 - 2\exp(-t). \tag{91}
$$

Therefore, with probability at least $1 - 2\exp(-4\mathscr{H}(\overline{\mathcal{X}}, r))$ we have

$$
|\mathbf{R}_{\mathbf{p},\mathbf{q}}| \le 5\big(m\,\mathsf{P}_{\mathbf{p},\mathbf{q}} + \mathscr{H}(\overline{\mathcal{X}}, r)\big). \tag{92}
$$

Therefore, we can combine (90) and (92), then take a union bound over $(\mathbf{p},\mathbf{q}) \in \mathcal{N}_{r,\mu_4}^{(2)}$, to obtain

$$
\begin{aligned}
T_3^{\mathbf{p},\mathbf{q}} &\le C_3 c^{(8)} r\eta\Delta \sqrt{\Big[\mathsf{P}_{\mathbf{p},\mathbf{q}} + \frac{\mathscr{H}(\overline{\mathcal{X}}, r)}{m} \Big] \Big[\frac{r^{-2}\omega^2(\mathcal{K}_{(r)}) + \mathscr{H}(\overline{\mathcal{X}}, r)}{m} \Big]} \\
&\quad + 5\eta\Delta \left[1 + \frac{\mathscr{H}(\overline{\mathcal{X}}, r)}{m\,\mathsf{P}_{\mathbf{p},\mathbf{q}}} \right] \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \mathsf{P}_{\mathbf{p},\mathbf{q}}\, \mathbb{E}(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w})), \quad \forall (\mathbf{p},\mathbf{q}) \in \mathcal{N}_{r,\mu_4}^{(2)}
\end{aligned} \tag{93}
$$

with probability at least $1 - 4\exp(-2\mathscr{H}(\overline{\mathcal{X}}, r))$. Moreover, by the linearity of $Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w})$ on $\mathbf{w}$ and $Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}) \sim \big(J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}} | E_{\mathbf{p},\mathbf{q}}^{(i)}\big)$ and (38b), we have

$$\begin{aligned}
&\sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \mathsf{P}_{\mathbf{p},\mathbf{q}}\,\mathbb{E}(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w})) \le r \sup_{\mathbf{w}\in\mathbb{S}^{n-1}} \big|\,\mathsf{P}_{\mathbf{p},\mathbf{q}}\,\mathbb{E}(Z_i^{\mathbf{p},\mathbf{q}}(\mathbf{w}))\big| \\
&= r \sup_{\mathbf{w}\in\mathbb{S}^{n-1}} \Big|\mathbb{P}(E_{\mathbf{p},\mathbf{q}}^{(i)})\mathbb{E}\big(J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}} | E_{\mathbf{p},\mathbf{q}}^{(i)}\big)\Big| = r \sup_{\mathbf{w}\in\mathbb{S}^{n-1}} \Big|\mathbb{E}\Big(J_{i,\mathbf{w}}^{\mathbf{p},\mathbf{q}} \mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big)\Big| \le \frac{r\varepsilon^{(4)}\|\mathbf{p}-\mathbf{q}\|_2}{\Delta \vee \Lambda}.
\end{aligned}$$

Substituting this into (93) yields the claim. □

E.5 The Proof of Proposition 5 (Bounding the First Term in (55))

Proof. By triangle inequality we decompose the first term in (55) into

$$\underbrace{|\eta\zeta| \sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}} \Big|\mathbf{w}^\top \Big(\mathbf{I}_n - \frac{1}{m}\sum_{i=1}^m \mathbf{a}_i\mathbf{a}_i^\top\Big)(\mathbf{u}-\mathbf{v})\Big|}_{:=\hat{T}_1} + r_1|\eta\zeta - 1|\|\mathbf{u}-\mathbf{v}\|_2.$$

To further control $\hat{T}_1$ above, we shall discuss "large-distance regime" and "small-distance regime".

Large-distance regime ($\|\mathbf{u}-\mathbf{v}\|_2 \ge r_1$): Uniformly for all $\mathbf{u}, \mathbf{v} \in \overline{\mathcal{X}} \subset 2\mathcal{K}$ obeying $\|\mathbf{u}-\mathbf{v}\|_2 \ge r_1$ we have

$$\frac{\mathbf{u}-\mathbf{v}}{\|\mathbf{u}-\mathbf{v}\|_2} \in \frac{2(\mathcal{K}-\mathcal{K})}{r_1} \cap \mathbb{B}_2^n \subset \frac{2\mathcal{K}_{(r_1)}}{r_1}.$$

Thus, we obtain

$$\hat{T}_1 \le \frac{2|\eta\zeta|\|\mathbf{u}-\mathbf{v}\|_2}{r_1} \sup_{\mathbf{w},\mathbf{s}\in\mathcal{K}_{(r_1)}} \Big|\frac{1}{m}\sum_{i=1}^m \mathbf{w}^\top\mathbf{a}_i\mathbf{a}_i^\top\mathbf{s} - \mathbf{w}^\top\mathbf{s}\Big|.$$

Small-distance regime ($\|\mathbf{u}-\mathbf{v}\|_2 < r_1$): Uniformly for all $\mathbf{u}, \mathbf{v}$ obeying $\|\mathbf{u}-\mathbf{v}\|_2 < r_1$, we can directly take the supremum over $\mathbf{u}-\mathbf{v} \in (2\mathcal{K})_{(r_1)} \subset 2\mathcal{K}_{(r_1)}$ and obtain

$$\hat{T}_1 \le 2|\eta\zeta| \sup_{\mathbf{w},\mathbf{s}\in\mathcal{K}_{(r_1)}} \Big|\frac{1}{m}\sum_{i=1}^m \mathbf{w}^\top\mathbf{a}_i\mathbf{a}_i^\top\mathbf{s} - \mathbf{w}^\top\mathbf{s}\Big|.$$

By Lemma 12, if $m \ge \frac{\omega^2(\mathcal{K}_{(r_1)})}{r_1^2}$, then for some absolute constants C_1, c_2, we have

$$\sup_{\mathbf{w},\mathbf{s}\in\mathcal{K}_{(r_1)}} \Big|\frac{1}{m}\sum_{i=1}^m \mathbf{w}^\top\mathbf{a}_i\mathbf{a}_i^\top\mathbf{s} - \mathbf{w}^\top\mathbf{s}\Big| \le \frac{C_1 r_1 \omega(\mathcal{K}_{(r_1)})}{\sqrt{m}} \tag{94}$$

with probability at least $1 - \exp(-c_2\omega^2(\mathcal{K}_{(r_1)})/r_1^2)$. Therefore, combining the two regimes and the bound (94) yield the claim. □

F The Proofs in Section 4 (Validating Assumptions)

F.1 The Proof of Fact 2 (Assumptions 6–7 for 1bCS)

Proof. Continuing from the proof of Fact 1, we have that $\mathbf{a}_i^\top\boldsymbol{\beta}_1 := a_1$ and $\mathbf{a}_i^\top\boldsymbol{\beta}_2 := a_2$ are independent $\mathcal{N}(0,1)$ variables.

(i) Validating Assumption 6

Recall that we can restrict our attention to $u_1 \leq \frac{1}{2}$ and $u_2 \geq \frac{1}{2}$ in the validation of (37a). For any $p \geq 2$ we have

$$\begin{aligned}
&\mathbb{E}\Big(|\mathbf{a}_i^\top\boldsymbol{\beta}_2|^p\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\Big)\\
&= \mathbb{E}\Big(|a_2|^p\mathbb{1}(u_1|a_1|>u_2|a_2|)\Big)\\
&\leq \mathbb{E}\Big(|a_2|^{p-1}\cdot 2u_1|a_1|\Big)\\
&= \|\mathbf{p}-\mathbf{q}\|_2\sqrt{\mathbb{E}[|a_2|^{2(p-1)}]}\\
&\leq \mathsf{P}_{\mathbf{p},\mathbf{q}}(C_1\sqrt{p})^p \leq C_2\,\mathsf{P}_{\mathbf{p},\mathbf{q}}\cdot\frac{p!}{2},
\end{aligned}$$

where in the first inequality we use the event $|a_2| < \frac{u_1|a_1|}{u_2} \leq 2u_1|a_1| = \|\mathbf{p}-\mathbf{q}\|_2|a_1|$. Therefore, (37a) holds with some absolute constant $c^{(7)} = C_2$.

In addition, (37b) holds with $\varepsilon^{(3)} = 0$ because

$$\mathbb{E}\Big(\operatorname{sign}(\mathbf{a}_i^\top\boldsymbol{\beta}_1)\mathbf{a}_i^\top\boldsymbol{\beta}_2\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\Big) = \mathbb{E}\Big(\operatorname{sign}(a_1)a_2\mathbb{1}(u_1|a_1|>u_2|a_2|)\Big) = 0,$$

where we use the randomness of $\operatorname{sign}(a_1)$.

(ii) Validating Assumption 7

For any $\mathbf{w}\in\mathbb{R}^n$ we have defined $J^{\mathbf{p},\mathbf{q}}_{i,\mathbf{w}} = \operatorname{sign}(\mathbf{a}_i^\top\boldsymbol{\beta}_1)[\mathbf{a}_i - (\mathbf{a}_i^\top\boldsymbol{\beta}_1)^\top\boldsymbol{\beta}_1 - (\mathbf{a}_i^\top\boldsymbol{\beta}_2)^\top\boldsymbol{\beta}_2]^\top\mathbf{w}$, and we further define

$$a_{\mathbf{w}} := [\mathbf{a}_i - (\mathbf{a}_i^\top\boldsymbol{\beta}_1)^\top\boldsymbol{\beta}_1 - (\mathbf{a}_i^\top\boldsymbol{\beta}_2)^\top\boldsymbol{\beta}_2]^\top\mathbf{w} = \mathbf{a}_i^\top(\mathbf{I}_n - \boldsymbol{\beta}_1\boldsymbol{\beta}_1^\top - \boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top)\mathbf{w}.$$

Because $(\mathbf{I}_n - \boldsymbol{\beta}_1\boldsymbol{\beta}_1^\top - \boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top)\mathbf{w}$ is orthogonal to $\boldsymbol{\beta}_1,\boldsymbol{\beta}_2$, the rotational invariance of $\mathbf{a}_i\sim\mathcal{N}(0,\mathbf{I}_n)$ implies that $a_1, a_2, a_{\mathbf{w}}$ are independent. In view of $E^{(i)}_{\mathbf{p},\mathbf{q}} = \{u_1|a_1|>u_2|a_2|\}$, $a_{\mathbf{w}}$ and $E^{(i)}_{\mathbf{p},\mathbf{q}}$ are independent. Note that $(\mathbf{I}_n - \boldsymbol{\beta}_1\boldsymbol{\beta}_1^\top - \boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top)\mathbf{w}\in\mathbb{B}_2^n$ when $\mathbf{w}\in\mathbb{S}^{n-1}$. Therefore, for $p\geq 2$ we have

$$\mathbb{E}\Big(|J^{\mathbf{p},\mathbf{q}}_{i,\mathbf{w}}|^p\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\Big) = \mathbb{E}\big(|a_{\mathbf{w}}|^p\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\big) = \mathsf{P}_{\mathbf{p},\mathbf{q}}\,\mathbb{E}(|a_{\mathbf{w}}|^p) \leq \mathsf{P}_{\mathbf{p},\mathbf{q}}(C_3\sqrt{p})^p,$$

yielding (38a) with some absolute constant $c^{(8)}$.

Moreover, (38b) holds with $\varepsilon^{(4)} = 0$ because we can use the randomness of $\operatorname{sign}(a_1)$ to obtain

$$\mathbb{E}\Big(J^{\mathbf{p},\mathbf{q}}_{i,\mathbf{w}}\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\Big) = \mathbb{E}\big(\operatorname{sign}(a_1)a_{\mathbf{w}}\mathbb{1}(u_1|a_1|>u_2|a_2|)\big) = 0,$$

as desired. $\square$

F.2 The Proof of Fact 3 (Assumptions 5–7 for D1bCS)

Proof. Recall that any two different points $\mathbf{p},\mathbf{q}$ in $\mathbb{B}_2^n$ are parameterized by some orthonormal $(\boldsymbol{\beta}_1,\boldsymbol{\beta}_2)$, and we have derived a two-sided bound in (64) that implies

$$-\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{2\lambda}\cdot\big[\mathbb{1}(|\mathbf{a}_i^\top\mathbf{p}|>\lambda)+\mathbb{1}(|\mathbf{a}_i^\top\mathbf{q}|>\lambda)\big]\le\mathbb{E}_{\tau_i}\big[\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\big]-\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{2\lambda}\le 0. \tag{95}$$

We introduce the notation

$$a_1:=\mathbf{a}_i^\top\boldsymbol{\beta}_1,\ a_2:=\mathbf{a}_i^\top\boldsymbol{\beta}_2,\ a_{\mathbf{w}}:=[\mathbf{a}_i-(\mathbf{a}_i^\top\boldsymbol{\beta}_1)\boldsymbol{\beta}_1-(\mathbf{a}_i^\top\boldsymbol{\beta}_2)\boldsymbol{\beta}_2]^\top\mathbf{w}=\mathbf{a}_i^\top(\mathbf{I}_n-\boldsymbol{\beta}_1\boldsymbol{\beta}_1^\top-\boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top)\mathbf{w}$$

with $\mathbf{w}\in\mathbb{S}^{n-1}$. Then by $\mathbb{E}(\mathbf{a}_i\mathbf{a}_i^\top)=\mathbf{I}_n$ it is easy to see $\mathbb{E}(a_1^2)=1$ and $\mathbb{E}(a_1a_2)=\mathbb{E}(a_1a_{\mathbf{w}})=0$. In this proof, we will first validate the moment bounds (36a), (37a) and (38a), and then show (36b), (37b) and (38b) via calculations.

(i) Validating (36a), (37a) and (38a)

We shall give a unified treatment. Let $A=a_1$, a_2 or $a_{\mathbf{w}}$, and note that when $\mathbf{w}\in\mathbb{S}^{n-1}$, for any choice of A and any $p\ge 2$, we have $\mathbb{E}|A|^p\le(\bar{c}_1\sqrt{p})^p$ as per (12b). By using the randomness of τ_i and the upper bound in (95), we obtain

$$\begin{aligned}&\mathbb{E}\Big(|A|^p\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\Big)\\&=\mathbb{E}\Big(|A|^p\mathbb{E}_{\tau_i}\big[E_{\mathbf{p},\mathbf{q}}^{(i)}\big]\Big)\\&\le\frac{\|\mathbf{p}-\mathbf{q}\|_2}{2\lambda}\mathbb{E}\Big(|A|^p|\mathbf{a}_i^\top\boldsymbol{\beta}_1|\Big)\\&\le\frac{\|\mathbf{p}-\mathbf{q}\|_2}{2\lambda}\sqrt{\mathbb{E}[|A|^{2p}]}\\&\le C_1\,\mathsf{P}_{\mathbf{p},\mathbf{q}}(2\bar{c}_1\sqrt{p})^p\le C_2\,\mathsf{P}_{\mathbf{p},\mathbf{q}}\cdot\frac{p!}{2}\end{aligned}$$

for some absolute constants C_1 and C_2, where we use Cauchy-Schwarz inequality and the verified $\mathsf{P}_{\mathbf{p},\mathbf{q}}\asymp\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\lambda}$. Therefore, (36a), (37a) and (38a) hold with absolute constants $c^{(5)}$, $c^{(7)}$ and $c^{(8)}$.

(ii) Validating (36b), (37b) and (38b)

Again we give a unified treatment and let A be one of the followings: a_1, a_2 or $a_{\mathbf{w}}$ with $\mathbf{w}\in\mathbb{S}^{n-1}$. Then, we shall calculate

$$\mathbb{E}\big(\operatorname{sign}(a_1)A\cdot\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\big)=\mathbb{E}\big(\operatorname{sign}(a_1)A\cdot\mathbb{E}_{\tau_i}\big[\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\big]\big)$$

for all choices of A. Our two-sided estimate (95) yields

$$\left|\mathbb{E}\big(\operatorname{sign}(a_1)A\cdot\mathbb{E}_{\tau_i}\big[\mathbb{1}(E_{\mathbf{p},\mathbf{q}}^{(i)})\big]\big)-\mathbb{E}\Big(\operatorname{sign}(a_1)A\cdot\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{2\lambda}\Big)\right|$$

$$\begin{aligned}
&\le \mathbb{E}\left[|A|\cdot\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{2\lambda}\Big(\mathbb{1}(|\mathbf{a}_i^\top\mathbf{p}|>\lambda)+\mathbb{1}(|\mathbf{a}_i^\top\mathbf{q}|>\lambda)\Big)\right]\\
&\le \frac{\|\mathbf{p}-\mathbf{q}\|_2}{\lambda}\sup_{\mathbf{u}\in\mathbb{B}_2^n}\mathbb{E}\left[|A|\cdot|\mathbf{a}_i^\top\boldsymbol{\beta}_1|\cdot\mathbb{1}(|\mathbf{a}_i^\top\mathbf{u}|>\lambda)\right]\\
&\le \frac{\|\mathbf{p}-\mathbf{q}\|_2}{\lambda}\Big[\mathbb{E}|A|^4\Big]^{1/4}\Big[\mathbb{E}|\mathbf{a}_i^\top\boldsymbol{\beta}_1|^4\Big]^{1/4}\sup_{\mathbf{u}\in\mathbb{B}_2^n}\sqrt{\mathbb{P}(|\mathbf{a}_i^\top\mathbf{u}|>\lambda)}\\
&\le \frac{\|\mathbf{p}-\mathbf{q}\|_2}{\lambda}\cdot(2\bar{c}_1)^2\cdot\sqrt{2\exp(-2\bar{c}_0\lambda^2)}\\
&\le 8(\bar{c}_1)^2\exp(-\bar{c}_0\lambda^2)\cdot\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\lambda},
\end{aligned}$$

where in the last line we use (12). Therefore, to establish (36b), (37b) and (38b) with $\varepsilon^{(2)}=\varepsilon^{(3)}=\varepsilon^{(4)}=8(\bar{c}_1)^2\exp(-\bar{c}_0\lambda^2)$, all that remains is to evaluate $\mathbb{E}(\mathrm{sign}(a_1)A\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{2\lambda})$:

$$\begin{aligned}
&\mathbb{E}\Big(\mathrm{sign}(a_1)A\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{2\lambda}\Big)\\
&=\frac{\|\mathbf{p}-\mathbf{q}\|_2}{2\lambda}\mathbb{E}\big(\mathrm{sign}(a_1)A|\mathbf{a}_i^\top\boldsymbol{\beta}_1|\big)\\
&=\frac{\|\mathbf{p}-\mathbf{q}\|_2}{2\lambda}\mathbb{E}(a_1A)=\begin{cases}\frac{\|\mathbf{p}-\mathbf{q}\|_2}{2\lambda}, & \text{if } A=a_1\\ 0\ , & \text{if } A=a_2,\ a_{\mathbf{w}}\end{cases}.
\end{aligned}$$

We come to the desired claim. $\square$

F.3 The Proof of Fact 4 (Assumptions 3, 8 for DMbCS)

Proof. Note that $\mathsf{P}_{\mathbf{u},\mathbf{v}}=\mathbb{P}(E_{\mathbf{u},\mathbf{v}}^{(i)})=\mathbb{E}\big(\mathbb{1}(E_{\mathbf{u},\mathbf{v}}^{(i)})\big)=\mathbb{E}_{\mathbf{a}_i}\big[\mathbb{E}_{\tau_i}\big(\mathbb{1}(E_{\mathbf{u},\mathbf{v}}^{(i)})\big)\big]$.

Expecting τ_i: By the definition of $\mathcal{Q}_{\delta,L}$ in (9), we have $\mathcal{Q}_{\delta,L}(a)=\mathcal{Q}_\delta(a)$ for $|a|<\frac{L\delta}{2}$, which together with $|\tau_i|\le\frac{\delta}{2}$ yields

$$\mathbb{1}(E_{\mathbf{u},\mathbf{v}}^{(i)})\le\mathbb{1}\big(\mathcal{Q}_\delta(\mathbf{a}_i^\top\mathbf{u}-\tau_i)\neq\mathcal{Q}_\delta(\mathbf{a}_i^\top\mathbf{v}-\tau_i)\big), \tag{96a}$$

$$\mathbb{1}(E_{\mathbf{u},\mathbf{v}}^{(i)})\ge\mathbb{1}\big(\mathcal{Q}_\delta(\mathbf{a}_i^\top\mathbf{u}-\tau_i)\neq\mathcal{Q}_\delta(\mathbf{a}_i^\top\mathbf{v}-\tau_i)\big)\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{u}|\vee|\mathbf{a}_i^\top\mathbf{v}|\le\frac{(L-1)\delta}{2}\Big) \tag{96b}$$

Therefore, we can utilize the randomness of $\tau_i\sim\mathscr{U}([-\frac{\delta}{2},\frac{\delta}{2}])$ to obtain from (96a)

$$\begin{aligned}
&\mathbb{E}_{\tau_i}\big(E_{\mathbf{u},\mathbf{v}}^{(i)}\big)\le\mathbb{E}_{\tau_i}\Big(\mathbb{1}\big(\mathcal{Q}_\delta(\mathbf{a}_i^\top\mathbf{u}-\tau_i)\neq\mathcal{Q}_\delta(\mathbf{a}_i^\top\mathbf{v}-\tau_i)\big)\Big)\\
&=\mathbb{E}_{\tau_i}\left[\mathbb{1}\big(|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|\ge\delta\big)+\mathbb{1}\big(|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|<\delta\big)\mathbb{1}\left(\tau_i\in\bigcup_{j\in\mathbb{Z}}\big(\mathbf{a}_i^\top\mathbf{u}\wedge\mathbf{a}_i^\top\mathbf{v}-j\delta,\mathbf{a}_i^\top\mathbf{u}\vee\mathbf{a}_i^\top\mathbf{v}-j\delta\big)\right)\right]\\
&=\mathbb{1}\big(|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|\ge\delta\big)+\mathbb{1}\big(|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|<\delta\big)\frac{|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|}{\delta}\le\frac{|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|}{\delta}
\end{aligned} \tag{97}$$

and similarly obtain from (96b)

$$\begin{aligned}
&\mathbb{E}_{\tau_i}\big(E^{(i)}_{\mathbf{u},\mathbf{v}}\big) \geq \mathbb{E}_{\tau_i}\Big(\mathbb{1}\big(\mathcal{Q}_\delta(\mathbf{a}_i^\top\mathbf{u}-\tau_i)\neq\mathcal{Q}_\delta(\mathbf{a}_i^\top\mathbf{v}-\tau_i)\big)\Big)\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{u}|\vee|\mathbf{a}_i^\top\mathbf{v}|\leq\frac{(L-1)\delta}{2}\Big)\\
&=\Big[\mathbb{1}\big(|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|\geq\delta\big)+\mathbb{1}\big(|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|<\delta\big)\frac{|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|}{\delta}\Big]\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{u}|\vee|\mathbf{a}_i^\top\mathbf{v}|\leq\frac{(L-1)\delta}{2}\Big) \qquad (98)\\
&\geq\frac{|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|}{\delta}\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{u}|\vee|\mathbf{a}_i^\top\mathbf{v}|\leq\frac{(L-1)\delta}{2},\quad|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|<\delta\Big) \qquad (99)
\end{aligned}$$

Expecting $\mathbf{a}_i$: We further take expectation regarding τ_i. By (97) we have

$$\mathsf{P}_{\mathbf{u},\mathbf{v}}\leq\frac{\mathbb{E}|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|}{\delta}\leq\frac{\|\mathbf{u}-\mathbf{v}\|_2}{\delta}.$$

Hence, Assumption 8 holds with $c^{(9)}=1$.

By continuing from (98) and some simple algebra, with $\mathbf{n}:=\frac{\mathbf{u}-\mathbf{v}}{\|\mathbf{u}-\mathbf{v}\|_2}$ we can lower bound $\mathsf{P}_{\mathbf{u},\mathbf{v}}$ as

$$\begin{aligned}
&\mathsf{P}_{\mathbf{u},\mathbf{v}}\geq\mathbb{E}\Big[\Big[\mathbb{1}\big(|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|\geq\delta\big)+\mathbb{1}\big(|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|<\delta\big)\frac{|\mathbf{a}_i^\top(\mathbf{u}-\mathbf{v})|}{\delta}\Big]\cdot\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{u}|\vee|\mathbf{a}_i^\top\mathbf{v}|\leq\frac{(L-1)\delta}{2}\Big)\Big]\\
&=\mathbb{E}\Big[\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{n}|\geq\frac{\delta}{\|\mathbf{u}-\mathbf{v}\|_2}\Big)+\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{n}|<\frac{\delta}{\|\mathbf{u}-\mathbf{v}\|_2}\Big)\frac{|\mathbf{a}_i^\top\mathbf{n}|\|\mathbf{u}-\mathbf{v}\|_2}{\delta}\Big] \qquad (100a)\\
&-\mathbb{E}\Big[\Big[\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{n}|\geq\frac{\delta}{\|\mathbf{u}-\mathbf{v}\|_2}\Big)+\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{n}|<\frac{\delta}{\|\mathbf{u}-\mathbf{v}\|_2}\Big)\frac{|\mathbf{a}_i^\top\mathbf{n}|\|\mathbf{u}-\mathbf{v}\|_2}{\delta}\Big]\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{u}|\vee|\mathbf{a}_i^\top\mathbf{v}|>\frac{(L-1)\delta}{2}\Big)\Big]. \qquad (100b)
\end{aligned}$$

It is not hard to see the following lower bound on (100a):

$$\text{The term in (100a)}\geq\min\Big\{\frac{\|\mathbf{u}-\mathbf{v}\|_2}{\delta},1\Big\}\cdot\mathbb{E}\Big[\mathbb{1}(|\mathbf{a}_i^\top\mathbf{n}|\geq1)+|\mathbf{a}_i^\top\mathbf{n}|\mathbb{1}(|\mathbf{a}_i^\top\mathbf{n}|<1)\Big].$$

By (13b), for some absolute constant L_0 we have

$$\mathbb{E}\big[\mathbb{1}(|\mathbf{a}_i^\top\mathbf{n}|\geq1)+|\mathbf{a}_i^\top\mathbf{n}|\mathbb{1}(|\mathbf{a}_i^\top\mathbf{n}|<1)\big]\geq\frac{1}{2}\mathbb{P}\Big(|\mathbf{a}_i^\top\mathbf{n}|\geq\frac{1}{2}\Big)\geq L_0,$$

hence we obtain the lower bound

$$\text{The term in (100a)}\geq L_0\min\Big\{\frac{\|\mathbf{u}-\mathbf{v}\|_2}{\delta},1\Big\}.$$

Next, we invoke Cauchy–Schwarz inequality to upper bound the term in (100b):

The term in (100b)

$$\begin{aligned}
&\leq\sqrt{\mathbb{E}\Big[\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{n}|\geq\frac{\delta}{\|\mathbf{u}-\mathbf{v}\|_2}\Big)+\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{n}|<\frac{\delta}{\|\mathbf{u}-\mathbf{v}\|_2}\Big)\frac{|\mathbf{a}_i^\top\mathbf{n}|^2\|\mathbf{u}-\mathbf{v}\|_2^2}{\delta^2}\Big]\cdot\mathbb{P}\Big(|\mathbf{a}_i^\top\mathbf{u}|\vee|\mathbf{a}_i^\top\mathbf{v}|>\frac{(L-1)\delta}{2}\Big)}\\
&\leq2\min\Big\{\frac{\|\mathbf{u}-\mathbf{v}\|_2}{\delta},1\Big\}\exp\Big(-\frac{\bar{c}_0L^2\delta^2}{16}\Big),
\end{aligned}$$

where in the last inequality we use (12a) and $\frac{L-1}{2}\geq\frac{L}{4}$; they imply

$$\sqrt{\mathbb{P}\Big(|\mathbf{a}_i^\top\mathbf{u}|\vee|\mathbf{a}_i^\top\mathbf{v}|>\frac{(L-1)\delta}{2}\Big)}\leq 2\exp\Big(-\frac{\bar{c}_0L^2\delta^2}{16}\Big).$$

Therefore, when $L\delta$ is sufficiently large such that $2\exp(-\frac{\bar{c}_0L^2\delta^2}{16})<\frac{L_0}{2}$, we arrive at

$$\mathsf{P}_{\mathbf{u},\mathbf{v}}\geq\frac{L_0}{2}\min\Big\{\frac{\|\mathbf{u}-\mathbf{v}\|_2}{\delta},1\Big\},$$

thus establishing Assumption 3 with $c^{(2)}=\frac{L_0}{2}$. The proof is complete. $\square$

F.4 The Proof of Fact 5 (Assumption 4 for DMbCS)

We shall first prove Lemma 1 to justify the last inequality in (63).

The proof of Lemma 1. By the definitions of $\mathcal{Q}_\delta$ and $\mathcal{Q}_{\delta,L}$ in (7) and (9), we have $|\mathcal{Q}_{\delta,L}(a)-\mathcal{Q}_{\delta,L}(b)|\geq\delta$ when $\mathcal{Q}_{\delta,L}(a)\neq\mathcal{Q}_{\delta,L}(b)$, and we always have

$$|\mathcal{Q}_{\delta,L}(a)-\mathcal{Q}_{\delta,L}(b)|\leq|\mathcal{Q}_\delta(a)-\mathcal{Q}_\delta(b)|\quad\text{and}\quad\mathbb{1}(\mathcal{Q}_{\delta,L}(a)\neq\mathcal{Q}_{\delta,L}(b))\leq\mathbb{1}(\mathcal{Q}_\delta(a)\neq\mathcal{Q}_\delta(b)).$$

Taken collectively, we arrive at

$$\big||\mathcal{Q}_{\delta,L}(a)-\mathcal{Q}_{\delta,L}(b)|-\delta\big|\cdot\mathbb{1}(\mathcal{Q}_{\delta,L}(a)\neq\mathcal{Q}_{\delta,L}(b))\leq\big(|\mathcal{Q}_\delta(a)-\mathcal{Q}_\delta(b)|-\delta\big)\cdot\mathbb{1}(\mathcal{Q}_\delta(a)\neq\mathcal{Q}_\delta(b)).$$

Further, by $|\mathcal{Q}_\delta(a)-a|\leq\frac{\delta}{2}$ $(\forall a\in\mathbb{R})$ we always have

$$|\mathcal{Q}_\delta(a)-\mathcal{Q}_\delta(b)|\leq|a-b|+\delta,\quad\forall\ a,b\in\mathbb{R}.\tag{101}$$

This leads to $|\mathcal{Q}_\delta(a)-\mathcal{Q}_\delta(b)|-\delta\leq|a-b|$. Also note that when $\mathcal{Q}_\delta(a)\neq\mathcal{Q}_\delta(b)$, we have $|\mathcal{Q}_\delta(a)-\mathcal{Q}_\delta(b)|\in\{\delta,2\delta,3\delta,\cdots\}$ and hence $|\mathcal{Q}_\delta(a)-\mathcal{Q}_\delta(b)|-\delta\geq 0$; and note that if $|\mathcal{Q}_\delta(a)-\mathcal{Q}_\delta(b)|-\delta>0$, we must have $|\mathcal{Q}_\delta(a)-\mathcal{Q}_\delta(b)|\geq 2\delta$, which combined with (101) implies $|a-b|\geq\delta$. Combining all these observations leads to

$$\big(|\mathcal{Q}_\delta(a)-\mathcal{Q}_\delta(b)|-\delta\big)\cdot\mathbb{1}(\mathcal{Q}_\delta(a)\neq\mathcal{Q}_\delta(b))\leq|a-b|\cdot\mathbb{1}(|a-b|\geq\delta),$$

as desired. $\square$

The Proof of Fact 5. Let us first show (31b).

Show (31b): Note that $\|\mathbf{p}-\mathbf{q}\|_2\leq r$ implies $\mathbf{p}-\mathbf{q}\in\mathcal{K}_{(r)}$, we continue from (63) to obtain

$$\begin{aligned}\eta\cdot\|\mathbf{h}(\mathbf{p},\mathbf{q})-\hat{\mathbf{h}}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}_{(r)}^\circ}&\leq\eta\sup_{\mathbf{w}\in\mathcal{K}_{(r)}}\sup_{\mathbf{v}\in\mathcal{K}_{(r)}}\frac{1}{m}\sum_{i=1}^m|\mathbf{a}_i^\top\mathbf{w}||\mathbf{a}_i^\top\mathbf{v}|\\&\leq\eta\sup_{\mathbf{w}\in\mathcal{K}_{(r)}}\frac{1}{m}\sum_{i=1}^m|\mathbf{a}_i^\top\mathbf{w}|^2\leq C_6\eta\Big(\frac{\omega(\mathcal{K}_{(r)})}{\sqrt{m}}+r\Big)^2\leq 4C_6\eta r^2\end{aligned}$$

with probability at least $1-\exp(-c_7 m)$, where C_6, c_7 are absolute constants; this is achieved by using Lemma 8 with $\ell = m$ and the sample complexity $m \geq \frac{\delta\omega^2(\mathcal{K}_{(r)})}{r^3} \geq \frac{\omega^2(\mathcal{K}_{(r)})}{r}$. Therefore, (31b) holds for some absolute constant $c^{(4)}$.

Show (31a): Note that we can restrict our attention on $(\mathbf{p},\mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4}$ obeying $\|\mathbf{p}-\mathbf{q}\|_2 \geq r$, since we can simply use the bound in (31b) if $\|\mathbf{p}-\mathbf{q}\|_2 \leq r$. The core idea is to control the number of effective contributors to the last line in (63) and then invoke Lemma 8.

We use $\mathbb{1}(|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})| \geq \delta)$ to control the number of non-zero contributors. For fixed $(\mathbf{p},\mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4}$ and a specific $i \in [m]$, by $\|\mathbf{p}-\mathbf{q}\|_2 \leq 2\mu_4 \leq 2c_1\delta$ and (12a) where $\bar{c}_0$ is an absolute constant, we have

$$\mathbb{P}\big(|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})| \geq \delta\big) \leq \mathbb{P}\Big(\Big|\frac{\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})}{\|\mathbf{p}-\mathbf{q}\|_2}\Big| \geq \frac{1}{2c_1}\Big) \leq 2\exp\Big(-\frac{\bar{c}_0}{2c_1^2}\Big) := p_0.$$

We assume $p_0 \in (0,1/2)$, which can be ensured by setting c_1 sufficiently small. By Chernoff bound and letting $c_2 = \frac{\bar{c}_0}{2}$, we obtain

$$\begin{aligned}
&\mathbb{P}\Big(\big|\{i \in [m] : |\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})| \geq \delta\}\big| \geq \frac{3mp_0}{2}\Big) \\
&\leq \mathbb{P}\Big(\mathrm{Bin}(m,p_0) \geq \frac{3mp_0}{2}\Big) \leq \exp\Big(-\frac{mp_0}{12}\Big) = \exp\Big(-\frac{m\exp(-c_2/c_1^2)}{6}\Big).
\end{aligned}$$

Because of the sample complexity $m \gtrsim \exp(\frac{c_2}{c_1^2})\mathscr{H}(\mathcal{X},r)$ and $|\mathcal{N}^{(2)}_{r,\mu_4}| \leq |\mathcal{N}(\mathcal{X},r)|^2$, we can take a union bound over $(\mathbf{p},\mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4}$ and obtain that the event

$$\big|\{i \in [m] : |\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})| \geq \delta\}\big| < \frac{3mp_0}{2}, \quad \forall (\mathbf{p},\mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4} \tag{102}$$

holds with the probability at least $1-\exp(-c_8\mathscr{H}(\mathcal{X},r))$.

Now we observe that any $(\mathbf{p},\mathbf{q}) \in \mathcal{N}^{(2)}_{r,\mu_4}$ obeying $\|\mathbf{p}-\mathbf{q}\|_2 \geq r$ satisfies $\frac{\mathbf{p}-\mathbf{q}}{\|\mathbf{p}-\mathbf{q}\|_2} \in \frac{\mathcal{K}-\mathcal{K}}{r} \cap \mathbb{B}_2^n = \frac{1}{r}\mathcal{K}_{(r)}$. On the event (102), we continue from (63) and take supremum over $\frac{r(\mathbf{p}-\mathbf{q})}{\|\mathbf{p}-\mathbf{q}\|_2} \in \mathcal{K}_{(r)}$ to reach

$$\begin{aligned}
&\eta \cdot \|\mathbf{h}(\mathbf{p},\mathbf{q}) - \hat{\mathbf{h}}(\mathbf{p},\mathbf{q})\|_{\mathcal{K}^\circ_{(r)}} \\
&\leq \frac{\eta\|\mathbf{p}-\mathbf{q}\|_2}{r} \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \sup_{\mathbf{v}\in\mathcal{K}_{(r)}} \max_{\substack{I\subset[m] \\ |I|\leq 3mp_0/2}} \frac{1}{m}\sum_{i\in I} |\mathbf{a}_i^\top\mathbf{w}||\mathbf{a}_i^\top\mathbf{v}| \qquad (103) \\
&\leq \frac{3\eta p_0\|\mathbf{p}-\mathbf{q}\|_2}{2r} \cdot \sup_{\mathbf{w}\in\mathcal{K}_{(r)}} \max_{\substack{I\subset[m] \\ |I|\leq 3mp_0/2}} \frac{1}{3mp_0/2}\sum_{i\in I} |\mathbf{a}_i^\top\mathbf{w}|^2 \\
&\leq C_9\eta r\|\mathbf{p}-\mathbf{q}\|_2 \cdot \Big(\frac{\omega^2(\mathcal{K}_{(r)})}{mr^2} + p_0\log(p_0^{-1})\Big), \qquad (104)
\end{aligned}$$

where the last step holds with probability at least $1-\exp(-c_{10}\mathscr{H}(\mathcal{X},r))$ for some absolute constants C_9, c_{10} due to Lemma 8 and $mp_0 = 2m\exp(-\frac{c_2}{c_1^2}) \gtrsim \mathscr{H}(\mathcal{X},r)$. By $\|\mathbf{p}-\mathbf{q}\|_2 \leq \mu_4 \leq 2c_1\delta$ and the sample complexity $m \geq \frac{\delta\omega^2(\mathcal{K}_{(r)})}{r^3}$, we have

$$\eta r\|\mathbf{p}-\mathbf{q}\|_2 \cdot \frac{\omega^2(\mathcal{K}_{(r)})}{mr^2} \lesssim \eta r\delta \cdot \frac{r}{\delta} = \eta r^2,$$

which can be accounted by $\frac{c^{(3)}\eta\Delta r}{\Delta\vee\Lambda}=c^{(3)}\eta r$ in (31a) for some absolute constant $c^{(3)}$. When c_1 is small enough, then $p_0=2\exp(-\frac{\bar{c}_0}{2c_1^2})=2\exp(-\frac{c_2}{c_1^2})$ is also small enough, and we have $p_0\log(p_0^{-1})\leq\frac{C_5}{2}\sqrt{p_0}\leq C_5\exp(-\frac{c_2}{2c_1^2})$ for some absolute constant C_5. Therefore, we have

$$\eta r\|\mathbf{p}-\mathbf{q}\|_2\cdot p_0\log(p_0^{-1})\leq C_5\exp\Big(-\frac{c_2}{2c_1^2}\Big)\eta r\|\mathbf{p}-\mathbf{q}\|_2$$

which can be accounted by $\frac{\varepsilon^{(1)}\phi\eta\Delta\|\mathbf{p}-\mathbf{q}\|_2}{\Delta\vee\Lambda}=\varepsilon^{(1)}\eta r\|\mathbf{p}-\mathbf{q}\|_2$ with $\varepsilon^{(1)}=C_5\exp(-\frac{c_2}{2c_1^2})$, as claimed. $\square$

F.5 The Proof of Fact 6 (Assumptions 5–7 for DMbCS)

Proof. We consider any $\mathbf{p},\mathbf{q}\in\mathbb{B}_2^n$ satisfying $\|\mathbf{p}-\mathbf{q}\|_2\leq 2\mu_4\leq 2c_1\delta$ for some small enough c_1. By Assumptions 3 and 8 already validated in Fact 4, we have $\mathsf{P}_{\mathbf{p},\mathbf{q}}\asymp\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\delta}$.

A useful technique is to first deal with the randomness of τ_i. By (97) and (99) we have the two sided bound

$$\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{\delta}\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{p}|\vee|\mathbf{a}_i^\top\mathbf{q}|\leq\frac{(L-1)\delta}{2},\ |\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|<\delta\Big)\leq\mathbb{E}_{\tau_i}(E^{(i)}_{\mathbf{p},\mathbf{q}})\leq\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{\delta}. \tag{105}$$

Recall that $\boldsymbol{\beta}_1=\frac{\mathbf{p}-\mathbf{q}}{\|\mathbf{p}-\mathbf{q}\|_2}$ and $\boldsymbol{\beta}_2$ are orthonormal. It is convenient to introduce the notation $\mathbf{a}_i^\top\boldsymbol{\beta}_1:=a_1$, $\mathbf{a}_i^\top\boldsymbol{\beta}_2:=a_2$ and $[\mathbf{a}_i-(\mathbf{a}_i^\top\boldsymbol{\beta}_1)\boldsymbol{\beta}_1-(\mathbf{a}_i^\top\boldsymbol{\beta}_2)\boldsymbol{\beta}_2]:=a_{\mathbf{w}}$ for $\mathbf{w}\in\mathbb{S}^{n-1}$. Moreover, we further use the generic notation A which can be a_1, a_2 or $a_{\mathbf{w}}$.

(i) Validating (36a), (37a) and (38a)

For any $p\geq 2$, by the upper bound $\mathbb{E}_{\tau_i}(E^{(i)}_{\mathbf{p},\mathbf{q}})\leq\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{\delta}=\frac{\|\mathbf{p}-\mathbf{q}\|_2|\mathbf{a}_i^\top\boldsymbol{\beta}_1|}{\delta}$ from (105), we have

$$\begin{aligned}&\mathbb{E}\big(|A|^p\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\big)\leq\mathbb{E}\Big(|A|^p\frac{\|\mathbf{p}-\mathbf{q}\|_2|\mathbf{a}_i^\top\boldsymbol{\beta}_1|}{\delta}\Big)\\&\leq\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\delta}\sqrt{\mathbb{E}[|A|^{2p}]\mathbb{E}|\mathbf{a}_i^\top\boldsymbol{\beta}_1|^2}\\&\leq C_1\,\mathsf{P}_{\mathbf{p},\mathbf{q}}(C_2\sqrt{p})^p\leq C_3\,\mathsf{P}_{\mathbf{p},\mathbf{q}}\,\frac{p!}{2}\end{aligned}$$

for some absolute constants C_i's. This establishes (36a), (37a) and (38a) with some absolute constants $c^{(5)}$, $c^{(7)}$ and $c^{(8)}$.

(ii) Validating (36b), (37b) and (38b)

This amount to evaluating $\mathbb{E}(\mathrm{sign}(a_1)A\cdot\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}}))$ for $A=a_1,\ a_2$ or $a_{\mathbf{w}}$. By (105), Cauchy–Schwarz inequality, the moment bound (12b) and the tail bound (12a), for all three choices of A we have

$$\begin{aligned}&\left|\mathbb{E}\big(\,\mathrm{sign}(a_1)A\cdot\mathbb{1}(E^{(i)}_{\mathbf{p},\mathbf{q}})\big)-\mathbb{E}\Big(\,\mathrm{sign}(a_1)A\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{\delta}\Big)\right|\\&\leq\mathbb{E}\left[\frac{|A||\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{\delta}\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{p}|\vee|\mathbf{a}_i^\top\mathbf{q}|>\frac{(L-1)\delta}{2}\Big)\right]+\mathbb{E}\left[\frac{|A||\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{\delta}\mathbb{1}\Big(|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|\geq\delta\Big)\right]\end{aligned}$$

$$= \frac{\|\mathbf{p}-\mathbf{q}\|_2}{\delta}\left(\mathbb{E}\left[|A||\mathbf{a}_i^\top\boldsymbol{\beta}_1|\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{p}|\vee|\mathbf{a}_i^\top\mathbf{q}|>\frac{(L-1)\delta}{2}\Big)\right]+\mathbb{E}\left[|A||\mathbf{a}_i^\top\boldsymbol{\beta}_1|\mathbb{1}\Big(|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|\geq\delta\Big)\right]\right)$$
$$\leq \frac{\|\mathbf{p}-\mathbf{q}\|_2}{\delta}\sqrt{\mathbb{E}\left[|A|^2|\mathbf{a}_i^\top\boldsymbol{\beta}_1|^2\right]}\left(\sqrt{\mathbb{P}\Big(|\mathbf{a}_i^\top\mathbf{p}|\vee|\mathbf{a}_i^\top\mathbf{q}|>\frac{L\delta}{4}\Big)}+\sqrt{\mathbb{P}\Big(|\mathbf{a}_i^\top\boldsymbol{\beta}_1|\geq\frac{1}{2c_1}\Big)}\right)$$
$$\leq \frac{\|\mathbf{p}-\mathbf{q}\|_2}{\delta}\cdot 8(\bar{c}_1)^2\Big[\exp\Big(-\frac{\bar{c}_0L^2\delta^2}{16}\Big)+\exp\Big(-\frac{\bar{c}_0}{4c_1^2}\Big)\Big]=\frac{\varepsilon'\|\mathbf{p}-\mathbf{q}\|_2}{\delta}.$$

All that remains is to compute $\mathbb{E}(\mathrm{sign}(a_1)A|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|/\delta)$:

$$\mathbb{E}\Big(\mathrm{sign}(a_1)A\frac{|\mathbf{a}_i^\top(\mathbf{p}-\mathbf{q})|}{\delta}\Big)=\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\delta}\mathbb{E}\big(\mathrm{sign}(a_1)A|a_1|\big)=\begin{cases}\frac{\|\mathbf{p}-\mathbf{q}\|_2}{\delta}, & \text{if } A=a_1\\ 0\ , & \text{if } A=a_1 \text{ or } a_{\mathbf{w}}\end{cases}.$$

The claim follows. □

F.6 The Proof of Proposition 6 (Bounding $\hat{T}$ for DMbCS)

We first restate [8, Corollary D.3] that will be used to bound $\hat{T}^{(1)}$ in (66).

Lemma 4 (Adapted from Corollary D.3 in [8])**.** *In our DMbCS setting, there exist absolute constants c_i's and C_i's, given $\rho\in(0,c_0\delta)$ for small enough c_0 and two sets $\mathcal{A},\mathcal{C}\subset\mathbb{B}_2^n$, if*

$$m\geq C_1\left[\frac{\delta^2(\mathscr{H}(\mathcal{A},\rho)+\omega^2(\mathcal{C}))}{\rho^2\log^2(\frac{\delta}{\rho})}+\frac{\delta\omega^2(\mathcal{A}_{(\rho)})}{\rho^3\log^{3/2}(\frac{\delta}{\rho})}\right],$$

then with probability at least $1-\exp(-c_2\mathscr{H}(\mathcal{A},\rho))$ we have

$$\sup_{\mathbf{a}\in\mathcal{A}}\sup_{\mathbf{c}\in\mathcal{C}}\Big|\frac{1}{m}\big\langle\mathcal{Q}_\delta(\mathbf{A}\mathbf{a}-\boldsymbol{\tau})-\mathbf{A}\mathbf{a},\mathbf{A}\mathbf{c}\big\rangle\Big|\leq C_3\rho\log\Big(\frac{\delta}{\rho}\Big).$$

Then we provide Lemma 5 that controls $\hat{T}^{(2)}$. We note that the fairly mild $\log^{1/2}(\frac{m}{\omega^2(\mathcal{X})})\lesssim L\delta\lesssim\omega(\mathcal{X})$ stems from certain technical treatment in our proof of Lemma 5.

Lemma 5. *In our setting of DMbCS, there exist some absolute constants c_i's and C_i's such that if $m\geq C_0\omega^2(\mathcal{X})$ and $C_1\log^{1/2}(\frac{m}{\omega^2(\mathcal{X})})\leq L\delta\leq C_2\omega(\mathcal{X})$ hold, then with probability at least $1-\exp(-\frac{c_3\omega^2(\mathcal{X})}{(L\delta)^2})$ we have*

$$\sup_{\mathbf{u}\in\overline{\mathcal{X}}}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\Big|\frac{1}{m}\langle\mathcal{Q}_\delta(\mathbf{A}\mathbf{u}-\boldsymbol{\tau})-\mathcal{Q}_{\delta,L}(\mathbf{A}\mathbf{u}-\boldsymbol{\tau}),\mathbf{A}\mathbf{w}\rangle\Big|\leq\frac{C_4\omega(\mathcal{X})\omega(\mathcal{K}_{(r_1)})}{m}. \tag{106}$$

Proof. By (7), (9) and $|\mathcal{Q}_\delta(a)|\leq|a|+\frac{\delta}{2}$, for any $a\in\mathbb{R}$ we have

$$|\mathcal{Q}_\delta(a)-\mathcal{Q}_{\delta,L}(a)|=\mathbb{1}\Big(|a|\geq\frac{L\delta}{2}\Big)\Big(|\mathcal{Q}_\delta(a)|-\frac{(L-1)\delta}{2}\Big)\leq|a|\mathbb{1}\Big(|a|\geq\frac{L\delta}{2}\Big).$$

Combining with $|\tau_i| \le \frac{\delta}{2}$ and $\frac{(L-1)\delta}{2} \ge \frac{L\delta}{4}$, it follows that

$$\begin{aligned}
&\sup_{\mathbf{u}\in\overline{\mathcal{X}}}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\left|\frac{1}{m}\sum_{i=1}^m\left[\mathcal{Q}_\delta(\mathbf{a}_i^\top\mathbf{u}-\tau_i)-\mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top\mathbf{u}-\tau_i)\right]\cdot\mathbf{a}_i^\top\mathbf{w}\right| \\
&\le \sup_{\mathbf{u}\in\overline{\mathcal{X}}}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\frac{1}{m}\sum_{i=1}^m\left|\mathcal{Q}_\delta(\mathbf{a}_i^\top\mathbf{u}-\tau_i)-\mathcal{Q}_{\delta,L}(\mathbf{a}_i^\top\mathbf{u}-\tau_i)\right||\mathbf{a}_i^\top\mathbf{w}| \\
&\le \sup_{\mathbf{u}\in\overline{\mathcal{X}}}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\frac{1}{m}\sum_{i=1}^m|\mathbf{a}_i^\top\mathbf{u}-\tau_i||\mathbf{a}_i^\top\mathbf{w}|\cdot\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{u}-\tau_i|\ge\frac{L\delta}{2}\Big) \\
&\le \sup_{\mathbf{u}\in\overline{\mathcal{X}}}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\frac{2}{m}\sum_{i=1}^m|\mathbf{a}_i^\top\mathbf{u}||\mathbf{a}_i^\top\mathbf{w}|\cdot\mathbb{1}\Big(|\mathbf{a}_i^\top\mathbf{u}|\ge\frac{L\delta}{4}\Big). \qquad (107)
\end{aligned}$$

Next, we bound the number of nonzero contributors to (107) by Lemma 8. Due to the factor $\mathbb{1}\big(|\mathbf{a}_i^\top\mathbf{u}|\ge\frac{L\delta}{4}\big)$, we wish to ensure

$$\left|\Big\{i\in[m]:|\mathbf{a}_i^\top\mathbf{u}|\ge\frac{L\delta}{4}\Big\}\right|\le r_0m,\quad \forall\mathbf{u}\in\overline{\mathcal{X}} \qquad (108)$$

for some $r_0\in(0,1)$ such that $r_0m\in[m]$, which is to be chosen later. To this end, we observe that it is sufficient to ensure

$$\sup_{\mathbf{u}\in\overline{\mathcal{X}}}\max_{\substack{I\subset[m]\\|I|\le r_0m}}\left(\frac{1}{r_0m}\sum_{i\in I}|\mathbf{a}_i^\top\mathbf{u}|^2\right)^{1/2}<\frac{L\delta}{8}. \qquad (109)$$

Further invoking Lemma 8 shows that, for some absolute constants c_1, c_2, with probability at least $1-2\exp(-c_1r_0m)$, we only need to ensure

$$\frac{\omega^2(\mathcal{X})}{r_0m}+\log\Big(\frac{e}{r_0}\Big)\le c_2(L\delta)^2. \qquad (110)$$

Hence, under the assumptions $m\gtrsim\omega^2(\mathcal{X})$ and $L\delta\gtrsim\log^{1/2}(\frac{m}{\omega^2(\mathcal{X})})$, we set $r_0:=\frac{C_3\omega^2(\mathcal{X})}{m(L\delta)^2}$ with sufficiently large C_3 to justify (110), which further leads to (108) with the promised probability.

On the event (108), we again apply Lemma 8 to obtain

$$\begin{aligned}
(107) &\le \sup_{\mathbf{u}\in\overline{\mathcal{X}}}\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\max_{\substack{I\subset[m]\\|I|\le r_0m}}\frac{2}{m}\sum_{i\in I}|\mathbf{a}_i^\top\mathbf{u}||\mathbf{a}_i^\top\mathbf{w}| \\
&\le 2r_0\left(\sup_{\mathbf{u}\in\overline{\mathcal{X}}}\max_{\substack{I\subset[m]\\|I|\le r_0m}}(r_0m)^{-1}\sum_{i\in I}|\mathbf{a}_i^\top\mathbf{u}|^2\right)^{1/2}\left(\sup_{\mathbf{w}\in\mathcal{K}_{(r_1)}}\max_{\substack{I\subset[m]\\|I|\le r_0m}}(r_0m)^{-1}\sum_{i\in I}|\mathbf{a}_i^\top\mathbf{w}|^2\right)^{1/2} \\
&\le C_4r_1\left(\frac{\omega(\mathcal{X})}{\sqrt{m}}+\sqrt{r_0\log(r_0^{-1})}\right)\left(\frac{\omega(\mathcal{K}_{(r_1)})/r_1}{\sqrt{m}}+\sqrt{r_0\log(r_0^{-1})}\right) \\
&\le \frac{C_5\omega(\mathcal{X})\omega(\mathcal{K}_{(r_1)})}{\sqrt{m}}
\end{aligned}$$

where in the last line, the first inequality holds with the promised probability, and the second inequality holds due to $r_0=\frac{C_3\omega^2(\mathcal{X})}{m(L\delta)^2}$ and $L\delta\gtrsim\log^{1/2}(\frac{m}{\omega^2(\mathcal{X})})$. $\square$

The Proof of Proposition 6. Under $r_1=c_1\delta$, we prove Proposition 6 by showing $\hat{T}^{(1)}\le\frac{r_1}{2}=\frac{c_1\delta}{2}$ and $\hat{T}^{(2)}\le\frac{r_1}{2}=\frac{c_1\delta}{2}$, which are sufficient because of (66). The constants in this proof can depend on c_1.

Show $\hat{T}_1\le\frac{c_1\delta}{2}$: We first write

$$\hat{T}^{(1)}=\sup_{\mathbf{u}\in\overline{\mathcal{X}}}\sup_{\mathbf{w}\in r_1^{-1}\mathcal{K}_{(r_1)}}\Big|\frac{1}{m}\big\langle\mathcal{Q}_\delta(\mathbf{Au}-\boldsymbol{\tau})-\mathbf{Au},\mathbf{Aw}\big\rangle\Big|.$$

We set $\rho=c_2\delta$ with small enough c_2, $\mathcal{A}=\overline{\mathcal{X}}$ and $\mathcal{C}=\frac{1}{r_1}\mathcal{K}_{(r_1)}$ in Lemma 4 and obtain that, if

$$m\ge C_3\left[\mathscr{H}(\overline{\mathcal{X}},c_2\delta)+\frac{\omega^2(\mathcal{K}_{(r_1)})}{r_1^2}+\frac{\omega^2(\overline{\mathcal{X}}_{(c_2\delta)})}{\delta^2}\right]\tag{111}$$

then with the promised probability we have $\hat{T}^{(1)}\le\frac{c_0\delta}{2}$. Note that $\overline{\mathcal{X}}\subset 2\mathcal{X}\subset 2\mathcal{K}$, hence (111) can be implied by $m\gtrsim\mathscr{H}(\mathcal{X},\frac{c_2\delta}{2})+\frac{\omega^2(\mathcal{K}_{(\delta)})}{\delta^2}$, which is assumed in the sample complexity (67).

Show $\hat{T}_2\le\frac{c_1\delta}{2}$: Next, by Lemma 5 and (67) that gives $m\ge\frac{C_5\omega(\mathcal{X})\omega(\mathcal{K}_{(r_1)})}{r_1\delta}$ with some large enough C_5 depending on c_0, we obtain $\hat{T}^{(2)}\le\frac{c_0r_1\delta}{2}$ with the promised probability. The proof is complete. $\square$

G Technical Lemmas

Lemma 6. *For random vectors $\mathbf{a}_i$ satisfying Assumption 1, there exists some absolute constant $L_0>0$ such that*

$$\mathbb{P}\Big(|\mathbf{a}_i^\top\mathbf{u}|\ge\frac{1}{2}\Big)\ge 2L_0,\quad\mathbb{E}|\mathbf{a}_i^\top\mathbf{u}|\ge L_0,\quad\forall\mathbf{u}\in\mathbb{S}^{n-1}.$$

Proof. We give a proof for this simple fact. Note that we only need to show $\mathbb{P}(|\mathbf{a}_i^\top\mathbf{u}|\ge 1/2)\ge 2L_0$ since this implies

$$\mathbb{E}|\mathbf{a}_i^\top\mathbf{u}|\ge\mathbb{E}[|\mathbf{a}_i^\top\mathbf{u}|\mathbb{1}(|\mathbf{a}_i^\top\mathbf{u}|\ge 1/2)]\ge\frac{\mathbb{P}(|\mathbf{a}_i^\top\mathbf{u}|\ge 1/2)}{2}\ge L_0.$$

For $\mathbf{a}_i$ satisfying Assumption 1, we have (12b) holds for some absolute constant $\bar{c}_1$. Hence by Paley-Zygmund inequality (e.g., Lemma 7.16 in [16]),

$$\mathbb{P}\big(|\mathbf{a}_i^\top\mathbf{u}|\ge 1/2\big)=\mathbb{P}\big(|\mathbf{a}_i^\top\mathbf{u}|^2\ge 1/4\big)\ge\frac{(3/4)^2}{\mathbb{E}|\mathbf{a}_i^\top\mathbf{u}|^4}\ge\frac{9}{256(\bar{c}_1)^4}.$$

Hence, we set $L_0:=\frac{9}{512(\bar{c}_1)^4}$ to yield the desired claim. $\square$

Lemma 7 (Bernstein's inequality, e.g., Theorem 2.10 in [4])**.** *Let $X_1,\cdots,X_n$ be independent real-valued random variables. Assume that there exist positive numbers v,c such that*

$$\sum_{i=1}^n\mathbb{E}|X_i|^q\le\frac{q!}{2}vc^{q-2}\quad\textit{for all integers } q\ge 2.$$

Then for any $t > 0$ we have

$$\mathbb{P}\left(\Big|\sum_{i=1}^{n}\left(X_i - \mathbb{E}(X_i)\right)\Big| \geq \sqrt{2vt} + ct\right) \leq 2\exp(-t).$$

Lemma 8 (See, e.g., [13]). *Let $\mathbf{a}_1, ..., \mathbf{a}_m$ be independent random vectors in $\mathbb{R}^n$ satisfying $\mathbb{E}(\mathbf{a}_i\mathbf{a}_i^\top) = \mathbf{I}_n$ and $\max_i \|\mathbf{a}_i\|_{\psi_2} \leq L$. For some given $\mathcal{W} \subset \mathbb{R}^n$ and $1 \leq \ell \leq m$, there exist constants C_1, c_2 depending only on L such that the event*

$$\sup_{\mathbf{w}\in\mathcal{W}} \max_{\substack{I\subset[m]\\|I|\leq\ell}} \left(\frac{1}{\ell}\sum_{i\in I}|\langle\mathbf{a}_i, \mathbf{w}\rangle|^2\right)^{1/2} \leq C_1\left(\frac{\omega(\mathcal{W})}{\sqrt{\ell}} + \mathrm{rad}(\mathcal{W})\log^{1/2}\left(\frac{em}{\ell}\right)\right)$$

holds with probability at least $1 - 2\exp(-c_2\ell\log(\frac{em}{\ell}))$, where $\mathrm{rad}(\mathcal{W}) = \sup_{\mathbf{w}\in\mathcal{W}}\|\mathbf{w}\|_2$.

Lemma 9 (Corollary 8.3 in [34]). *Let $\mathcal{K}$ be a star-shaped set, then for any $\mathbf{x} \in \mathcal{K}$ and $\mathbf{u} \in \mathbb{R}^n$ we have*

$$\|\mathcal{P}_{\mathcal{K}}(\mathbf{u}) - \mathbf{x}\|_2 \leq \max\left\{\phi, \frac{2}{\phi}\|\mathbf{u} - \mathbf{x}\|_{\mathcal{K}^\circ_{(\phi)}}\right\}, \quad \forall\phi > 0.$$

Lemma 10 (A Useful Parameterization, Lemma H.4 in [10]). *Given two distinct points $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, we let $\boldsymbol{\beta}_1 = \frac{\mathbf{u}-\mathbf{v}}{\|\mathbf{u}-\mathbf{v}\|_2}$ and can further find $\boldsymbol{\beta}_2$ satisfying $\|\boldsymbol{\beta}_2\|_2 = 1$ and $\langle\boldsymbol{\beta}_1, \boldsymbol{\beta}_2\rangle = 0$, such that*

$$\mathbf{u} = u_1\boldsymbol{\beta}_1 + u_2\boldsymbol{\beta}_2 \quad \textit{and} \quad \mathbf{v} = v_1\boldsymbol{\beta}_1 + u_2\boldsymbol{\beta}_2$$

hold for some (u_1, u_2, v_1) satisfying $u_1 > v_1$ and $u_2 \geq 0$. We have $v_1 = -u_1$ when $\|\mathbf{u}\|_2 = \|\mathbf{v}\|_2$ holds.

Lemma 11 (See, e.g., Exercise 8.6.5 in [38]). *Let $(R_{\mathbf{u}})_{\mathbf{u}\in\mathcal{W}}$ be a random process (that is not necessarily zero-mean) on a subset $\mathcal{W} \subset \mathbb{R}^n$. Assume that $R_0 = 0$, and $\|R_{\mathbf{u}} - R_{\mathbf{v}}\|_{\psi_2} \leq K\|\mathbf{u} - \mathbf{v}\|_2$ holds for all $\mathbf{u}, \mathbf{v} \in \mathcal{W} \cup \{0\}$. Then, for every $t \geq 0$, the event*

$$\sup_{\mathbf{u}\in\mathcal{W}} \left|R_{\mathbf{u}}\right| \leq CK\big(\omega(\mathcal{W}) + t\cdot\mathrm{rad}(\mathcal{W})\big)$$

with probability at least $1 - 2\exp(-t^2)$, where $\mathrm{rad}(\mathcal{W}) = \sup_{\mathbf{w}\in\mathcal{W}}\|\mathbf{w}\|_2$.

Lemma 12 (Concentration of Product Process, [26]; see also [19]). *Let $\{g_{\mathbf{u}}\}_{\mathbf{u}\in\mathcal{U}}$ and $\{h_{\mathbf{v}}\}_{\mathbf{b}\in\mathcal{V}}$ be stochastic processes indexed by two sets $\mathcal{U} \subset \mathbb{R}^p$ and $\mathcal{V} \subset \mathbb{R}^q$, both defined on a common probability space $(\Omega, A, \mathbb{P})$. We assume that there exist $K_{\mathcal{U}}, K_{\mathcal{V}}, r_{\mathcal{U}}, r_{\mathcal{V}} \geq 0$ such that*

$$\|g_{\mathbf{u}} - g_{\mathbf{u}'}\|_{\psi_2} \leq K_{\mathcal{U}}\|\mathbf{u} - \mathbf{u}'\|_2,\ \|g_{\mathbf{u}}\|_{\psi_2} \leq r_{\mathcal{U}},\ \forall\mathbf{u}, \mathbf{u}' \in \mathcal{U};$$
$$\|h_{\mathbf{v}} - h_{\mathbf{v}'}\|_{\psi_2} \leq K_{\mathcal{V}}\|\mathbf{v} - \mathbf{v}'\|_2,\ \|h_{\mathbf{v}}\|_{\psi_2} \leq r_{\mathcal{V}},\ \forall\mathbf{v}, \mathbf{v}' \in \mathcal{V}.$$

Suppose that $\mathbf{a}_1, ..., \mathbf{a}_m$ are independent copies of a random variable $\mathbf{a} \sim \mathbb{P}$, then for every $t \geq 1$ the *event*

$$\begin{aligned}
&\sup_{\mathbf{u}\in\mathcal{U}}\sup_{\mathbf{v}\in\mathcal{V}} \left|\frac{1}{m}\sum_{i=1}^{m} g_{\mathbf{u}}(\mathbf{a}_i)h_{\mathbf{v}}(\mathbf{a}_i) - \mathbb{E}[g_{\mathbf{u}}(\mathbf{a}_i)h_{\mathbf{v}}(\mathbf{a}_i)]\right| \\
&\leq C\left(\frac{(K_{\mathcal{U}}\cdot\omega(\mathcal{U}) + t\cdot r_{\mathcal{U}})\cdot(K_{\mathcal{V}}\cdot\omega(\mathcal{V}) + t\cdot r_{\mathcal{V}})}{m} + \frac{r_{\mathcal{U}}\cdot K_{\mathcal{V}}\cdot\omega(\mathcal{V}) + r_{\mathcal{V}}\cdot K_{\mathcal{U}}\cdot\omega(\mathcal{U}) + t\cdot r_{\mathcal{U}}r_{\mathcal{V}}}{\sqrt{m}}\right)
\end{aligned}$$

holds with probability at least $1 - 2\exp(-ct^2)$.